

IBM Z NetView
Version 6 Release 3

Installation: Getting Started



Note

Before using this information and the product it supports, read the information in [“Notices” on page 163.](#)

This edition applies to version 6, release 3 of IBM Z NetView (product number 5697-NV6) and to all subsequent versions, releases, and modifications until otherwise indicated in new editions.

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About this publication

The IBM Z® NetView® product provides advanced capabilities that you can use to maintain the highest degree of availability of your complex, multi-platform, multi-vendor networks and systems from a single point of control. This publication, *IBM Z NetView Installation: Getting Started*, provides information about installing the NetView program after creating the target libraries using SMP/E.

Intended audience

This publication is for system programmers who install the NetView program.

Publications

This section lists publications in the IBM Z NetView library and related documents. It also describes how to access NetView publications online and how to order NetView publications.

IBM Z NetView library

The following documents are available in the IBM Z NetView library:

- *Administration Reference*, SC27-2869, describes the NetView program definition statements required for system administration.
- *Application Programmer's Guide*, SC27-2870, describes the NetView program-to-program interface (PPI) and how to use the NetView application programming interfaces (APIs).
- *Automation Guide*, SC27-2846, describes how to use automated operations to improve system and network efficiency and operator productivity.
- *Command Reference Volume 1 (A-N)*, SC27-2847, and *Command Reference Volume 2 (O-Z)*, SC27-2848, describe the NetView commands, which can be used for network and system operation and in command lists and command procedures.
- *Installation: Configuring Additional Components*, GC27-2851, describes how to configure NetView functions beyond the base functions.
- *Installation: Configuring the NetView Enterprise Management Agent*, GC27-2853, describes how to install and configure the IBM Z NetView Enterprise Management Agent.
- *Installation: Getting Started*, GI11-9443, describes how to install and configure the base NetView program.
- *Installation: Migration Guide*, GC27-2854, describes the new functions that are provided by the current release of the NetView product and the migration of the base functions from a previous release.
- *IP Management*, SC27-2855, describes how to use the NetView product to manage IP networks.
- *Messages and Codes Volume 1 (AAU-DSI)*, GC27-2856, and *Messages and Codes Volume 2 (DUI-IHS)*, GC27-2857, describe the messages for the NetView product, the NetView abend codes, the sense codes that are included in NetView messages, and generic alert code points.
- *Programming: Pipes*, SC27-2859, describes how to use the NetView pipelines to customize a NetView installation.
- *Programming: REXX and the NetView Command List Language*, SC27-2861, describes how to write command lists for the NetView product using the Restructured Extended Executor language (REXX) or the NetView command list language.
- *Security Reference*, SC27-2863, describes how to implement authorization checking for the NetView environment.

- *Troubleshooting Guide*, GC27-2865, provides information about documenting, diagnosing, and solving problems that occur in the NetView product.
- *Tuning Guide*, SC27-2874, provides tuning information to help achieve certain performance goals for the NetView product and the network environment.
- *User's Guide: Automated Operations Network*, SC27-2866, describes how to use the NetView Automated Operations Network (AON) component, which provides event-driven network automation, to improve system and network efficiency. It also describes how to tailor and extend the automated operations capabilities of the AON component.
- *User's Guide: NetView*, SC27-2867, describes how to use the NetView product to manage complex, multivendor networks and systems from a single point.
- *User's Guide: NetView Enterprise Management Agent*, SC27-2876, describes how to use the NetView Enterprise Management Agent.
- *Using Tivoli System Automation for GDPS/PPRC HyperSwap Manager with NetView*, GI11-4704, provides information about the Tivoli® System Automation for GDPS®/PPRC HyperSwap® Manager with NetView feature, which supports the GDPS and Peer-to-Peer Remote Copy (PPRC) HyperSwap Manager services offering.
- *Licensed Program Specifications*, GC31-8848, provides the license information for the NetView product.
- *Program Directory for IBM Z NetView US English*, GI11-9444, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Japanese*, GI11-9445, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Enterprise Management Agent*, GI11-9446, contains information about the material and procedures that are associated with installing the IBM Z NetView Enterprise Management Agent.

The following books are archived:

- *Customization Guide*, SC27-2849, describes how to customize the NetView product and points to sources of related information.
- *Data Model Reference*, SC27-2850, provides information about the Graphic Monitor Facility host subsystem (GMFHS), SNA topology manager, and MultiSystem Manager data models.
- *Installation: Configuring Graphical Components*, GC27-2852, describes how to install and configure the NetView graphics components.
- *Programming: Assembler*, SC27-2858, describes how to write exit routines, command processors, and subtasks for the NetView product using assembler language.
- *Programming: PL/I and C*, SC27-2860, describes how to write command processors and installation exit routines for the NetView product using PL/I or C.
- *Resource Object Data Manager and GMFHS Programmer's Guide*, SC27-2862, describes the NetView Resource Object Data Manager (RODM), including how to define your non-SNA network to RODM and use RODM for network automation and for application programming.
- *SNA Topology Manager Implementation Guide*, SC27-2864, describes planning for and implementing the NetView SNA topology manager, which can be used to manage subarea, Advanced Peer-to-Peer Networking, and TN3270 resources.
- *User's Guide: NetView Management Console*, SC27-2868, provides information about the NetView management console interface of the NetView product.

Related publications

You can find additional product information on the IBM Z NetView web site at <https://www.ibm.com/us-en/marketplace/ibm-tivoli-netview-for-zos>.

For information about the NetView Bridge function, see *Tivoli NetView for OS/390® Bridge Implementation*, SC31-8238-03 (available only in the V1R4 library).

Terminology in this Library

The following terms are used in this library:

CNMCMD

For the CNMCMD member and the members that are included in it using the %INCLUDE statement

CNMSTYLE

For the CNMSTYLE member and the members that are included in it using the %INCLUDE statement

DSIOPF

For the DSIOPF member and the members that are included in it using the %INCLUDE statement

IBM® Tivoli Netcool®/OMNIbus

For either of these products:

- IBM Tivoli Netcool/OMNIbus
- IBM Tivoli OMNIbus and Network Manager

MVS™

For z/OS® operating systems

MVS element

For the base control program (BCP) element of the z/OS operating system

NetView

For the following products:

- IBM Z NetView version 6 release 3
- IBM Tivoli NetView for z/OS version 6 release 2 modification 1
- NetView releases that are no longer supported

PARMLIB

For SYS1.PARMLIB and other data sets in the concatenation sequence

VTAM®

For Communications Server - SNA Services

Unless otherwise indicated, topics to programs indicate the latest version and release of the programs. If only a version is indicated, the topic is to all releases within that version.

When a topic is made about using a personal computer or workstation, any programmable workstation can be used.

Using IBM Z NetView online help

The following types of IBM Z NetView mainframe online help are available, depending on your installation and configuration:

- General help and component information
- Command help
- Message help
- Sense code information
- Recommended actions

Accessing publications online

IBM posts publications for this and all other products, as they become available and whenever they are updated, to the IBM Knowledge Center at <https://www.ibm.com/support/knowledgecenter>. You can find IBM Z NetView documentation on [IBM Z NetView Knowledge Center](#).

Note: If you print PDF documents on other than letter-sized paper, set the option in the **Print** window that enables Adobe Reader to print letter-sized pages on your local paper.

Ordering publications

You can order many Tivoli publications online at <http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss>

You can also order by telephone by calling one of these numbers:

- In the United States: 800-426-4968
- In Canada: 800-879-2755

In other countries, contact your software account representative to order Tivoli publications. To locate the telephone number of your local representative, perform the following steps:

1. Go to <http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss>.
2. Select your country from the list and click the grey arrow button beside the list.
3. Click **About this site** to see an information page that includes the telephone number of your local representative.

Accessibility

Accessibility features help users with a physical disability, such as restricted mobility or limited vision, to use software products successfully. Standard shortcut and accelerator keys are used by the product and are documented by the operating system. Refer to the documentation provided by your operating system for more information.

For additional information, see the Accessibility appendix in the *User's Guide: NetView*.

Tivoli user groups

Tivoli user groups are independent, user-run membership organizations that provide Tivoli users with information to assist them in the implementation of Tivoli Software solutions. Through these groups, members can share information and learn from the knowledge and experience of other Tivoli users.

Support information

If you have a problem with your IBM software, you want to resolve it quickly. IBM provides the following ways for you to obtain the support you need:

Online

Please follow the instructions located in the support guide entry: <https://www.ibm.com/support/home/pages/support-guide/?product=4429363>.

Troubleshooting information

For more information about resolving problems with the IBM Z NetView product, see the *IBM Z NetView Troubleshooting Guide*. You can also discuss technical issues about the IBM Z NetView product through the NetView user group located at <https://groups.io/g/NetView>. This user group is for IBM Z NetView customers only, and registration is required. This forum is also monitored by interested parties within IBM who answer questions and provide guidance about the NetView product. When a problem with the code is found, you are asked to open an official case to obtain resolution.

Conventions used in this publication

This section describes the conventions that are used in this publication.

Typeface conventions

This publication uses the following typeface conventions:

Bold

- Lowercase commands and mixed case commands that are otherwise difficult to distinguish from surrounding text
- Interface controls (check boxes, push buttons, radio buttons, spin buttons, fields, folders, icons, list boxes, items inside list boxes, multicolumn lists, containers, menu choices, menu names, tabs, property sheets), labels (such as **Tip:**, and **Operating system considerations:**)
- Keywords and parameters in text

Italic

- Citations (examples: titles of publications, diskettes, and CDs)
- Words defined in text (example: a nonswitched line is called a *point-to-point line*)
- Emphasis of words and letters (words as words example: "Use the word *that* to introduce a restrictive clause."; letters as letters example: "The LUN address must start with the letter *L*.")
- New terms in text (except in a definition list): a *view* is a frame in a workspace that contains data.
- Variables and values you must provide: ... where *myname* represents...

Monospace

- Examples and code examples
- File names, programming keywords, and other elements that are difficult to distinguish from surrounding text
- Message text and prompts addressed to the user
- Text that the user must type
- Values for arguments or command options

Operating system-dependent variables and paths

For workstation components, this publication uses the UNIX convention for specifying environment variables and for directory notation.

When using the Windows command line, replace *\$variable* with *%variable%* for environment variables and replace each forward slash (/) with a backslash (\) in directory paths. The names of environment variables are not always the same in the Windows and UNIX environments. For example, %TEMP% in Windows environments is equivalent to \$TMPDIR in UNIX environments.

Note: If you are using the bash shell on a Windows system, you can use the UNIX conventions.

Syntax diagrams

The following syntax elements are shown in syntax diagrams. Read syntax diagrams from left-to-right, top-to-bottom, following the horizontal line (the main path).

- [“Symbols” on page xiii](#)
- [“Parameters” on page xiv](#)
- [“Punctuation and parentheses” on page xiv](#)
- [“Abbreviations” on page xv](#)

For examples of syntax, see [“Syntax examples” on page xv](#).

Symbols

The following symbols are used in syntax diagrams:



Marks the beginning of the command syntax.



Marks the end of the command syntax.



Indicates that the command syntax is continued on the next line.



Indicates that a statement is continued from the previous line.



Marks the beginning and end of a fragment or part of the command syntax.

Parameters

The following types of parameters are used in syntax diagrams:

Required

Required parameters are shown on the main path.

Optional

Optional parameters are shown below the main path.

Default

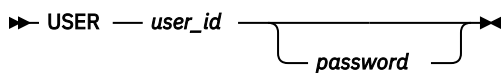
Default parameters are shown above the main path. In parameter descriptions, default parameters are underlined.

Syntax diagrams do not rely on highlighting, brackets, or braces. In syntax diagrams, the position of the elements relative to the main syntax line indicates whether an element is required, optional, or the default value.

When you issue a command, spaces are required between the parameters unless a different separator, such as a comma, is specified in the syntax.

Parameters are classified as keywords or variables. Keywords are shown in uppercase letters. Variables, which represent names or values that you supply, are shown in lowercase letters and are either italicized or, in NetView help, displayed in a differentiating color.

In the following example, the `USER` command is a keyword, the `user_id` parameter is a required variable, and the `password` parameter is an optional variable.



Punctuation and parentheses

You must include all punctuation that is shown in the syntax diagram, such as colons, semicolons, commas, minus signs, and both single and double quotation marks.

When an operand can have more than one value, the values are typically enclosed in parentheses and separated by commas. For a single value, the parentheses typically can be omitted. For more information, see [“Multiple operands or values” on page xvi](#).

If a command requires positional commas to separate keywords and variables, the commas are shown before the keywords or variables.

When examples of commands are shown, commas are also used to indicate the absence of a positional operand. For example, the second comma indicates that an optional operand is not being used:

```
COMMAND_NAME opt_variable_1,,opt_variable_3
```

You do not need to specify the trailing positional commas. Trailing positional and non-positional commas either are ignored or cause a command to be rejected. Restrictions for each command state whether trailing commas cause the command to be rejected.

Abbreviations

Command and keyword abbreviations are listed in synonym tables after each command description.

Syntax examples

The following examples show the different uses of syntax elements:

- “Required syntax elements” on page xv
- “Optional syntax elements” on page xv
- “Default keywords and values” on page xv
- “Multiple operands or values” on page xvi
- “Syntax that is longer than one line” on page xvi
- “Syntax fragments” on page xvi

Required syntax elements

Required keywords and variables are shown on the main syntax line. You must code required keywords and variables.

➤ REQUIRED_KEYWORD — *required_variable* ➤

A required choice (two or more items) is shown in a vertical stack on the main path. The items are shown in alphanumeric order.

➤ REQUIRED_OPERAND_OR_VALUE_1
 REQUIRED_OPERAND_OR_VALUE_2 ➤

Optional syntax elements

Optional keywords and variables are shown below the main syntax line. You can choose not to code optional keywords and variables.

➤ _____ ➤
 OPTIONAL_OPERAND

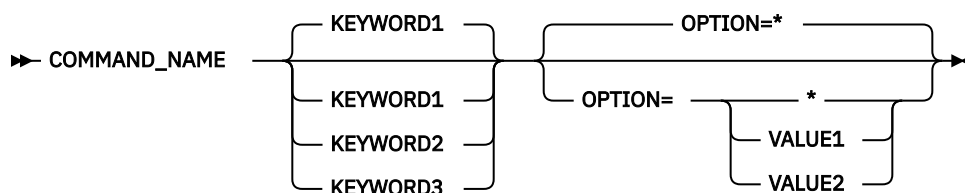
A required choice (two or more items) is shown in a vertical stack below the main path. The items are shown in alphanumeric order.

➤ _____ ➤
 OPTIONAL_OPERAND_OR_VALUE_1
 OPTIONAL_OPERAND_OR_VALUE_2

Default keywords and values

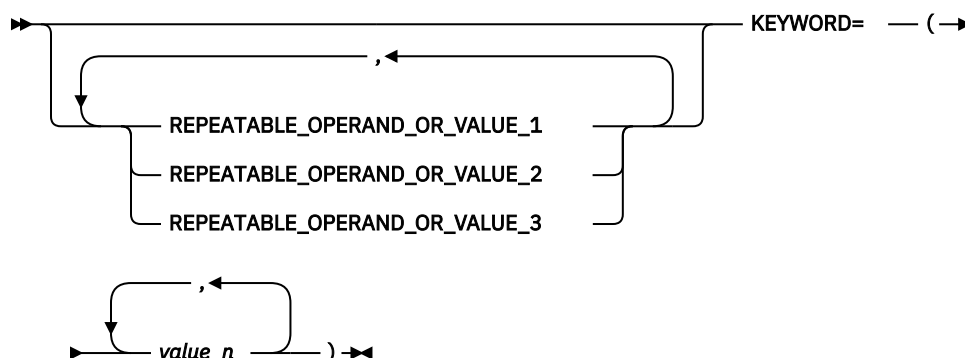
Default keywords and values are shown above the main syntax line in one of the following ways:

- A default keyword is shown only above the main syntax line. You can specify this keyword or allow it to default. The following syntax example shows the default keyword KEYWORD1 above the main syntax line and the rest of the optional keywords below the main syntax line.
- If an operand has a default value, the operand is shown both above and below the main syntax line. A value below the main syntax line indicates that if you specify the operand, you must also specify either the default value or another value shown. If you do not specify the operand, the default value above the main syntax line is used. The following syntax example shows the default values for operand OPTION=* above and below the main syntax line.



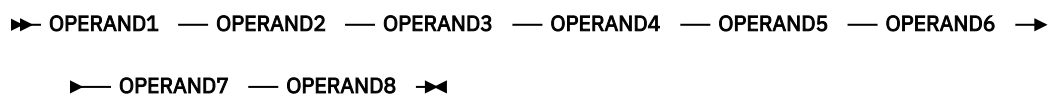
Multiple operands or values

An arrow returning to the left above a group of operands or values indicates that more than one can be selected or that a single one can be repeated.



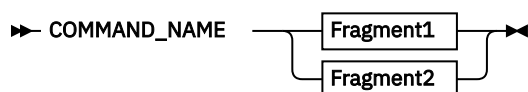
Syntax that is longer than one line

If a diagram is longer than one line, each line that is to be continued ends with a single arrowhead and the following line begins with a single arrowhead.



Syntax fragments

Some syntax diagrams contain syntax fragments, which are used for lengthy, complex, or repeated sections of syntax. Syntax fragments follow the main diagram. Each syntax fragment name is mixed case and is shown in the main diagram and in the heading of the fragment. The following syntax example shows a syntax diagram with two fragments that are identified as `Fragment1` and `Fragment2`.



Fragment1



Fragment2



Chapter 1. Understanding the NetView Installation

Before starting the installation and configuration procedures described in this book, perform the SMP/E installation of the NetView product that is listed in the NetView program directory.

This document is designed to guide you through the installation, configuration, and verification of the NetView program in a minimum amount of time. The assumption for this document is that this is the first time that the NetView program is being installed on this system. Initially, make only essential changes to the sample definitions provided with the NetView program. Your first objective is a fast and error-free installation.

This chapter describes:

- The organization of the NetView program
- Installation of the NetView program

Migrating Users: Refer to the *IBM Z NetView Installation: Migration Guide* for information on new and deleted functions from prior releases of the NetView product, as well as migration considerations for base functions.

NetView and Network Organization

A sample network comes with the NetView program. It contains definitions for both SNA and non-SNA resources. You can use this sample network as a model for building a new network or for expanding your existing network. For more information about the sample network, see [Appendix B, “NetView Samples Overview,”](#) on page 101.

Organization of the NetView Program

You can use the NetView program to manage your systems and networks from *one console* and can benefit users of small or large networks and small or large systems. Small installations with a limited staff can take advantage of NetView graphics and automation functions to simplify operator jobs and improve productivity. Graphics and automation can reduce, simplify, and in some cases even eliminate operator intervention, enabling faster and more reliable problem diagnosis and resolution.

These same advantages apply to large installations, contending with a higher volume of activity, and greater complexity. These users can *automate* repetitive and syntax-sensitive tasks to improve reaction time and reduce human errors. They can also automate knowledge-intensive tasks, such as problem determination, and develop expert systems, using the NetView programming interfaces.

[Figure 1 on page 2](#) shows the relationship between NetView components.

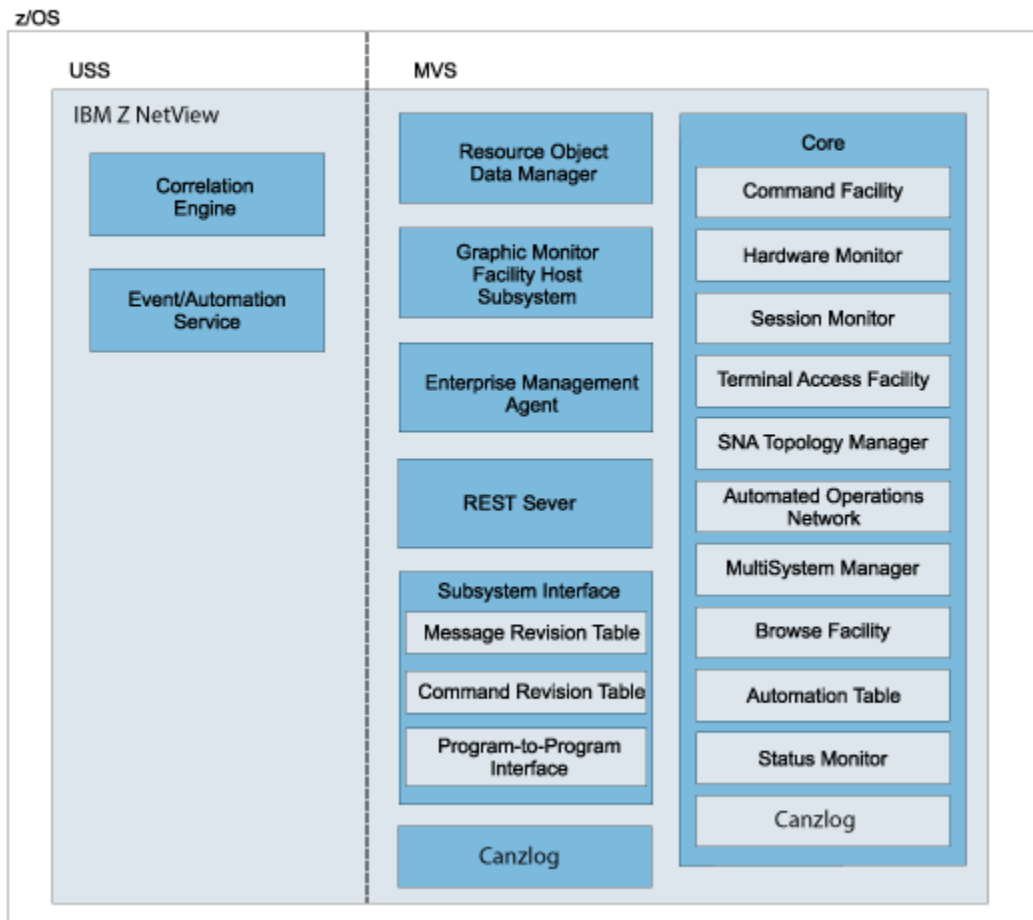


Figure 1. NetView Program Host Components

The NetView program includes the following components:

- The command facility is used to send commands and receive messages. The command facility also provides base functions and services for other components such as intercomponent communication, presentation services, database services, and automation facilities.
- The hardware monitor component collects and displays events and statistical data for both hardware and software to identify failing resources in a network. It provides probable cause and recommended actions so that operators can perform problem determination more efficiently.
- The session monitor component provides information about SNA sessions (subarea and Advanced Peer-to-Peer Networking) including session partner identification, session status, connectivity of active sessions, and response time data. The session monitor also provides session trace data, route data, and Virtual Telecommunications Access Method (VTAM) sense code information for problem determination.
- The terminal access facility (TAF) provides operator control of any combination of CICS®, IMS, TSO, and other subsystems from one terminal. The operator does not have to log off or use a separate terminal for each subsystem. The subsystem can be in the same domain or in another domain.
- The SNA topology manager dynamically collects topology and status of Advanced Peer-to-Peer Networking and subarea resources. This data is stored in the Resource Object Data Manager (RODM) for display by the NetView management console.

The topology agent supplies information consisting of the SNA nodes in a network, the Advanced Peer-to-Peer Networking transmission groups (TGs) between them, and the underlying logical links and ports supporting the TGs, in response to requests from the manager application.

- Automated Operations Network (AON) uses NetView automation facilities to automate the monitoring and recovery of both TCP/IP and SNA network resources. AON can monitor messages and alerts, and then automatically perform recovery actions. AON also provides an automated help desk to assist with resolving network problems, and generates reports so that you can monitor how well your automation is working.

AON provides default policy definitions that enable automation, without lengthy configuration, as soon as AON is enabled.

- MultiSystem Manager provides for the management of distributed resources from the NetView program. The NetView operator can use MultiSystem Manager to view and manage resources that are identified and managed locally by products. The topology and status of these resources are dynamically managed through RODM and the graphical workstation components of the NetView program.
- The browse facility is used to view local or remote NetView data set members including the Canzlog log, NetView log, NetView parameter files, and NetView panels.
- With the NetView automation table, you can specify processing options for incoming messages and MSUs and issue automatic responses. The table contains a sequence of statements that define the actions that the NetView program can take in various circumstances. The automation table is one of several components that provide automation capabilities.
- The status monitor component provides status information about SNA subarea network resources.
- Resource Object Data Manager (RODM) is an object-oriented data cache. Objects in RODM can represent resources in your network. The data cache is located entirely in the memory of the host processor for fast access to data and high transaction rates. RODM can contain approximately 2 million objects, providing support for large and growing networks.

MultiSystem Manager provides for the management of distributed resources from the NetView program. The NetView operator can use MultiSystem Manager to view and manage resources that are identified and managed locally by an Open agent created by you or a vendor. The topology and status of these resources are dynamically managed through RODM and the graphical workstation components of the NetView program.

Additionally, authorized operators can use the RODMVIEW command to display, create, update, and delete classes, objects, fields, and relationships in RODM.

- The NetView Graphic Monitor Facility host subsystem component maintains the status of resources in RODM and supplies the NetView Management Console with information about RODM resources. It works with RODM and the NetView Management Console to display graphic views of networks and to issue commands to resources that you select from a NetView Management Console view.

The Graphic Monitor Facility host subsystem works with the SNA topology manager and the NetView Management Console to manage SNA resources. It works with MultiSystem Manager and the NetView Management Console to manage non-SNA resources.

- The Z NetView Enterprise Management Agent provides information that you can use to manage your network from the Tivoli Enterprise Portal using sampled and real-time data. The data provides information about network resources and outages, using situations and expert advice, as well as trends in your network through the use of historical data. You can issue NetView, VTAM, and z/OS commands directly from the Tivoli Enterprise Portal to provide instant display and troubleshooting capabilities. The NetView agent enables you to manage both availability and performance data from the Tivoli Enterprise Portal using cross-product links to selected z/OS OMEGAMON® XE agents.
- The subsystem interface is used to receive system messages and to enter system commands. With extended multiple console support (EMCS) consoles, the subsystem interface is used to receive commands, but not messages. In a single system, multiple NetView programs can use the subsystem interface. Each NetView program that uses the subsystem interface requires a NetView subsystem address space in addition to the NetView application address space.

You can use the message revision table to intercept z/OS messages before they are displayed, logged, automated, or routed through your sysplex. With this table, you can make decisions about a message based on its message ID, job name, and other properties and can revise or suppress a message or take certain actions. The message revision table is one of several components that provide automation capabilities.

You can use the command revision table to intercept z/OS commands and make simple modifications inline, without needing to transfer the command to the NetView application address space. Commands can be deleted; parameters and keywords can be added, removed, or modified; nicknames can be expanded (such as creating new command or parameter synonyms); and explanatory WTO messages can be issued. The command revision table is one of several components that provide automation capabilities.

The program-to-program interface (PPI) is an address space provided by the NetView program to enable application programs to communicate with the NetView program and other applications running in the same host. When an application calls the PPI using its application programming interface (API), the request is synchronous.

- The correlation engine correlates multiple events over time, based on duplicates, thresholds, presence or absence of specific events, and other user-specified criteria. The correlation engine is one of several components that provide automation capabilities.
- The Event/Automation Service (E/AS) serves as a gateway for event data between the IBM Z NetView management environment, IBM Tivoli Netcool/OMNIbus probes, and SNMP managers and agents. With this gateway function, you can manage all network events from the management platform of your choice.
- The NetView REST Server is an OpenAPI-conforming server that handles front-end requests from applications and provides RESTful APIs that are suited for industry standardization to interact with automation and network resources.

The server can be accessed within Zowe™, hosted by the Open Mainframe Project, or independently from an application. The server runs in UNIX System Services (USS). APIs are available in the following areas:

- Canzlog messages and message attributes
- Automation table statements and automation members
- Sysplex connection distribution statistics for distributed dynamic virtual IP addresses (DDVIPAs) and ports
- NetView task health
- NetView domains accessible to an Enterprise Master NetView program or a Sysplex Master NetView program
- NetView commands

Swagger documentation can be accessed through the Zowe API Mediation Layer or directly from the server.

IBM Server Management Unite Automation V1.1.7 is available with IBM Z NetView V6.3.

For more information about NetView components, see [Appendix A, “NetView Functions and Tasks,” on page 89](#).

Naming Conventions

The NetView program uses the following types of data sets:

- System data sets
- NetView user partitioned data sets
- NetView VSAM data sets
- NetView sequential data sets
- UNIX system services (USS) file system data sets

System data sets use the high-level qualifier NETVIEW.V6R3M0 as the default naming convention. This high-level qualifier is specified as &SQ1 in the NetView Job Control Language (JCL) samples.

NetView user partitioned data sets use the following default naming convention:

- The high-level qualifier is NETVIEW.V6R3USER, specified as &Q1 in the NetView JCL.
- The middle-level qualifier is the host NetView domain.
- The low-level qualifier is the library name.

For example, with data set NETVIEW.V6R3USER.CNM01.DSIPARM, the high-level qualifier is NETVIEW.V6R3USER, the middle-level qualifier is CNM01, and the low-level qualifier is DSIPARM.

NetView VSAM data sets have the following default naming convention:

- The high-level qualifier is NETVIEW, specified as &VQ1 in the NetView JCL.
- The middle-level qualifier, when specified, is the host NetView domain.
- In most cases, the low-level qualifier is defined by the DSTINIT primary VSAM DD name (PDDNM) and secondary VSAM DD name (SDDNM) for the initialization member associated with a specific NetView component.

For example, with NETVIEW.CNM01.DSILOGP, the high-level qualifier is NETVIEW, the middle-level qualifier is CNM01, and the low-level qualifier is DSILOGP.

NetView sequential data sets have the following default naming convention:

- The high-level qualifier is NETVIEW, specified as &TQ1 in the CNMPROC (CNMSJ009) procedure.
- The middle-level qualifier is the host NetView domain.

For example, with NETVIEW.CNM01.CNMDVIPP, the high-level qualifier is NETVIEW, the middle-level qualifier is CNM01, and the low-level qualifier is CNMDVIPP.

You can substitute your own values for &SQ1, &Q1, &VQ1, and &TQ1 in the JCL samples and procedures that are supplied with the NetView product.

The NetView program also uses zSeries file system (zFS) data sets relative to z/OS UNIX system services as described in the NetView program directory. The NetView program allocates the following directories:

```
./usr/lpp/netview/v6r3/bin
./usr/lpp/netview/v6r3/mibs
./usr/lpp/netview/v6r3/lib
./usr/lpp/netview/v6r3/samples
./usr/lpp/netview/v6r3/samples/at-tls
./usr/lpp/netview/v6r3/samples/properties
./usr/lpp/netview/v6r3/www
./usr/lpp/netview/v6r3/www/img
```

Installation Package

If you are installing the NetView program for the first time, you will find familiar IBM fulfillment methods (such as Shop zSeries), installation tools (such as SMP/E or InstallShield), and installation documentation, including a program directory.

Product media (or their electronic representation) are in the standard IBM relfile format that IBM software manufacturing uses to create the images for installation systems such as ServerPac and CBPDO. If you receive your product through the CBPDO system, maintenance is included on the media for you to install. If you receive your product as part of a ServerPac or SystemPac system, maintenance is preinstalled.

For detailed information about the installation package contents, see the NetView program directory.

Workstation-based NetView code is shipped in two formats:

- DVD

- Tivoli website:

<http://www.ibm.com/software/tivoli/products/netview-zos/>

For service updates to workstation-based NetView code, see the Tivoli website.

NetView Program Directory

The *Program Directory for IBM Z NetView, GI11-9444 (English) or GI11-9445 (Japanese)* is shipped with the product and is also available on the IBM Z NetView Knowledge Center. The Program Directory provides instructions to install the NetView product to the SMP/E libraries. In addition, it contains helpful hints for a problem-free installation. It also lists the program temporary fixes (PTFs) that enable other products to work properly with NetView. Read the NetView program directory to help ensure a problem-free installation.

Preventive Service Planning (PSP)

Before you install the NetView program, make sure that you review the current Preventive Service Planning (PSP) information. For information on how to access PSP information, see the *Program Directory for IBM Z NetView*.

Applying a Required PTF

The NetView program directory lists APARs from the previous release that have been created for z/OS products to work with this NetView release.

The program directory also lists APARs from the previous NetView releases that are incorporated into the release. Check with the IBM Support Portal to obtain the latest available service.

Estimating Required Resources

Refer to the *IBM Z NetView Tuning Guide* and the NetView program directory for information about estimating the hardware and software resources required to install NetView. If you cannot determine how to estimate the resource types, call IBM Software Support at 1-800-237-5511 or contact your IBM Software Support account representative.

Installation and Configuration Procedures

Installation and configuration steps described in this book follow this order:

1. Installing the NetView program from the distribution media using z/OS SMP/E with the help of the NetView program directory
2. Preparing your MVS system for the NetView program
3. Customizing and running sample jobs to install the basic functions of the NetView program and the sample network
4. Preparing VTAM system definitions
5. Starting the NetView program and verifying your installation.

Use the sample network as a model for building a new network or for expanding your existing network. All VTAM, NetView, and NCP definition statements are supplied as samples. The CNMSAMP library provides

the JCL and data necessary to build the entire network in subarea 01. Also included, is network NETB which shows a multiple-network environment. Network NETB has one host subarea, 01.

This book assumes that you are installing NetView using the default values provided in this manual and in the sample files (installation in subarea 01 and using domain CNM01 for network NETA). If you are installing the NetView program in a subarea other than 01 or a domain other than CNM01, see the sample data set members for directions on what to change.

Installation Results

When you complete the procedures described in this book, the following functions are operational:

- Base NetView components including the command facility, hardware monitor, and session monitor
- Automation table to respond automatically to messages and MSUs, including alerts
- Canzlog log that stores NetView and system messages
- NetView log that stores messages and commands
- Save / Restore VSAM facility to save critical information across NetView restarts
- NetView to NetView remote command processing (RMTCMD)
- TCP/IP connection management
- Discovery manager
- XCF services

When the NetView program is installed and functioning correctly, many NetView functions are not yet active.

If you want information about...	Refer to...
How to adapt the NetView system to your own production environment	<i>IBM Z NetView Installation: Configuring Additional Components</i>
How to complete your installation for graphics support	<i>IBM Z NetView Installation: Configuring Graphical Components</i>
How to install the NetView agent	<i>IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent</i>

Chapter 2. Preparing the MVS System

Follow these steps to prepare your MVS system if you are installing the NetView program:

- Use SMP/E to load the NetView V6.3 libraries.
- Update the number of Language Processor (REXX) environments.
- Update SYS1.PARMLIB members.
- Modify the IEFUSI exit.
- Review the attributes assigned to extended multiple console support (EMCS) consoles.

The NetView program consists of the following address spaces:

- The base NetView program including the following components:
 - Command facility
 - Session monitor
 - Status monitor
 - Hardware monitor
 - Automated Operations Network (AON)
- NetView subsystem address space, which may include the program-to-program interface (PPI)
- Event/Automation Service
- Graphic Monitor Facility host subsystem (GMFHS)
- Resource Object Data Manager (RODM)
- Z NetView Enterprise Management Agent
 - The Z NetView Enterprise Management Agent is a separately orderable feature.

This book guides you through the installation and configuration of base NetView and the NetView subsystem address application.

If you want information about...	Refer to...
RODM and GMFHS installation (including applications dependent on GMFHS such as the MultiSystem Manager and the SNA Topology Manager)	<i>IBM Z NetView Installation: Configuring Graphical Components</i>
Z NetView Enterprise Management Agent installation	<i>IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent</i>
Event/Automation Service installation	<i>IBM Z NetView Installation: Configuring Additional Components</i>
AON configuration	<i>IBM Z NetView Installation: Configuring Additional Components</i>

Using SMP/E to Load the Libraries

Use the information in the NetView program directory to load the NetView libraries from the distribution media.

Modifying the Maximum Number of Language Processor (REXX) Environments for NetView

One of the most powerful features of the NetView program is the ability to run REXX code in the NetView environment. Many NetView components and base NetView functions exercise code that has been written in REXX. The NetView program also contains several parts that make use of the Data REXX function. Use the Data REXX function to include REXX instructions and functions in data files. To initialize the NetView program, you might need to adjust the maximum number of language processor environments defined by TSO that the system can initialize for the NetView address space.

If you want information about...	Refer to...
Language Processor (REXX) Environments for NetView	<i>IBM Z NetView Installation: Configuring Additional Components</i>

Updating SYS1.PARMLIB Members

Review the following members in the SYS1.PARMLIB data set:

- [“LPALSTxx” on page 10](#)
- [“PROGxx” on page 11](#)
- [“IEFSSNxx” on page 12](#)
- [“IEASYSxx” on page 14](#)
- [“IEASYMxx” on page 15](#)
- [“IKJTSOxx” on page 16](#)
- [“COUPLExx” on page 16](#)
- [“SMFPRMxx” on page 17](#)

The xx that is used in the member names in this section corresponds with the xx specified in the IEASYSxx member of the SYS1.PARMLIB library.

LPALSTxx

The LPALST member contains the names of program libraries you want the system to concatenate to SYS1.LPALIB. This concatenation is called the LPALST concatenation, which the system uses to build the pageable link pack area (PLPA).

Add the SCNMLPA1 NetView library to the LPALST member.

Data sets to be concatenated to the SYS1.LPALIB data set must be cataloged in the system master catalog or in a user catalog identified in the LPALST member. If the SCNMLPA1 target data set is not cataloged in the system master catalog but is cataloged in a user catalog, specify (in parenthesis immediately following the data set name) the 1- to 6-character VOLSER of the pack on which the SCNMLPA1 data set resides, for example:

```
NETVIEW.V6R3M0.SCNMLPA1(NVPTFM)
```

In this case, the system verifies that the NETVIEW.V6R3M0.SCNMLPA1 data set has a user catalog entry pointing to the NVPTFM volume.

If you want information about...	Refer to...
LPALSTxx	<i>z/OS MVS Initialization and Tuning Reference</i>

PROGxx

The PROGxx member contains the names of program libraries that you want the system to concatenate to the SYS1.LINKLIB data set and define as authorized with the Authorized Program Facility (APF). The PROGxx member can also add exit routines that are used by the NetView program to the MVS dynamic exits facility.

Usage note: Use PROGxx instead of using both LNKSTxx (to concatenate program libraries to SYS1.LINKLIB) and IEAAPFxx (to authorize program libraries to APF).

LNKLST concatenation

If you are using LNKST instead of a STEPLIB DD statement in any of your NetView JCL members (for example, your startup procedure), update your PROGxx member in the SYS1.PARMLIB data set to include the following program libraries as extensions to the SYS1.LINKLIB data set:

- NETVIEW.V6R3M0.SCNMMJPN (if the Japanese NetView program is being installed)
- NETVIEW.V6R3M0.SAQNLINK
- NETVIEW.V6R3M0.CNMLINK
- NETVIEW.V6R3M0.SCNMLNKN

Usage notes:

1. If you installed the IBM Z NetView for Continuous Availability product, make sure that the modules in SAQNLINK library are loaded prior to those in the CNMLINK library.
2. The SCNMLNKN data set is used for RODM trace, MVS command exit, and DSIPHONE (which is used by CMDSERV and by the UNIX, TSO command, and REST servers). The SCNMLNKN data set is not referenced in the NetView program samples; JCL for those samples must specify the SCNMLNKN data set on the STEPLIB DD statement if it is not included in the PROGxx member.

APF authorization

The following libraries must be authorized with the APF facility. To be authorized, a library name and volume ID must be in the list of authorized libraries in the PROGxx member of SYS1.PARMLIB.

1. If these are not already authorized, authorize all the libraries from the //STEPLIB, //VTAMLIB, and //NCPLOAD concatenations in your VTAM start procedure. The VTAM start procedure, CNMSJ008 (CNMNET), in NETVIEW.V6R3M0.CNMSAMP is the default member shipped with the NetView program.
2. Authorize all the libraries from the //STEPLIB concatenation in the following procedures:
 - NetView start procedure, CNMSJ009 (CNMPROC)
 - NetView procedure to start the subsystem interface (SSI) address space, CNMSJ010 (CNMPSSI)
 - RODM start procedure, EKGXRODM
 - GMFHS start procedure, CNMSJH10 (CNMGMFHS)
 - Event/Automation Service (E/AS) start procedure, IHSAEVNT

These procedures are located in the NETVIEW.V6R3M0.CNMSAMP samples library.

3. Authorize the following NetView libraries:
 - NETVIEW.V6R3M0.SAQNLINK
 - NETVIEW.V6R3M0.SCNMLNK1
 - NETVIEW.V6R3M0.SCNMLPA1
 - NETVIEW.V6R3M0.CNMLINK
 - NETVIEW.V6R3M0.SCNMLNKN
4. If you are installing the NetView Japanese program, authorize the SCNMMJPN data set. This data set is listed in the //STEPLIB concatenation of CNMPROC.

5. Add the SEAGLPA data set (REXX/370 library) or the SEAGALT data set (REXX alternate library) to the list of APF-authorized libraries. For information about which data set to use, see the comments in the NetView start procedure, CNMPROC.
6. If you plan to use the NetView agent, authorize the following additional libraries:
 - RKANMOD
 - RKANMODL
 - RKANMODU
 - Any runtime libraries that are concatenated in the STEPLIB DDNAME and in the RKANMODL DDNAME of the IBMNA and IBMDS started tasks
7. If you plan to use the Web Services Gateway function, add the SIXMLOD1 runtime library (XML toolkit) to the list of APF-authorized libraries.

Usage note: If you specify an unauthorized library on a STEPLIB or concatenate unauthorized libraries with authorized libraries, all libraries are treated as if they are unauthorized.

Exit routines

Update the PROGxx member if you use the following exit routines:

1. The MVS IEFACRT SMF installation exit receives control from the system when a job or job step ends, either normally or abnormally. The NetView program provides an IEFACRT sample exit (CNMSMF3E) that passes data across the PPI to a receiver which issues a message that can be automated using the NetView automation facilities. If you plan to use the CNMSMF3E sample as an IEFACRT exit routine, associate the CNMSMF3E sample to the IEFACRT exit by adding the following statement to your PROGxx member:

```
EXIT ADD EXITNAME(SYS.IEFACRT) MODNAME(CNMSMF3E)
```

More than one exit routine can be defined for the IEFACRT exit, so there might be more than one EXIT statement for the SYS.IEFACRT exit in your PROGxx member.

2. You can use the Canzlog log to view the system log messages that were issued before the NetView subsystem was initialized. To enable these preinitialization messages to be written to the Canzlog log, add this statement to the PROGxx member:

```
EXIT ADD EXITNAME(CNZ_MSGTOSYSLOG) MODNAME(DSI4LCUI)
```

The preinitialization messages are logged with an ASTYPE value of E.

Note: CNZ_MSGTOSYSLOG exit routines receive control only when SYSLOG is enabled as a hardcopy device. If SYSLOG is not enabled as a hardcopy device, such as if only OPERLOG is used, no preinitialization messages are written to the Canzlog log.

Additional information

If you want information about...	Refer to...
APF authorization	<i>z/OS library</i>
CNMSMF3E sample	<i>IBM Z NetView Installation: Configuring Additional Components</i>
NetView agent	<i>IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent</i>

IEFSSNxx

The IEFSSNxx member in the SYS1.PARMLIB data set contains parameters that define the secondary subsystems during MVS system initialization. Each SUBSYS statement in the IEFSSNxx member contains parameters that define a single subsystem.

Add the names in [Table 1 on page 13](#) to the IEFSSNxx member in the SYS1.PARMLIB data set.

Table 1. Subsystem Names		
Subsystem Name	Used by:	Procedures Name
CNMP	NetView subsystem (SSI)	CNMPSSI
EKGX	RODM subsystem	EKGXRODM

Usage notes:

1. The name of the subsystem for the NetView program is 4 characters and must be the first 4 characters of the procedure names that you use to start the following address spaces:

- NetView subsystem (sample CNMSJ010)
- NetView application (sample CNMSJ009)

For example, if both procedure names begin with CNMP, you can code the following entry in the IEFSSNxx member:

```
SUBSYS SUBNAME(CNMP) INITRTN(DSI4LSIT)
```

Create one entry for each instance of the NetView program that you plan to start on this system. Make sure that you specify the INITRTN value (DSI4LSIT) for each entry.

2. Any subsystem name that you define for the NetView program can be used only for the NetView program. If you choose a subsystem name that was previously used, restart MVS before using the subsystem name. If the NetView program is started as a job rather than as a task, the job name must have the same first four characters as the subsystem interface and startup procedure.
3. Consider the following conditions before deciding where to place the NetView subsystem name in the IEFSSNxx member:
 - If you place the NetView subsystem name after other subsystem names in the IEFSSNxx member, all MVS messages and commands received by the NetView subsystem are affected by the changes made by the other subsystems that are listed before the NetView subsystem.
 - If you are defining more than one NetView subsystem (SSI) in the IEFSSNxx member, the first NetView subsystem must come before the BEGINPARALLEL keyword. Any remaining NetView subsystems can be defined following the BEGINPARALLEL keyword. If you are defining only one NetView subsystem, the NetView subsystem can be placed before or after the BEGINPARALLEL keyword in the IEFSSNxx member.
 - By default, the NetView program does not send unsolicited foreign messages (those originating at another MVS LPAR in the same sysplex that are not specifically destined for an EMCS console owned by a NetView operator) to Canzlog. To have foreign messages eligible to be logged in Canzlog, specify the following parameter on the first SUBSYS statement that contains INITRTN(DSI4LSIT):

```
INITPARM('LOGFORN')
```

Note that the NetView subsystem initialization routine ignores any value specified on INITPARM on a SUBSYS statement containing INITRTN(DSI4LSIT) that is not the first such statement read by MVS.

4. The DSI4LSIT initialization routine allocates the CNMCANZO data space where MVS and NetView messages are stored. The data space wraps when it reaches its current usable size. CNMCANZO is the main Canzlog data space, and by default is allocated with a static data space size of 2 GB. If the data space is specified as dynamic, it will allocate 128 MB of storage and grow by 8 MB as necessary accommodate message processing until it reaches its maximum size.

To explicitly define the data space size to be a maximum of 2GB, specify the following value on the INITPARM parameter on the first SUBSYS statement that contains INITRTN(DSI4LSIT):

```
INITPARM('FULLDS')
```

To restrict the data space size to a maximum of 512 MB, specify the following value on the INITPARM parameter on the first SUBSYS statement that contains INITRTN(DSI4LSIT):

```
INITPARM('TINYDS')
```

To explicitly define the data space size to be static, which means the entire data space storage will be used before wrapping occurs, specify the following value on the INITPARM parameter on the first SUBSYS statement that contains INITRTN(DSI4LSIT):

```
INITPARM('STATIC')
```

To explicitly define the data space size to be dynamic, which means only 128 MB of data space storage will be used before wrapping occurs, unless more space is needed to accommodate situations where logging of messages is much faster than automation of messages, specify the following value on the INITPARM parameter on the first SUBSYS statement that contains INITRTN(DSI4LSIT):

```
INITPARM('DYNAMIC')
```

Note that when DYNAMIC is used and Canzlog data is not being archived, the amount of messages that can be browsed is less than when the Canzlog data space is defined as STATIC.

The NetView subsystem initialization routine ignores any value specified on the INITPARM parameter on a SUBSYS statement that contains INITRTN(DSI4LSIT) that is not the first such statement read by MVS.

Multiple values can be specified on the INITPARM parameter. For example, if the TINYDS, DYNAMIC, and LOGFORN values are required, specify these values on the same INITPARM parameter:

```
INITPARM('TINYDS,DYNAMIC,LOGFORN')
```

Separate the values with a comma.



Warning: Proper planning for real and auxiliary storage must be performed before enabling the Canzlog data space, as system slowdown and hangs can occur with insufficient storage.

5. If you do not add the NetView subsystem to the subsystem name table, you can dynamically add it before starting the NetView subsystem. For information on dynamically adding the subsystem name to the table, see [“Starting the NetView Program Without Restarting the MVS System” on page 67.](#)

If you want information about...	Refer to...
IEFSSNxx	z/OS library
How to restart MVS	z/OS library
NetView agent	<i>IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent</i>

IEASYSxx

The IEASYSxx member in the SYS1.PARMLIB data set contains system parameters.

If you are using the status monitor, system automation, or Automatic Restart Manager (ARM) support, update the MAXUSER and RSVNONR values using the following steps:

- For the status monitor, determine the number of times you expect to stop and restart the NetView program over the life of the MVS IPL.
- For system automation, determine the number of times you expect to stop and restart the NetView subsystem interface (SSI) address space over the life of the MVS IPL.
- If both the preceding sets of conditions apply, add the values together that you computed.

Increase the number of address spaces available to MVS by this value by changing the MAXUSER or RSVNONR value, or both:

- Set MAXUSER to the number of ASIDs you want available at any one time.
- Set RSVNONR to the replacement values.

Usage notes:

1. The MVS program does not reuse any address space identifiers (ASIDs) that have established a program call interface. The NetView subsystem interface address space does reuse ASIDs that have established a program call interface. The RSVNONR value provides a way to specify replacement ASIDs for the ones lost.
2. The total of the values of MAXUSER, RSVNONR, and RSVSTRT, cannot exceed 32767. If you want a low MAXUSER value, be sure to provide a reasonably large value for RSVNONR.

If you are adding this system to a sysplex to enable the NetView program to use XCF Services, see *z/OS MVS Setting Up a Sysplex*.

If you are using ARM support, use the COUPLE system parameter to identify the COUPLExx member containing your DATA statements.

Set the PLEXCFG operand as appropriate for your environment.

If you want information about...	Refer to...
IEASYSxx	z/OS library

IEASYMxx

Add a SYSDEF statement in member IEASYMxx to identify user-defined system symbolic variables for the NetView program, including the TCP/IP application name, RODM name, and network ID.

Setting these symbolic variables can alleviate modification of many of the NetView initialization members unless some default parameter such as a TCP/IP port needs to be changed.

For example, you can define the following symbolic variables (these are the default names used in the NetView program):

```
SYSDEF SYMDEF(&CNMTCPN='tcpip_name')
SYSDEF SYMDEF(&CNMRODM='rodm_name')
SYSDEF SYMDEF(&CNMNETID='network_id')
```

Usage note:

If you need to produce an actual ampersand (&) in the text of members that support symbolic substitution, one easy method is to define the ampersand as follows in your system substitution tables.

```
SYSDEF SYMDEF(&AMP.='&')
```

You can use the symbol **&.** to specify an ampersand in this statement anywhere without risking an inadvertant symbol substitution.

The initialization members that use the symbolic variables are shown in this table.

Table 2. Symbolic Variable Usage by Initialization Members				
Member	Task	TCP/IP NAME &CNMTCPN	RODM NAME &CNMRODM	NETID &CNMNETID
CNMSTYLE	NetView initialization	X	X	X
DUIGINIT	GMFHS (Graphics)		X	
FLBSYSD	SNA topology manager		X	

Note: After you run job CNMSJBUP (see “Creating the Installation JCL Samples Library Using Job CNMSJBUP” on page 21), you can use sample CNMSJM12 in data set NETVIEW.V6R3USER.INSTALL as

an alternate method to do symbol substitution. CNMSJM12 replaces symbolic variables in NetView members.

If you do not define these symbolic variables in the IEASYMxx member, you can specify their values in the CNMSTYLE member. Additionally, you can use the CNMSJM12 sample to replace other symbolic variables that are used in your data set members.

Additional considerations:

1. You can use the &CNMDOMN variable to define the NetView domain to applications outside of the main NetView address space. If you use the A01APPLS (CNMS0013) sample, you can use the &CNMDOMN variable to provide the NetView domain to VTAM. If you do not set this variable, replace all occurrences of the &CNMDOMN variable in the A01APPLS sample with the NetView domain. Within the NetView address space, use the &DOMAIN local symbolic variable to specify the NetView domain.
2. RODM looks for an optional EKGCUST DD statement in the jobstream. The EKGCUST statement names the input customization member. If you defined a customization member, RODM reads it. If not, RODM provides defaults. You can use symbols to represent operands, values, and comments on the EKGCUST statement, if symbolic substitution is enabled on your system. Ensure that these symbols are defined in the IEASYMxx member.
3. The SNA topology manager is shipped with an initialization file (FLBSYSD). In addition, the FLBOSIDS, FLBSRT, and FLBEXV files can be used to further customize SNA topology manager views. If symbolic substitution is enabled on your system, you can use symbols in these initialization and customization files. Ensure that the symbols are defined in the IEASYMxx member.

If you want information about...	Refer to...
IEASYMxx	z/OS library
Use of symbols in RODM EKGCUST file	<i>IBM Z NetView Installation: Configuring Graphical Components</i>
Use of symbols in SNA topology manager initialization files	<i>IBM Z NetView Installation: Configuring Graphical Components</i>

IKJTSOxx

The IKJTSOxx member contains TSO/E settings, including command and program authorization information.

If you plan to use the NetView agent, add the KPDDSCO command in the authorized program (AUTHPGM) section:

```
AUTHPGM NAMES(KPDDSCO,pgm2,...)
```

IBM Tivoli Monitoring components require the KPDDSCO command to be authorized in the IKJTSOxx member for persistent data store processing.

If you want information about...	Refer to...
NetView agent	<i>IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent</i>

COUPLExx

The COUPLExx member contains cross-system coupling facility (XCF) parameters.

Add the following DATA statement to identify the automatic restart manager (ARM) couple data to XCF:

```
DATA TYPE(ARM) PCOUPLE(primary-dsname) ACOUPLE(alternate-dsname)
```

Initialize the primary and alternate ARM couple data sets after you create them.

If you are adding this system to a sysplex to enable the NetView program to use XCF Services, see *z/OS MVS Setting Up a Sysplex*.

If you want information about...	Refer to...
Initializing the primary and alternate ARM couple data sets	<i>z/OS library</i>

SMFPRMxx

To write the following records to the System Management Facility (SMF) log, ensure that member SMFPRMxx in the SYS1.PARMLIB data set is set up to collect type 37, type38, and type 39 SMF records accordingly:

- **SMF record type 37:** hardware monitor records
- **SMF record type 38:** authorization, task resource utilization, and command resource utilization records
- **SMF record type 38:** session monitor records

The SMF 38 subtype 4 record is written on an interval. Set the SMF interval value, and if desired, the SMF synchronization value.

If you plan to use the CNMSMF3E sample as an IEFACRT exit routine, verify that type 30 SMF records are set up to be collected and that the EXITS operand of the SYS specification includes the IEFACRT exit.

For example, the following statement causes SMF record types 0 through 255 to be collected and the IEFACRT exit to be called:

```
SYS(TYPE(0:255),EXITS(IEFU83,IEFU84,IEFU85,IEFACRT,IEFUJV,IEFUSI,  
IEFUJP,IEFUS0,IEFUJI,IEFUTL,IEFU29,IEFUAV),NOINTERVAL,NODETAIL)
```

If you want information about...	Refer to...
SMF record formats	<i>IBM Z NetView Application Programmer's Guide</i>
SMF record types	<i>z/OS library</i>
CNMSMF3E sample	<i>IBM Z NetView Installation: Configuring Additional Components</i>

Modifying the IEFUSI exit

You can use the IEFUSI exit to limit the region size and region limit, and to limit the size of data spaces and hiperspaces for jobs started on an MVS system. If you have the IEFUSI exit on your system, modify it as needed to accommodate the region sizes for the various address spaces of the NetView program. See [Table 3 on page 17](#) for a list of the data spaces that are created by NetView address spaces.

Table 3. Data spaces created by the NetView program	
Address Space	Data Spaces Created
CONNSEC	TCP/IP connection security (CONNSEC) creates a data space for each TCP/IP stack on your system.

Table 3. Data spaces created by the NetView program (continued)

Address Space	Data Spaces Created
Master Scheduler	Subsystem initialization for the NetView program creates a data space to contain Canzlog data. This data space is, by default, defined with 2GB of virtual storage, which can significantly impact real and auxiliary storage requirements. For more information about defining this data space, see the subsection named "IEFSSNxx" in the section named "Updating SYS1.PARMLIB Members" in this chapter.  Warning: Proper planning for real and auxiliary storage must be performed before enabling the Canzlog data space, as system slowdown and hangs can occur with insufficient storage.
NetView	• The NetView program creates an internal trace data space at initialization that is managed through the TRSIZE parameter in the CNMPROC startup procedure. • The Canzlog archiving function creates data spaces CNMCANZ1, CNMCANZ2, ..., CNMCANZ9, CNMCANZA that are used for browsing archived Canzlog data. The number of data spaces that are created is limited by the specification on the ARCHIVE.BROWSE.DATASPACEs or ARCHIVE.BROWSE.MAXDSPSIZE statement in the CNMSTYLE member. The data space(s) will be created as necessary to accommodate usage requirements, and their sizes can increase up to the defined limit. • TCP/IP connection management (TCPCONN) creates a data space for each TCP/IP stack on your system. • The IP packet trace (PKTS) function creates a data space for each TCP/IP stack on your system. • The OSA packet trace (OPKT) function creates a data space for each TCP/IP stack on your system. • If the DISCOVERY tower is enabled in the CNMSTYLE member, the real-time SMF data network management interface (SYSTCPSM) creates a data space for each TCP/IP stack on your system. • As of V6R3, if LOGSMF is enabled in the Command Statistics function, a data space is created to store Command Statistics data before being written into SMF. The size of the data space is specified by the CMDMON.INIT.SMFDSSIZE CNMSTYLE statement.
RODM	RODM creates and utilizes three data spaces. Note that RODM allocates a 2 gigabyte data space at initialization.
NetView agent	• The NetView agent creates a data space for each NetView program with which it communicates. • A second data space is created if the value of the NACMD.PERSIST statement in the CNMSTYLE member is greater than zero.

The IEFUSI exit is passed the address of the program name on input.

If you want information about...	Refer to...
Modifying the IEFUSI step initiation exit	z/OS library

Using EMCS Consoles within the NetView Program

The NetView program uses extended multiple console support (EMCS) consoles for tasks that issue MVS system operator commands.

EMCS consoles are dynamically defined. Any console names that you specify in the CONSOLxx member are not available for use as EMCS console names. Therefore, if you specify any console names in CONSOLxx, make sure that they are different from console names that you plan to use within the NetView program. For information on how to construct console names for use within the NetView program, see [“Assigning Unique EMCS Console Names” on page 55](#).

EMCS consoles can be assigned attributes. For more information about these attributes, see the *IBM Z NetView Security Reference*.

Planning to Restart the Target System

If your business policy and schedule allow it, restart the MVS system now to activate the changes. When restarting the MVS system, use the CLPA option, which enables the NetView program to access the new LPALIB members from NETVIEW.V6R3M0.SCNMLPA1.

If you cannot restart the MVS system at this time, you can manually activate the changes that are required to start the NetView program. For more information on manually activating the changes, see [“Starting the NetView Program Without Restarting the MVS System” on page 67](#). Plan to restart the MVS system at a later time to activate all the changes that you made to the MVS system.

Important: If you are implementing a user-defined RACF® resource class and a user-defined RACF router table for RODM, you must restart MVS.

If you want information about...	Refer to...
----------------------------------	-------------

Defining a user-defined RACF resource class and a user-defined RACF router table	<i>IBM Z NetView Security Reference</i> .
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Chapter 3. Preparing the NetView Program

This chapter describes the steps that help you prepare the NetView program. Each step gives you explicit directions for installing an operational NetView system. Unless otherwise stated, each step is required for a NetView system to be functional.

Creating the Installation JCL Samples Library Using Job CNMSJBUP

The CNMSJBUP job in the NETVIEW.V6R3M0.CNMSAMP data set copies the installation members and JCL samples in the CNMSAMP data set into the NETVIEW.V6R3USER.INSTALL data set. Not all of the NetView sample library is copied.

The installation members that are copied are described in this chapter. Use the USER.INSTALL data set to edit and submit installation JCL (from a data set other than an SMP/E-managed data set).

Follow these steps to create the installation JCL samples library:

1. Edit CNMSJBUP to match your system naming conventions, if necessary.
2. Review the variables in the comments supplied in this job. Change the &UNIT and &SER variables to match your installation, if needed.
3. Run CNMSJBUP.
4. Verify your return codes before continuing with the next step.

Using the Installation JCL Samples Library

For the remaining steps, edit and run the sample JCL in NETVIEW.V6R3USER.INSTALL.

Warning: If instead you use the sample JCL in the NETVIEW.V6R3M0.CNMSAMP data set to run the installation jobs described in this chapter, you might lose your existing customization when you apply maintenance.

Defining the User Catalog Using the CNMSJ001 Job

Job CNMSJ001 creates an ICF catalog and defines the ALIAS name NETVIEW as the high-level qualifier for the NetView data sets.

Run job CNMSJ001 if you did not define this alias name during the NetView program directory installation and you are going to use this high-level qualifier. To define the user catalog, perform the following steps:

1. Review the comments in the JCL. If you are using one of your existing catalogs, comment out the first step in CNMSJ001 and change the RELATE value in the second step accordingly. You can also change the &UNIT and &SER JCL variables to reflect a different DASD device type.
2. Run CNMSJ001.
3. Verify your return codes before continuing with the next step.

A return code of 12 can indicate a duplicate data set name exists from a previous run or release. The duplicate name might not require correction.

Allocating Data Sets Using the CNMSJ002 Job

The CNMSJ002 job allocates partitioned and sequential data sets. These data sets are used by the NetView system that you are installing. The data set names reflect the values for &DOMAIN and &Q1 that are supplied by the NetView product.

The partitioned data sets that are allocated by the CNMSJ002 job are listed in [Table 4 on page 22](#).

<i>Table 4. Partitioned Data Sets for the NetView Program</i>	
Data Set Name	Function
NETVIEW.V6R3USER.CNM01.CNMCLST	Contains user-modified command lists and REXX Execs
NETVIEW.V6R3USER.CNM01.CNMPNL1	Contains user-modified panels
NETVIEW.V6R3USER.CNM01.DSIARPT	Stores the output report that is produced from running tests of the automation table.
NETVIEW.V6R3USER.CNM01.DSIASRC	Contains the members to be used when testing the automation table. This file contains source messages and MSUs for testing the automation table.
NETVIEW.V6R3USER.CNM01.DSILIST	Stores the NetView reports, listings, files, and output from the security migration tool in addition to the reports from the CNMSTYLE report generator.
NETVIEW.V6R3USER.CNM01.DSIPARM	Contains user-modified NetView system definitions.
NETVIEW.V6R3USER.CNM01.DSIPRF	Contains user-modified NetView operator profiles.
NETVIEW.V6R3USER.CNM01.SDSIOPEN	Stores non-protected data set members.
NETVIEW.V6R3USER.CNM01.VTAMLST	Contains VTAM source definitions for the sample network.
NETVIEW.V6R3USER.VTAMLIB	Contains VTAM load modules for the sample network. Add the VTAMLIB data set to the list of authorized libraries in PROGxx or IEAAPFxx in SYS1.PARMLIB.

The sequential data sets that are allocated by the CNMSJ002 job are listed in [Table 5 on page 23](#).

Table 5. Sequential Data Sets for the NetView Program

Data Set Name	Function
NETVIEW.CNM01.CNMDVIPP NETVIEW.CNM01.CNMDVIPS	Contains distributed DVIPA statistics that are kept each time data collection is run for the DVIPA.DVTAD subtower. CNMDVIPP is the primary data set and CNMDVIPS is the secondary data set. Usage notes: • Use the default (or larger) size allocations for these data sets so that the default number of records specified by the DVIPA.STATS.Pri.MAXR and DVIPA.STATS.Sec.MAXR statements in the CNMSTYLE member do not cause the data sets to reach capacity and lose data. • If you use the CNMSJ002 job to allocate new CNMDVIPP and CNMDVIPS data sets to a system that already has a previous NetView program installed, the data sets are not cataloged.

The data sets allocated by the CNMSJ002 job are used to hold user data. Initially, they are empty data sets. In some cases, you might need to make modifications to the data set members provided with the NetView program. To do this, copy the members that you need to modify from the NETVIEW.V6R3M0 data set to the NETVIEW.V6R3USER.*domain* data set and then make modifications to the copied members in the user data set. This preserves the original member and protects your modifications when maintenance is applied. For example, to make changes to the CNMSTUSR member, copy the CNMSTUSR member from the NETVIEW.V6R3M0.DSIPARM data set to the NETVIEW.V6R3USER.CNM01.DSIPARM data set, and then make your modifications to the CNMSTUSR member in the user DSIPARM data set.

Although data set NETVIEW.V6R3USER.CNM01.DSIPRF is not used for operator profiles if an SAF security product is used exclusively for operator definitions, it is required to be present when the NetView address space is started. Because this data set is used for other members besides operator profiles, such as DSITCPRF.

Allocate the data sets in the following way:

1. Review the variables and associated comments in the CNMSJ002 job. Change the &UNIT and &SER variables to match your installation, if needed.
2. Change the &DOMAIN variable to match the NetView domain name you are installing.
3. Run CNMSJ002.
4. Verify your return codes before continuing with the next step.

Changing Sample Installation Jobs Using the CNMSJ000 Job

You can use the CNMSJ000 job to do the following tasks:

- Change the domain
- Create a VTAM APPL major node
- Change VSAM volume serial information

If you are installing the NetView program in a domain other than CNM01 for network NETA, or a subarea other than 01, CNMSJ000 converts the NetView samples in NETVIEW.V6R3M0.CNMSAMP and

NETVIEW.V6R3M0.DSIPARM and places them into NETVIEW.V6R3USER.*domain*.DSIPARM, NETVIEW.V6R3USER.INSTALL, and NETVIEW.V6R3USER.*domain*.VTAMLST.

Job CNMSJ000 also creates a VTAM APPL major node to define the NetView program to VTAM. If one of the sample network domains and subareas is chosen (CNM01, CNM02, CNM99 in NETA, or B01NV in NETB), the VTAM APPL major node already exists as part of the VTAM samples shipped with the NetView program. If the domain and subarea chosen are not one from the sample network, member NETVAPPL is created in data set NETVIEW.V6R3USER.*domain*.VTAMLST.

NETVAPPL can be used as part of your existing VTAM definitions if the resource naming convention used by NETVAPPL matches your system definitions. If the resource naming convention does not match your system definitions, you can edit NETVAPPL and your NetView definitions so that your resource names are consistent.

To change the domain, subarea numbers, and VSAM volume serial information:

1. Edit job CNMSJ000 and specify the domain ID (&DOMAIN) and subarea number (&SANUM) you want to use during your installation.
2. Update the parameters in [Table 6 on page 24](#) with your volume serial specifications:

Table 6. VSAM VOLSER Parameters		
Parameter	VSAM Cluster	Sample to be updated
V1	Base NetView	CNMSI101
V2 (obsolete)		
V3	Central Site Control Facility	CNMSI501
V4	RODM	EKGS1101
V5	AON	EZLSI101
V6	Saved Packet Trace	FKXSI101

Usage notes:

- a. These samples are used as input for job CNMSJ004. For more information, see [“Allocating VSAM Clusters Using the CNMSJ004 Job” on page 26](#).
 - b. If you specify NOVOLSER, the volume specification is removed in the VSAM allocation member. This allows the VSAM cluster to be allocated to any available volume.
3. Run CNMSJ000.
 4. Verify your return codes before continuing with the next step.

The remaining steps assume that you are installing the NetView program in subarea 01 and domain CNM01. If you are installing the NetView program in a subarea other than 01 and domain CNM01, make the appropriate substitutions throughout this installation.

Loading Members of Partitioned Data Sets Using the CNMSJ003 Job

The CNMSJ003 job copies selected members from the following data sets:

- NETVIEW.V6R3USER.INSTALL
- NETVIEW.V6R3M0.CNMSAMP
- NETVIEW.V6R3M0.DSIPARM
- NETVIEW.V6R3M0.CNMPNL1

With the exception of the SYS1.PROCLIB data set, the copied members are placed into the user data sets described in [“Allocating Data Sets Using the CNMSJ002 Job” on page 21](#).

[Table 7 on page 25](#) lists the steps in CNMSJ003.

Table 7. Copy Steps in CNMSJ003				
Step	Data set to which members are copied	IEBCOPY sample used by CNMSJ003	Procedure	Description
//PDS1	NETVIEW.V6R3USER. CNM01.VTAMLST	CNMSJI03		VTAM samples for network NETA and NETB.
//PDS2	SYS1.PROCLIB	CNMSJI10	CNMNET	Sample VTAM start procedure
			CNMPROC	NetView start procedure
			CNMPSSI	NetView subsystem interface start procedure
			CNMPRT	Print network procedure
			EKGXRODM	RODM start procedure
			EKGLOADP	Load RODM data cache procedure
			CNMNDEF	Status monitor preprocessor procedure
			CNMGMFHS	GMFHS start procedure
			IHSAEVNT	E/AS start procedure
//PDS3	NETVIEW.V6R3USER. CNM01.DSIPARM	EZLSI300		AON system definitions
//PDS4	NETVIEW.V6R3USER. CNM01.CNMPNL1	EZLSI301		AON panels

To copy members:

1. If necessary, edit the CNMSJ003 job to match your system naming conventions. You can change the specification for the Q2 variable on the PDS2 step to use a different high-level qualifier for PROCLIB.
2. Step PDS2 is initially commented out to prevent the overlaying of existing members in the target PROCLIB data set. Before running this step, ensure that the members that are copied do not exist in the PROCLIB data set, so they do not get replaced (see [Table 7 on page 25](#)).
3. Run the CNMSJ003 job.
4. Verify your return codes before continuing with the next step.

Reviewing Canzlog Archiving Allocation Parameters

The NetView Canzlog log is a NetView logging facility that contains NetView, system, and some job messages. Archiving of the Canzlog log is active by default. This archiving function copies the Canzlog log into sequential data sets that are dynamically allocated by the NetView program after NetView initialization.

The parameters used by the NetView program to allocate the Canzlog archive data sets are defined in the NetView CNMSTYLE configuration member. Allocation parameters are provided for the following sequential data sets:

- Message data sets
- Index data sets and Primary Index data set

Review the allocation parameters before starting the NetView program. For information on updating the CNMSTYLE member, see [“Configuring Allocation Parameters for Canzlog Archive Data Sets”](#) on page 54.

Usage note: If you decide to update the CNMSTYLE member at a later time, the following BNH803I message is issued during NetView initialization:

```
BNH803I VALUE *NONE* FOR KEYWORD ARCHIVE.HLQ CONFLICTS WITH VALUE YES FOR KEYWORD ARCHIVE.WRITE
```

The Canzlog archiving function is disabled and Canzlog data is not archived. By not setting up the archive data sets, you can lose message data when the Canzlog log wraps and writes over previously stored message data.

Allocating Data Sets for Canzlog Print

NetView provides functionality for printing Canzlog messages to a sequential data set or a member of a partitioned or partitioned extended data set by using the PRINT command. The data sets need to be allocated before you use PRINT. The following example shows attributes for defining a sequential data set:

```
Organization . . : PS
Record format . . : FB
Record length . . : 134
Block size . . . : 27872
1st extent cyls . . : 15
Secondary cyls . . : 2
```

The next example shows sample attributes for a partitioned or partitioned extended data set:

```
Organization . . . . : PO
Record format . . . . : FB
Record length . . . . : 134
Block size . . . . . : 27872
1st extent cylinders.. : 15
Secondary cylinders .. : 2
```

A default data set for use with the PRINT command can be defined in CNMSTYLE. For information about configuring CNMSTYLE to support Canzlog printing, see *Configuring parameters for Canzlog Print* in *Installation: Configuring Additional Components*.

Allocating VSAM Clusters Using the CNMSJ004 Job

The CNMSJ004 job allocates the VSAM clusters for various NetView facilities. [Table 8 on page 27](#) lists the VSAM cluster names associated with these facilities, and the names of the members containing information for each VSAM cluster.

Note: The records needed for the system to view databases as active data sets are added during NetView component initialization.

Consider the following information before running the CNMSJ004 job:

1. Review CNMSID01 and each of the VSAM allocate members listed in [Table 8 on page 27](#). These sample input members provide the name, volume, catalog, and password specifications for the VSAM clusters that are being defined or deleted. Ensure that these values are correct for your system and that the VSAM cluster names match the names you use in your NetView startup procedure, CNMPROC. The supplied default volume for the NetView VSAM clusters is VOL(CPDLB2), unless you have run job CNMSJ000 to change the volume information for each of the VSAM allocate members.

For the RODM data sets, ensure that the VSAM cluster names match the names you use in your RODM startup procedure, EKGXRODM. If you add additional data window checkpoint data sets (for example, EKGCK003), also define them to EKGXRODM.

<i>Table 8. VSAM Clusters for NetView Facilities</i>		
NetView Facility	VSAM Allocate Member	VSAM Clusters
Network log	CNMSI101	NETVIEW.CNM01.DSILOGP NETVIEW.CNM01.DSILOGS
Session monitor	CNMSI101	NETVIEW.CNM01.AAUVSPL NETVIEW.CNM01.AAUVSSL
Hardware monitor	CNMSI101	NETVIEW.CNM01.BNJLGPR NETVIEW.CNM01.BNJLGSE
Save/restore	CNMSI101	NETVIEW.CNM01.DSISVRT
TCP/IP connection management	CNMSI101	NETVIEW.CNM01.DSITCONP NETVIEW.CNM01.DSITCONS
Trace log	CNMSI101	NETVIEW.CNM01.DSITRCP NETVIEW.CNM01.DSITRCS
Central site control facility (CSCF)	CNMSI501	NETVIEW.CNM01.DSIKP NL
RODM log	EKGS I101	NETVIEW.CNM01.EKGLOGP NETVIEW.CNM01.EKGLOGS
RODM checkpoint	EKGS I101	NETVIEW.CNM01.EKGMAST NETVIEW.CNM01.EKGTRAN NETVIEW.CNM01.EKGCK001 NETVIEW.CNM01.EKGCK002
AON	EZLS I101	NETVIEW.CNM01.STATS NETVIEW.CNM01.LOGP NETVIEW.CNM01.LOGS NETVIEW.CNM01.PASSWORD
Saved Packet Trace	FKXS I101	NETVIEW.CNM01.FKXP KTS

Note: All members referenced by the CNMSJ004 job for deleting or allocating VSAM data sets are found in the NETVIEW.V6R3USER.INSTALL data set.

2. If you want to define a security password for the VSAM data sets, add the password to the CNMSTPWD member in the DSIPARM data set. Also add this password to the VSAM allocate members (for example CNMSI101). Specify a password on the catalog that contains the entry for the data set before the system verifies the password for that data set.

To run the CNMSJ004 job to allocate or reallocate VSAM clusters, follow these steps:

1. If you are reallocating VSAM clusters, edit member CNMSID01 to uncomment the DELETE statements for the VSAM clusters to be reallocated.
2. Run the CNMSJ004 job.
3. Verify that all steps ran correctly. If you are reallocating VSAM clusters and uncommented only some of the names in member CNMSID01, the DEFINE statements for the remaining VSAM clusters might fail. This is acceptable because those VSAM clusters were not being reallocated.

If you are operating in an environment such as an SMS environment and receive an open error such as the DSI556I message with a return code of X'08' and an ACB error field value of X'A8', you can delete and redefine the database, moving it to a different volume. To accomplish these tasks:

1. Free the database.
2. Delete the database.

3. Reallocate the database.

You can use the NetView FREE and ALLOCATE commands to free and reallocate the database.

Optional: You can use the CNMSJ005 job to initialize the AAUVSPL and BNJLGPR data sets that are allocated by the CNMSI101 sample. These data sets are used when you verify the NetView installation.

If you want information about...	Refer to...
Defining passwords for the network log	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Printing the network log	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Maintaining the session monitor database	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Allocating RODM VSAM clusters	<i>IBM Z NetView Installation: Configuring Graphical Components</i>
Allocating AON VSAM clusters	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Enabling packet trace data collection	<i>IBM Z NetView Installation: Configuring Additional Components</i>

Initializing the Saved Packet Trace Database Using the CNMSJ033 Job

The CNMSJ033 job loads an initialization record into the Saved Packet Trace database that was allocated using the CNMSJ004 job.

To initialize the Saved Packet Trace database:

1. In the REPRO step, review and update as necessary the OUTDATASET high-level qualifier for your local NetView VSAM catalog. You defined the catalog using the CNMSJ001 job.
2. Run the CNMSJ033 job.
3. Verify your return codes.

If you need to rerun the CNMSJ033 job for any reason, first update and rerun the job you ran in [“Allocating VSAM Clusters Using the CNMSJ004 Job” on page 26](#) to delete and reallocate the Saved Packet Trace database.

Chapter 4. Preparing VTAM

To define the NetView program to VTAM, you need to:

- Modify the application (APPL) major node.
- Review VTAM and NCP definitions for mode table and unformatted system service (USS) table changes.
- Review VTAM start options.

Modifying the Application (APPL) Major Node

The APPL statements for the NetView tasks are located in the A01APPLS (CNMS0013) member. If you ran the CNMSJ000 job (described in [“Changing Sample Installation Jobs Using the CNMSJ000 Job”](#) on page 23) to create the VTAM APPL major node, then most of the modifications to your A01APPLS major node are already done.

Add this member to your VTAM configuration list so it is included by the status monitor preprocessor and started. The VTAM configuration list used in the samples is ATCCON01 (CNMS0003). You might need to change some of these statements or rename this member.

If you change the APPL statement names, add the ACBNAME operand to each statement that you are changing. The ACBNAME must be the original APPL name that is supplied by IBM of the statement being changed. For example, you can rename:

```
DSICRTR APPL AUTH=CNM,PRTCT=&CNMDOMN.
```

to

```
HN2CRTR APPL AUTH=CNM,PRTCT=&CNMDOMN.,ACBNAME=DSICRTR
```

Usage notes:

1. You cannot change the corresponding task ID on the TASK definition in the CNMSTASK member in DSIPARM.
2. All APPL statements with the CNM01 name prefix must follow the existing DOMAINID constraints.
3. An APPL name prefixed with CNM01 cannot have its suffix changed. For example, CNM01LUC must retain the LUC suffix.
4. Verify the original APPL name with the TASK statement in the CNMSTASK member. The MEM operand specifies the startup initialization member in DSIPARM.
5. Command help panels reference the original APPL names.
6. You cannot change BNJHWMON because of its downward compatibility with the BNJDSERV task.
7. You cannot change ALIASAPL because it is necessary for the VTAM Alias Name Translation function to work.
8. The ACBNAME keyword for CNM01VPD in A01APPLS must match the name in DSIVPARM.

Changing the Domain Name (DOMAINID)

Set the domain ID (system variable &CNMDOMN) in SYS1.PARMLIB member IEASYMxx. If you do not set a system variable, change *every* occurrence of &CNMDOMN. in A01APPLS (CNMS0013) to the current domain ID. For example, if you changed the domain ID to CNM12, then change:

```
&CNMDOMN.000 APPL AUTH=(NVPACE,SPO,ACQ,PASS),PRTCT=PW006,EAS=4,X
```

to:

CNM12000	APPL	AUTH=(NVPACE,SPO,ACQ,PASS),PRTCT=PW006,EAS=4,	X
----------	------	---	---

Note:

1. If you code the optional ACBNAME operand on the APPL statement, it must match the APPL name in column 1. Therefore, if you change the DOMAIN name, you must also change ACBNAME.

Changing the Password

The original password on the ACBpassword keyword in the CNMSTPWD member is CNM01. If the value of the &CNMDOMN variable in the IEASYMxx member is different from CNM01 or if you change this password, change every occurrence of PRTCT in the A01APPLS (CNMS0013) member to the same value. For example, if you change the password to PW006, then change:

&CNMDOMN.000	APPL	AUTH=(NVPACE,SPO,ACQ,PASS),PRTCT=&CNMDOMN.,EAS=4,	X
--------------	------	---	---

to:

&CNMDOMN.000	APPL	AUTH=(NVPACE,SPO,ACQ,PASS),PRTCT=PW006,EAS=4,	X
--------------	------	---	---

Defining the NetView Management Console to VTAM (LU 6.2 only)

You can use TCP/IP or LU 6.2 to communicate between the NetView program and the NetView management console. If you use LU 6.2, the following APPL statement defines both the NetView system and the NetView management console:

&CNMDOMN.	APPL	AUTH=(VSPACE,ACQ,PASS),PRTCT=&CNMDOMN.,	X
		MODETAB=AMODETAB,DLOGMOD=DSIL6MOD,	X
		APPC=YES,PARSESS=YES,	X
		DMINWNL=4,DMINWNR=4,DSESLIM=8,VPACING=10,	X
		AUTOSES=2	

Defining the SNA Topology Manager to VTAM

VTAM uses APPL definitions for the SNA topology manager application. Sample definitions are included in member A01APPLS (CNMS0013). The APPL name for the SNA topology manager must match the APPLNAME specified in the VTAM section of the FLBSYSD initialization file used by the manager application.

If you want information about...

Refer to...

VTAM requirements

the description of the VTAM MIBConnect parameters contained in the FLBSYSD initialization file or refer to *IBM Z NetView SNA Topology Manager Implementation Guide*

Defining the VTAM Primary Program Operator Interface Task

The primary program operator interface (POI) task is defined with the statement:

&CNMDOMN.PPT	APPL	AUTH=(NVPACE,PPO),PRTCT=&CNMDOMN.,EAS=1,	X
		MODETAB=AMODETAB,DLOGMOD=DSILGMOD	
*		STATOPT='NETVIEW PPT'	

VTAM does not open more than one access method control block with AUTH=PPO specified. If you are migrating and running a previous release of the NetView program concurrently with the NetView program for verification or testing, change this statement from AUTH=(NVPACE,SPO) back to AUTH=(NVPACE,PPO).

Changing the Logmode Table (LOGMODE)

The sample network has a logmode table named AMODETAB (CNMS0001) that includes logmode entries for NetView sessions. In A01APPLS (CNMS0013), the DLOGMOD operand for an application points to an entry in this logmode table. For example, the following operands are for the application &CNMDOMN.000:

```
MODETAB=AMODETAB,DLOGMOD=DSILGMOD
```

Check the DLOGMOD operands for your applications and ensure that they are pointing to the proper entries for your installation.

Defining Resources to the Status Monitor

The A01APPLS (CNMS0013) sample includes STATOPT statements that define resources to the status monitor. For example, the following statement is for the application &CNMDOMN.000:

```
*      STATOPT='NETVIEW 000'
```

If you want to redefine the name of this resource, or any other resource, change the STATOPT statement.

If you want information about...	Refer to...
STATOPT statements, CNMS0013 sample	<i>IBM Z NetView Installation: Configuring Additional Components</i>

Defining Multiple Concurrent NetView Operators

You can define tasks for multiple concurrent NetView operators using APPL statements, for example:

```
&CNMDOMN.000 APPL AUTH=(NVPACE,SPO,ACQ,PASS),PRTCT=&CNMDOMN.,EAS=4,X
               MODETAB=AMODETAB,DLOGMOD=DSILGMOD
*               STATOPT='NETVIEW 000'
```

You must specify an APPL statement for each concurrent NetView operator. Each APPL statement name contains a 3-character suffix which is a hexadecimal number in the range of 000–FFF. If you are starting autotasks before VTAM, you must have enough consecutively numbered APPL statements to allow each autotask to obtain one. The numbers after those reserved for the autotasks do not need consecutive suffixes.

To enable takeover or reconnect, include PASS value in your AUTH= definition. If PASS is not specified and a takeover or reconnect is attempted, the authorized receiver receives message DSI133I, and the operator attempting to log on receives the following message:

```
DSI213I ACCESS TO 'CLSDST PASS' IS NOT AUTHORIZED
```

If you are not using the status monitor preprocessor, define one APPL statement using wildcard characters (??) for multiple concurrent NetView operators. See the following example:

```
&CNMDOMN.0?? APPL AUTH=(NVPACE,SPO,ACQ,PASS),PRTCT=&CNMDOMN.
               EAS=4,MODETAB=AMODETAB,DLOGMOD=DSILGMOD
```

The maximum number of concurrent users of NetView application LUs is 4096. Users of the NetView application LUs include the following operators and tasks:

- Started hardcopy tasks (for LUs defined on HARDCOPY statements)
- Started autotasks
- Operators logged on through real or simulated terminals, the web browser, or the NetView management console client

Factors such as local system queue area (LSQA) storage and the number of tasks can affect the maximum limit for the number of concurrent users of NetView application LUs to be less than 4096.

Using the REFRESH command, refresh DSIOPF to add or delete NetView operators and operator profiles and to change operator profile definitions while the NetView program is running. Changes to operator profile information (either NetView profiles or SAF NETVIEW segment) require the operator to log off and log back on for the changes to take effect.

If not enough APPL statements are available for dynamically added operators after NetView is started, create a new APPL major node similar to A01APPLS. In this new member, you can define as many additional APPL statements as you need for new operators. You do not have to stop and restart the NetView program to use the new operators you defined. When you start this new major node, the application definition statement for the new operator becomes available for immediate use. If you are using alert forwarding, you need to define additional APPL statements.

Allowing for Additional TAF LU1 Operators

Tasks for five concurrent terminal access facility (TAF) users of operator-control sessions are defined with APPL statements. The following example shows the first one:

```
TAF01000 APPL  MODETAB=AMODETAB,EAS=9,          X
                DLOGMOD=M3767
*              STATOPT='TAFAPPL 000'
```

If you have more than five concurrent TAF users of operator control sessions, you need to add APPL statements.

Note: If the status monitor preprocessor is not used, you can define one APPL statement using the wildcard character (?). For example:

```
TAF010?? APPL  MODETAB=AMODETAB,EAS=9,          X
                DLOGMOD=M3767
```

Allowing for Additional TAF LU2 Operators

Tasks for 20 concurrent TAF users of full-screen sessions are defined with APPL statements. The following example shows the first one:

```
TF01#000 APPL  MODETAB=AMODETAB,EAS=9,          X
                DLOGMOD=M2SDLCNQ
*              STATOPT='DYNAMIC TAF
000'
```

If you have more than 20 concurrent TAF users of full-screen sessions, you need to add APPL statements.

Note: If the status monitor preprocessor is not used, you can define one APPL statement using the wildcard character (?). For example:

```
TF01#0?? APPL  MODETAB=AMODETAB,EAS=9,          X
                DLOGMOD=M2SDLCNQ
```

Defining Alias Names

If you are using cross-network communication, you might need to use alias names to resolve duplicate resource names. With alias names, the name of the resource (such as a logical unit, a class of service, an SRCLU, or a logon mode table) from the sending network is translated to a name that is unique to the receiving network.

If you want to use the alias name translation facility provided by the NetView program, include an APPL definition statement for that application program in the application program major node. The name on the APPL definition statement must be ALIASAPL.

The following example shows the APPL statement in A01APPLS:

```
ALIASAPL APPL  AUTH=(CNM),PRTCT=&CNMDOmn.
*              STATOPT='ALIAS TASK'
```


The AUTH=CNM operand shows that this application program can use the communication network management (CNM) interface. The PRTCT=CNM01 operand specifies the NetView program password.

Member DSIALATD contains the statement:

```
LABEL      ALIASMEM DSIALTAB
```

This statement points to DSIALTAB, which contains examples of alias translation statements.

To add an alias data set member that defines one of your networks, add the member name to the statement in DSIALATD in the following way:

```
LABEL      ALIASMEM DSIALTAB,YOURLMEM
```

Create a data set member of the same name following the format of DSIALTAB to define your alias translations. You can add more than one member name to the same ALIASMEM statement. Ensure that you either modify DSIALTAB with valid aliases or comment out the example statements provided.

Note: Do not use alias names to translate NetView APPL names, including the domain ID, and any names that are constructed from it; for example, A01M, A01M001, A02M, and A02M002.

If you want information about...	Refer to...
Alias Name Translation Facility	z/OS Communications Server library

DSIALTAB provides examples of the following statements. Modify these statements for your own environment.

ORIGNET Statement

For each member name you code on an ALIASMEM statement, code at least one ORIGNET definition statement for each network that requires alias name translations. The ORIGNET statement has the following format:

```
ORIGNET    netid
```

Where:

netid

Is the network name as coded on the NETWORK statement for the CDRM major node, or on the VTAM NETID start option. This name always represents the network that knows the resource by its alias name.

The following example shows this statement:

```
ORIGNET    NETA
```

LU Statement

If you have LUs with the same name in more than one network, code the LU name translation statement. With this name translation, the LU can be known in its owning network by its real name and in other networks by the assigned alias name. The LU statement has the following format:

```
LABEL      LU    realname,ownernet,  
aliasname[,ownersscp]
```

Where:

realname

Is the LU name as it is defined in the domain where the LU resides.

ownernet

Is the 1–8 character network name of the network where the LU is known by its *realname*.

aliasname

Is the alias name for this LU that is used by the network specified on the ORIGNET definition statement.

ownersscp

Is the SSCP that owns the LU in the *ownernet* network. If you do not code *ownersscp*, the SSCP name is taken from the VTAM CDRSC definition statement.

The following example shows this statement:

```
ORIGNET  NETA
LABEL    LU  TERM1,NETB,TERM1B
```

Class of Service Statement

In interconnected networks, the class-of-service (COS) names used in one network can be different from the names used in another network. However, two COS definitions can provide the same type of service. For example, NETA calls its fastest class of routes FAST, while NETB calls its fastest class of routes QUICK. The COS names are different, but the result in each case is the same: the fastest route is chosen. With alias names, you can make the COS name used in one network equivalent to a name in another network. Each network can use the name it knows and still get the class of service it expects.

The COS statement has the following format:

```
LABEL    COS    adjacnme,adjacnet,localnme
```

Where:**adjacnme**

Is the COS name defined in the adjacent or destination network that provides the class-of-service equivalent to the COS entry specified in the local-name class of service.

adjacnet

Is the 1–8 character NETID of the adjacent network where the equivalent class of service is defined.

localnme

Is the name of the class of service defined in the network specified on the ORIGNET definition statement.

The following example shows this statement:

```
ORIGNET  NETA
LABEL    COS    QUICK,NETB,FAST
```

You can use the same local name (or alias) within one network for different adjacent networks. Also, each adjacent name can be given any number of aliases or local names.

MODE Statement

In interconnected networks, the set of logon mode entry names used in one network can be different from the names used in another network. However, two logon mode entries with different names can result in the session parameters being the same. With the name translation facility, you can make a logon mode entry name used in one network equivalent to a name in another network. In this way, you can access the information in a logon mode table from another network without changing the logon procedures used in either network.

The MODE statement has the following format:

```
LABEL    MODE    destname,destnet,localnme
```

Where:**destname**

Is the name of the logon mode entry defined in the destination network.

destnet

Is the 1- to 8-character NETID of the destination network where the logon mode entry is known by its equivalent name.

localnme

Is the name of the logon mode entry defined in the network specified on the ORIGNET statement.

The following example shows this statement:

```
ORIGNET  NETA
LABEL    MODE  LOG1B,NETB,LOG1A
ORIGNET  NETB
LABEL    MODE  LOG1A,NETA,LOG1B
```

The logon mode entry name defined to NETA as LOG1A is known in NETB as LOG1B. The logon mode entry name defined to NETB as LOG1B is known in NETA as LOG1A.

For more information about rules for name translation, refer to the z/OS Communications Server library.

Reviewing VTAM and NCP Definitions for Mode Table and USS Table Changes

The sample definitions include the A01SNA (CNMS0073) statement for SNA locally attached terminals, the A01LOCAL (CNMS0016) statement for non-SNA locally attached terminals, and your VTAM and NCP definitions for any mode table changes. Update your USS table for the NetView program and reflect that change in your definitions. The USS table is AUSSTAB (CNMS0011) in the samples.

Note: If you modify AUSSTAB, run CNMSJ006 to reassemble and link edit the table and carry out the changes.

Assembling VTAM Tables Using Job CNMSJ006

Job CNMSJ006 assembles and link-edits the following tables and places them in NETVIEW.V6R3USER.VTAMLIB:

- AMODETAB, which is the logmode table supporting miscellaneous devices defined in the sample network
- AUSSTAB, which is the USS table supporting SNA synchronous data link control
- ISTSDCOS, which is the class-of-service table containing routes.

To assemble VTAM tables:

1. Edit CNMSJ006 in the NETVIEW.V6R3USER.INSTALL data set.

Note:

- a. The ISTMGC00 table is included with VTAM as part of the ISTMGC01 table. The ISTMGC01 table is a default routing table included with VTAM. This table tells VTAM to route unsolicited request units (RUs) to the NetView program so they can be received and routed to components such as the hardware monitor and the session monitor.
 - b. Verify that the mode entries for DSIL6MOD, PARALLEL, and RBRIDGE from CNMS0001 (AMODETAB) are in your current AMODETAB member in VTAMLST.
 - c. The sample job CNMSJ006 provided in &PGMNAME uses the High Level Assembler ASMA90. To use Assembler H, change the program name to PGM=IEV90.
2. Run CNMSJ006.
 3. NETVIEW.V6R3USER.CNM01.VTAMLST has two members that define local terminal support:
 - A01LOCAL (CNMS0016) for non-SNA channel-attached terminals
 - A01SNA (CNMS0073) for SNA channel-attached terminals

Verify that in the member you use, each CUADDR parameter is the correct address for your host local terminals. Also, be sure that the node name references these addresses. These members reference the AMODETAB and AUSSTAB tables.

4. Verify your return codes before continuing with the next step listed for your version and release.

The changes to the tables take effect the next time you start VTAM or issue the VTAM command:

```
MODIFY NET, TABLE, NEWTAB=AMODETAB, OPTION=LOAD
```

If you want information about...	Refer to...
----------------------------------	-------------

MODIFY command	z/OS Communications Server library
----------------	------------------------------------

Reviewing VTAM Start Options

If you specify the VTAM start option MSGMOD, turn it off while running NetView command lists and NetView automation.

The NetView program supplies a sample VTAM startup member CNMS0010 which is copied to NETVIEW.V6R3USER.CNM01.VTAMLST as ATCSTR00. This sample specifies PPOLOG=NO because the PPOLOG=YES statement is not needed to keep the status monitor panels accurate. However, if you want all VTAM messages kept in the NetView log, take one of the following actions:

- Specify PPOLOG=YES in the ATCSTR00 member to activate the PPOLOG feature of the VTAM program. This causes the VTAM program to send copies of all messages to the Programmed Operator Interface in the NetView program. These copies are logged to both netlog and Canzlog. (Note that the original messages are already in Canzlog.)

The PPOLOG copies of the messages are *not* automated; the original messages are subject to automation. To reduce duplication, code NETLOG(NO) for any automation of VTAM messages when you use the PPOLOG feature.

- Specify PPOLOG=NO and force the unsolicited VTAM messages to the NetView log using the automation table or the ASSIGN command. If you only specify PPOLOG=NO, VTAM commands issued outside the NetView program are not recorded in the NetView log unless a match in the automation table or an ASSIGN command for the message exists.

If you want information about...	Refer to...
----------------------------------	-------------

Start options	z/OS Communications Server library
---------------	------------------------------------

Modifying the VTAM Start Procedure

The sample VTAM start procedure provided with the NetView program is located in the CNMNET (CNMSJ008) member. Review this sample procedure and make the following updates to the VTAM start procedure that you are using:

1. Add the following data set to the concatenated list of data sets in the VTAMLST DD statement (add this data set before SYS1.VTAMLST):

```
//          DD DISP=SHR,DSN=NETVIEW.V6R3USER.CNM01.VTAMLST
```

The VTAMLST data set that is shipped with the NetView program contains data set members that you might have already customized. Protect any members that you customized. Otherwise, the VTAM program does not perform as expected.

2. Add the following data sets to the VTAMLIB DD statement:

- Add the USER.VTAM data set before the SYS1.VTAMLIB data set

```
// DD DISP=SHR,DSN=NETVIEW.V6R3USER.VTAMLIB
```

- If you plan to use the NetView Application Management Instrumentation, (AMI) or status monitor functions, add the SCNMLNK1 data set after the SYS1.VTAMLIB data set

```
// DD DISP=SHR,DSN=NETVIEW.V6R3M0.SCNMLNK1
```

The new VTAMLIB DD concatenation looks similar to the following example:

```
//VTAMLIB DD DISP=SHR,DSN=NETVIEW.V6R3USER.VTAMLIB  
// DD DISP=SHR,DSN=SYS1.VTAMLIB  
// DD DISP=SHR,DSN=NETVIEW.V6R3M0.SCNMLNK1
```

Usage notes:

1. Sample CNMSJ002 allocates the VTAMLST and VTAMLIB data sets. For more information, see [“Allocating Data Sets Using the CNMSJ002 Job” on page 21](#).
2. If you added the SCNMLNK1 data set to the VTAMLIB DD statement, make sure that it is APF-authorized. For information about authorizing this data set, see [“PROGxx” on page 11](#).
3. If you added the VTAMLIB data set to the VTAMLIB DD statement, make sure that it is APF-authorized. For information about authorizing this data set, see [“PROGxx” on page 11](#).
4. The modifications you made to your VTAM start procedure go into effect when you restart the VTAM program.

Starting VTAM ACBs and Corresponding NetView Tasks

If a VTAM ACB and a corresponding NetView task are both active, and the ACB is deactivated and then reactivated, the NetView task typically becomes active. Some NetView tasks remain active when you end and restart VTAM without bringing NetView down. Other tasks become inactive but reactivate automatically when you restart VTAM.

DSICRTR, DSIGDS, BNJDSERV, ALIASAPL, and AAUTCNMI are examples of NetView tasks that remain active when the corresponding ACB becomes inactive.

VPDTASK, DSIAMLUT, CNM01VMT, CNM01BRW, and CNM01LUC are NetView tasks that become inactive when the corresponding ACB becomes inactive, but reactivate automatically when you restart VTAM. If you use a VTAM VARY INACT command to deactivate the ACB, you need to restart these NetView tasks after the ACB is reactivated. If you use a Z NET,QUICK command to deactivate the ACB, these NetView tasks are automatically reactivated when VTAM is restarted.

When NetView recognizes that VTAM is active, it invokes CLIST CNMEVTAM. This runs under the PPT. At this time the PPT ACB and the main NetView ACB are open.

Chapter 5. Getting Ready to Start NetView

Before starting the NetView program, review the following START procedures:

- CNMPROC (CNMSJ009) for the NetView program
- CNMPSSI (CNMSJ010) for the NetView subsystem

The CNMSTYLE member in the DSIPARM data set is used during NetView initialization. Review the CNMSTYLE member and make any changes necessary for your installation.

NetView and Subsystem Application Procedures

Review and update the CNMPROC (CNMSJ009) sample and the CNMPSSI (CNMSJ010) sample:

- The name of the PROCLIB member and the PROC statement must begin with the 4-character subsystem name that you defined for running the NetView program. The associated CNMPSSI (CNMSJ010) start procedure must also begin with this same subsystem name. CNMP is used in the sample network.

If the PROCLIB member name does not match an entry in IEFSSNxx, use the SUB= parameter with the START command to specify a subsystem other than the MASTER subsystem. Specify a subsystem where SYSIN and SYSOUT are not supported.

- If you start a second copy of the NetView program in the same host, use a procedure name that begins with a 4-character subsystem name that is different from the one you already started. Add the 4-character subsystem name to the IEFSSNxx member in the SYS1.PARMLIB data set and specify DSI4LSIT as INITRTN.
- If you plan to start the NetView application PROC under the master subsystem before you start JES, see [“Starting the NetView Program Before Starting JES” on page 66](#).

If you start the NetView program before starting the NetView subsystem (SSI) or stop the SSI before stopping the NetView program, messages similar to the following are issued:

```
CNM563I NETVIEW SUBSYSTEM IS NOT ACTIVE
```

Modifying the NetView Subsystem Interface Procedure

Update the NetView subsystem interface procedure, CNMPSSI (CNMSJ010) member in the SYS1.PROCLIB data set, as required for your environment:

- [“Changing Symbolic Variables” on page 39](#)
- [“Running the NetView Subsystem Interface as a Job” on page 40](#)

Changing Symbolic Variables

Review and modify the following symbolic variables as required for your installation:

ARM

Enables the NetView subsystem for MVS automatic restart management (ARM) and supplies a name (up to 16 characters) by which this NetView subsystem is to be known to ARM.

The CNMPSSI ARM parameter can have the following values:

***ARM**

Specifies that the NetView subsystem is to be enabled. The NetView program generates a NetView ARM element name.

name

Specifies that the NetView subsystem is to be enabled and identifies the NetView ARM element name. This name can be 1 - 16 alphanumeric characters in length. The first character must be alphabetic. The name can contain the special characters #, @, and \$.

***NOARM**

Specifies that the NetView subsystem is not to be enabled. *NOARM is the default value.

PPIOPT

Specifies whether you want to initialize the PPI facility. The PPIOPT parameter can have the following values:

PPI

Specifies that the PPI facility is to be started and initialized for the NetView subsystem address space. You cannot request this option for more than one subsystem address space. If the PPI is already active on another subsystem address space, it does not initialize for additional requests. PPI is the default value.

NOPPI

Specifies that the PPI facility is not initialized for the subsystem address space.

P256BUF

Specifies the number of 256-byte PPI buffers. The default value is 300.

P4000BUF

Specifies the number of 4000-byte PPI buffers. The default value is 0.

REG

Specifies the region size for the NetView subsystem buffer address space in KB (the number specified times 1024). The default value is 16600.

The default value allows for 4200 total message and command buffers. To calculate the correct region size for your network, refer to *IBM Z NetView Tuning Guide*.

ROUTECD

Specifies the route code to be used for messages issued by the SSI address space.

Messages that can be issued before this parameter is processed use the default route code 1, regardless of the value set here. Valid values are in the range 1 – 128.

SQ1

Specifies the high-level qualifiers for NetView system data sets. The default value is NETVIEW.V6R3M0.

You can also adjust the CNMPSSI parameters using the SSI statements in the CNMSTYLE member.

Running the NetView Subsystem Interface as a Job

To run the NetView subsystem interface as a job, remove the asterisk (*) from the first two lines and the final two lines. Also update the JOB statement to reflect your installation requirements. The first two lines are shown here:

```
//*CNMSJ010 JOB 'ACCOUNTING INFORMATION', 'NETVIEW SUBSYSTEM',
//* CLASS=A MSGCLASS=A,MSGLEVEL=(1,1)
```

The final two lines are shown here:

```
//*          PEND
//*CNMPSSI EXEC PROC=CNMPSSI
```

Make sure that the PROC name you specify in this statement agrees with the one you specify in CNMPROC (CNMSJ009). Both of these procedure names must match the NetView subsystem name that you specified in your SYS1.PARMLIB data set member IEFSSNxx.

Modifying the NetView Startup Procedure

Update the NetView startup procedure, the CNMPROC (CNMSJ009) member in the SYS1.PROCLIB data set, as required for your environment. These changes do not involve CNMPSSI (CNMSJ010).

Changing Symbolic Variables

The following symbolic variables are used for user-defined data sets:

Variable	Default Value
&Q1	NETVIEW.V6R3USER
&DOMAIN	CNM01
&SQ1	NETVIEW.V6R3M0

You can change these values as required for your environment.

Specifying the Allocation for the NetView Internal Trace Table

The internal trace starts early during NetView initialization, as long as the TRSIZE value specified on the startup procedure is not 0. The trace starts with all options for the following reasons:

- Documentation is available to diagnose a problem that might occur during NetView initialization.
- Trace options are not interpreted during NetView initialization until the CNMSTYLE statements are read. When the CNMSTYLE TRACE statements are interpreted, the options, size, and mode of the tracing are changed to reflect what is specified on those statements.

You can specify the number of pages allocated for this initial NetView internal trace table. The default value is 4000 pages. Because the trace is allocated out of a data space, the maximum value that can be specified is 524286, the size of a data space. The syntax is

```
TRSIZE=[pages | 0]
```

If 0 is specified, no trace table is allocated, and the NetView internal trace does not start early during initialization. The CNMSTYLE trace options take effect regardless of whether the NetView internal trace starts early. Leave early tracing on unless it significantly degrades NetView performance.

Specifying the Japanese Version

If you are installing the Japanese version of the NetView program, uncomment the appropriate statements in CNMPROC and comment out the corresponding English statements. Refer to the comments in CNMPROC.

Setting System Variables

You can set several system variables in CNMPROC.

The start procedure contains the specification of a two-character identifier NV2I. This value becomes a local system variable &NV2I, and is used to construct names that need to be unique to each NetView within an MVS image, a sysplex, or a network. If no value is specified for &NV2I, the value of NM is used.

If you specify a value for &NV2I (xx), the NetView program reads CxxSTYLE in DSIPARM for initialization parameters. If this member is not found, the NetView program reads the CNMSTYLE member instead.

Note:

1. If you plan to have one NetView program for each z/OS system, consider using the value of the &SYSCONE variable for the value of &NV2I.

2. For more information, refer to the comments in CNMPROC (CNMSJ009).

Setting the Dispatch Priority

The procedure to start the NetView applications contains the following parameter:

```
//          DPRTY=(13,13)
```

Set the DPRTY parameter to the *highest dispatching priority possible* below the priority set for VTAM to get the best performance and to help prevent records from being overlaid.

Modifying the Region Size, Buffer Size and Slot Size

The following sample statements specify the region size, buffer size, and slot size:

```
//          REG=256,          ** REGION SIZE(IN K) FOR NETVIEW
//          **              (16M-SYSTEM OVERHEAD) = 24-BIT REGION
//          **              64M = 31-BIT REGION
//          **              NOTE: AVOID USING A REGION SIZE <16M.
//          **              SEE THE NETVIEW TUNING GUIDE AND HELP
//          **              FOR THE "RESOURCE" COMMAND.
//          BFSZ=24,          ** BUFFER SIZE(IN K)
//          SLSZ=200,          **
//          **
```

The default region size provided with the NetView program might not be adequate for all environments. If you enable additional components in the NetView program, for example the SA tower in the CNMSTYLE member, or the size of your network is large, consider increasing the region size. For more information on calculating the region size, refer to the *IBM Z NetView Tuning Guide*.

The buffer size parameter specifies the amount of storage allocated to the global buffer, the local buffer, and the router buffer. Each buffer is allocated the amount of storage specified by this parameter. The acceptable values are 8, 12, 16, 20, and 24 KB.

The slot size parameter refers to a fixed portion of a buffer that is used to contain incoming error records. As these error records increase in number, you can increase the size of the buffer. Acceptable values are 200, 250, or 300 bytes. If the slot size is too low, the SVC76 mapper truncates error records of greater length than the slot size. If the slot size is too great, the number of slots available is reduced and records can be lost unless you also define a larger buffer size.

Defining the REXX Runtime Library to STEPLIB

Essential NetView functions use REXX runtime programs that are compiled with the ALTERNATE option. If you access the REXX/370 runtime library from the NetView program, these REXX runtime programs run in compiled mode. Otherwise, the REXX alternate library is used and the REXX programs run in interpreted mode. If the REXX/370 runtime library or REXX alternate library is not accessible from the link pack area (PLPA), modify the NetView start procedure to access one of these libraries:

- To use the REXX/370 runtime library, ensure that the following statement is uncommented:

```
//          DD    DSN=&REXX..SEAGLPA,DISP=SHR
```

- To use the REXX alternate library, ensure that the following statement is uncommented:

```
//          DD    DSN=&REXX..SEAGALT,DISP=SHR
```

For more information about which REXX library to use, refer to the comments in the CNMPROC start procedure.

Defining Command Lists

The NetView command lists are defined in the following statements:

```
//DSICLD DD DSN=&Q1..&DOMAIN..CNMCLST,DISP=SHR
//      DD DSN=&SQ1..SAQNCLST,DISP=SHR
```

```
//      DD DSN=&SQ1. .CNMCLST,DISP=SHR
//      DD DSN=&SQ1. .CNMSAMP,DISP=SHR
//*     DD DSN=SYS1.PROCLIB,DISP=SHR
```

If you write additional command lists, add them to the following data set:

```
NETVIEW.V6R3USER.&domain.CNMCLST
```

Each command list is a separate member of the command list data set. The name of the command list is the data set member name.

You can have several data sets for different types of command lists. Add the names of any data sets that you have defined for your command lists before those of the NetView command lists, for example:

```
//DSICLD DD DSN=DATASETNAME1,DISP=SHR
//      DD DSN=DATASETNAME2,DISP=SHR
//      DD DSN=&Q1. .&DOMAIN. .CNMCLST,DISP=SHR
```

If you allocate your command list library with secondary space, you might have to recycle the NetView program to update the library.

Specifying Definition Members and Sense Codes

The NetView definition data set members and sense codes are defined in the following statements:

```
//DSIPARM DD DSN=&Q1. .&DOMAIN. .DSIPARM,DISP=SHR
//      DD DSN=&SQ1. .SAQNPARM,DISP=SHR
//      DD DSN=&SQ1. .DSIPARM,DISP=SHR
```

If you have other definition and sense code libraries, concatenate them after this statement.

Defining Data Sets to the NetView Program

The NetView program uses data sets to read information and write output while running various functions. The following list identifies some of these data sets and how they are used by the NetView program.

- If you are installing the Japanese version of the NetView program, change the STEPLIB statement as follows:

```
//STEPLIB DD DSN=&SQ1. .SCNMMJPN,DISP=SHR
//      DD DSN=&SQ1. .SAQNLINK,DISP=SHR
//      DD DSN=&SQ1. .CNMLINK,DISP=SHR
```

- The following statements define the data sets that contain VTAM definitions:

```
//DSIVTAM DD DSN=&Q1. .&DOMAIN. .VTAMLST,DISP=SHR
//*      DD DSN=SYS1.VTAMLST,DISP=SHR
```

- The following statements define the data sets that contain your NetView operator profiles:

```
//DSIPRF DD DSN=&Q1. .&DOMAIN. .DSIPRF,DISP=SHR
//      DD DSN=&SQ1. .DSIPRF,DISP=SHR
```

- The following statements define the English online help panels:

```
//CNMPNL1 DD DSN=&Q1. .&DOMAIN. .CNMPNL1,DISP=SHR
//      DD DSN=&SQ1. .CNMPNL1,DISP=SHR
```

- The following statement defines the English online help panels for the hardware monitor:

```
//BNJPNL1 DD DSN=&SQ1. .BNJPNL1,DISP=SHR
```

For online help panels in the Japanese language, replace the BNJPLN1 statement with the following statement:

```
//BNJPNL1 DD DSN=&SQ1. .SBNJPNL3,DISP=SHR
```

- The following statement defines the data set used for the hardware monitor color map:

```
//BNJPNL2 DD DSN=&SQ1..BNJPNL2,DISP=SHR
```

- Uncomment the following statement for online help panels in Japanese:

```
//* DD DSN=&SQ1..SCNMPNL2,DISP=SHR
```

- The following statement defines the data set used for user-defined message members and translation members:

```
//DSIMSG DD DSN=&SQ1..SDSIMSG1,DISP=SHR
```

- The following statements define the NetView non-secured data sets:

```
//DSIOPEN DD DSN=&Q1..&DOMAIN..SDSIOPEN,DISP=SHR
// DD DSN=&SQ1..SDSIOPEN,DISP=SHR
```

- The following statement defines the data set used for storing NetView reports, listings, and other output:

```
//DSILIST DD DSN=&Q1..&DOMAIN..DSILIST,DISP=SHR
```

- The following statement defines the data set containing members to be used for testing the automation table:

```
//DSIASRC DD DSN=&Q1..&DOMAIN..DSIASRC,DISP=SHR
```

- The following statement defines the data set used for storing the report produced from running tests of the automation table:

```
//DSIARPT DD DSN=&Q1..&DOMAIN..DSIARPT,DISP=SHR
```

Running with Information/Management

If you use Information/Management to log problems from the hardware monitor, the NetView program must have access to the Information/Management load library and session members. The NetView program can accomplish this if you install Information/Management in a library accessible to the NetView program, such as NETVIEW.V6R3M0.CNMLINK. If you did not install into CNMLINK, concatenate the Information/Management libraries into the NetView STEPLIB DD statement. Information/Management libraries must be APF-authorized.

If you want information about...

Refer to...

Installing Information/Management with the NetView program

Information/Management library

Enabling Network Logs to Print Automatically

The CNMPROC (CNMSJ009) procedure includes statements that are required to print the network logs automatically whenever the start procedure ends. Statements are provided for both the primary and secondary logs.

To enable automatic printing of the network primary and secondary logs, uncomment the statements in [Figure 2 on page 45](#) and [Figure 3 on page 45](#).

```

//***PRNTLOGP EXEC PGM=DSIPRT,COND=EVEN
//***STEPLIB DD DSN=&SQ1..CNMLINK,DISP=SHR
//***DSILT DD DSN=&VQ1..&DOMAIN..DSILOGP,
//*** DISP=SHR,AMP=AMORG
//***SYSPRINT DD SYSOUT=&SOUTA
//***DSILST DD SYSOUT=&SOUTA,DCB=(LRECL=121,RECFM=F,BLKSIZE=121)
//***DSIINP DD
DUMMY,DCB=(LRECL=80,RECFM=F,BLKSIZE=80)

```

Figure 2. Primary Network Log Print Step

```

//***PRNTLOGS EXEC PGM=DSIPRT,COND=EVEN
//***STEPLIB DD DSN=&SQ1..CNMLINK,DISP=SHR
//***DSILT DD DSN=&VQ1..&DOMAIN..DSILOGS,
//*** DISP=SHR,AMP=AMORG
//***SYSPRINT DD SYSOUT=&SOUTA
//***DSILST DD SYSOUT=&SOUTA,DCB=(LRECL=121,RECFM=F,BLKSIZE=121)
//***DSIINP DD
DUMMY,DCB=(LRECL=80,RECFM=F,BLKSIZE=80)

```

Figure 3. Secondary Network Log Print Step

For more information, see *IBM Z NetView Installation: Configuring Additional Components*.

Omitting Some of the Logs from Automatic Printing

The sample start procedure, CNMPROC (CNMSJ009), includes JCL steps that pertain to the automatic printing of logs when the start procedure ends. Each step is identified by a comment explaining its function. If you want automatic printing of some of the logs but not all of them, alter only those statements about the logs you want to print. Read the comments to determine which steps pertain to the logs you want. The first step ensures that VSAM databases are properly closed before printing. Activate the statements in the first step, then activate the statements in any other steps that pertain to logs you want to print automatically.

Defining TCP/IP to the NetView Program

The NetView program provides services that rely on TCP/IP to communicate with remote applications. To communicate with TCP/IP, each of these services use a program function library, referred to as the TCP/IP socket library. This makes the NetView application a TCP/IP socket application.

Any TCP/IP socket application must reference TCP/IP configuration data. The method of accessing this data is defined by z/OS Communications Server.

An example SYSTCPD DD statement is provided in the NetView startup procedure to identify the location of TCP/IP configuration data. A SYSTCPD statement is not required for the NetView program, but any TCP/IP socket application must be able to locate TCP/IP configuration data.

You also need access to z/OS TCP/IP data sets from the NetView start procedure. If the z/OS TCP/IP data sets are not contained in the LNKST concatenation, add the following z/OS TCP/IP data set (which must be APF-authorized) to the STEPLIB DD concatenation:

SEZALOAD

Executable load modules for concatenation to LINKLIB

To optimize performance, make these data sets available from the LNKST concatenation.

Ensure that the user ID that is associated with the NetView started task or job is defined to your SAF product with an OMVS segment. This is required so that the NetView program can use z/OS UNIX System Services (USS) functions, which include TCP/IP functions.

Usage notes:

1. For each of the TCP/IP services that is provided by the NetView program, the stack affinity is specified in the CNMSTYLE member using the TCPname statement. Using this name, NetView sets the stack affinity by specifying it on an INITAPI socket call.

2. Some NetView applications that are not a part of the NetView address space rely on TCP/IP to communicate with remote applications. Some of these applications use the z/OS UNIX socket library. These applications are therefore z/OS UNIX socket applications. Information on how these applications reference TCP/IP configuration data is discussed in the books describing those applications. Examples of z/OS UNIX socket applications are the Event/Automation Service and the UNIX command server.

If you want information about...	Refer to...
TCP/IP socket applications, SYSTCPD DD statement	<i>z/OS Communications Server IP Configuration Guide</i>

Running the NetView Application as a Job

To run the NetView application as a job, remove the asterisk (*) from the first two lines and the final two lines of the CNMSJ009 job. Also, update the JOB statement to reflect your installation. The first two lines are shown here:

```
//*CNMSJ009 JOB 'ACCOUNTING INFORMATION','NETVIEW STARTUP PROC',  
/* CLASS=A,MSGCLASS=A,MSGLEVEL=(1,1)
```

The final two lines are shown here:

```
/*  
/*CNMPROC EXEC PROC=CNMAPROC
```

Make sure the PROC name that you specify in this statement agrees with the one you specify in CNMPSSI (CNMSJ010). Both of these procedure names must match the NetView subsystem name that you specified in your SYS1.PARMLIB data set member IEFSSNxx.

Updating the DSITBL01 member

The DSITBL01 member in the DSIPARM data set is the default NetView automation table. The DSITBL01 automation table is required by the NetView program and provides basic automation statements to respond to messages and management services units (MSUs). If the DSITBL01 automation table is not activated during NetView initialization, the DWO093W warning message is issued by the NetView program.

You can make modifications to the DSITBL01 automation table. Make sure that you do not disable vital functions of the NetView program. You can add your own automation table by using the AUTOCMD statement in the CNMSTYLE member. Place your automation table before the DSITBL01 automation table and code CONTINUE (YES) on any local customization to make sure that messages flow to the DSITBL01 automation table.

Updating the CNMSTYLE Member

The CNMSTYLE member in the DSIPARM data set is read by the NetView program at initialization and controls many customizable properties of the NetView program. The sample CNMSTYLE member contains descriptive comments about the types of statements that are included in the member. Read the comments and review the default values. For more information about CNMSTYLE statements, see *IBM Z NetView Administration Reference*.

To change a CNMSTYLE statement, copy the statement to either the CNMSTUSR or CxxSTGEN member and then make any necessary updates there.

Update the CNMSTYLE member as required for your environment:

- [“Changing the CNMSTYLE Member Name” on page 47](#)
- [“Using %INCLUDE Members” on page 47](#)

- [“Using System Symbols” on page 48](#)
- [“Customizing the CNMSTYLE Member” on page 49](#)
- [“Activating NetView Components” on page 49](#)
- [“Creating Tower Statements” on page 52](#)
- [“Defining Variables” on page 53](#)
- [“Configuring Allocation Parameters for Canzlog Archive Data Sets” on page 54](#)
- [“Setting up Security” on page 55](#)
- [“Assigning Unique EMCS Console Names” on page 55](#)
- [“Disabling XCF Group Participation” on page 56](#)
- [“Specifying Commands to Run Automatically When the NetView Program Is Started” on page 56](#)
- [“Starting the NetView Subsystem Interface” on page 57](#)
- [“Including Japanese Support” on page 57](#)
- [“CNMSTYLE Processing” on page 58](#)
- [“Using the Report Generator” on page 60](#)

Changing the CNMSTYLE Member Name

The member name for CNMSTYLE is controlled by the string &NV2I in the NetView start procedure and is resolved during fetch processing. It becomes the two characters specified for NV2I on the start command. If no value is specified, the default value is NM. If you specify xx as a value for &NV2I, then the NetView program reads CxxSTYLE in DSIPARM for initialization parameters. If that member cannot be found, the NetView program reads CNMSTYLE instead. The included member CxxSTGEN is also resolved using the value of &NV2I for xx. For more information on NV2I, see [“Configuring Allocation Parameters for Canzlog Archive Data Sets” on page 54](#).

Using %INCLUDE Members

The following members are included when the CNMSTYLE member initializes:

Member Usage

CNMSTACT

Includes the CNMSTAAL member if the IBM Z NetView for Continuous Availability product is installed. The CNMSTACT member is used for the GDPS Continuous Availability solution.

CNMSTPWD

If needed, you can use this member to include VPD, VSAM, and ACB passwords. You can use the READSEC command to protect the CNMSTPWD member from being displayed by the BROWSE command.

CNMSTNXT

Includes modifiable CNMSTYLE statements by release. The CNMSTNXT member is commented out in the CNMSTYLE member. It is provided for documentation purposes only.

CNMSTASK

Task statements that are provided with the NetView program. Do not modify this member. Instead, include any task statements that you want to include directly in the CNMSTUSR or CxxSTGEN member. The task statements in the CNMSTUSR and CxxSTGEN member override those that are provided in the CNMSTASK member.

CNMSTIDS

Includes Intrusion Detection Services (IDS) statements. Review this member if you are enabling IDS support.

CNMSTTWR

Includes statements from non-NetView towers. Do not edit this member unless instructed by documentation for a tower you are installing.

CNMSTWBM

Includes webmenu statements.

CNMSTUSR

You can include global (enterprise) definition statements that override statements in the CNMSTYLE member. Use this member to customize the CNMSTYLE member. You can use Data REXX logic.

Note: You can also use the %INCLUDE statement in this member to include other members of your choosing.

C&NV2I.STGEN

You can include system-specific definition statements in this member, including Data REXX logic.

Note: You can also use the %INCLUDE statement in this member to include other members of your choosing.

Using System Symbols

The NetView program uses MVS system symbols to store various names including the RODM name, TCP/IP stack name, and the network ID. You can initially set these system symbols in the IEASYMxx member in the SYS1.PARMLIB data set:

- &CNMRODM - RODM name
- &CNMTCPN - TCP/IP stack name
- &CNMNETID - network ID

You can use these MVS system symbols to set values in the CNMSTYLE member. During CNMSTYLE processing, NetView symbols with the same name are created and can subsequently be used by the NetView program where symbolic values are allowed. [Table 9 on page 48](#) lists the CNMSTYLE statements that set the values for the given NetView symbol. If you set any of these values in the CNMSTYLE member, the CNMSTYLE value becomes the symbol value for that instance of the NetView program.

Table 9. NetView symbols defined in the CNMSTYLE member		
NetView symbol	Description	CNMSTYLE Statement
&CNMNETID	Network ID	NetID =&CNMNETID.
&CNMRODM	RODM Name	RODMname = &CNMRODM. Note: This statement is ignored if you are not using RODM.
&CNMTCPN	TCP name	TCPname =&CNMTCPN.
&DOMAIN	NetView domain	DOMAIN =C&NV2I.01 (default value is CNM01) Note: This identifier is the access method control block (ACB) name that is listed on the VTAM APPL statement.

The &DOMAIN NetView symbol stores the NetView domain name. The &NV2I NetView symbol is set in the CNMPROC (CNMSJ009) procedure and has a default value of NM if it is not specified.

If you specify the NetView domain ID and password in the CNMPROC (CNMSJ009) procedure, the DOMAIN statement in the CNMSTYLE member and the ACBpassword statement in the CNMSTPWD member are not used. They are ignored unless the parameters that are passed by the CNMPROC procedure are null. If the domain password is not specified in the CNMPROC (CNMSJ009) procedure or in the CNMSTYLE member, the domain name becomes the password.

To find the value of a system symbol or NetView symbol while the NetView program is running, use the SUBSYM REXX function or the SUBSYM pipe stage. If you set a system symbol for the network ID, it must be the same as the value returned by VTAM when the NetView program opens its ACB. The NetView program uses the &CNMTCPN value for certain functions prior to communication with VTAM.

An MVS system symbol can also be specified using MVS commands. In this case, the value is substituted before the command is run

If you want information about...	Refer to...
Defining system symbols	<i>MVS Initialization and Tuning Reference</i>

Customizing the CNMSTYLE Member

Customize the CNMSTYLE member by making global (enterprise) changes to the CNMSTUSR member, and then copying the modified CNMSTUSR member to each NetView system.

You can make system-specific changes to the CxxSTGEN member that is included in the CNMSTYLE member using the %INCLUDE statement (where xx is the value of &NV2I, which is initially set to NM). Code all override statements for the CNMSTYLE and CNMSTUSR members in this member. Duplicate statements found in the CxxSTGEN member override earlier statements in the CNMSTYLE and CNMSTUSR members. You can specify the value of NV2I in the NetView start procedure.

You can use Data REXX in the members that are included in the CNMSTYLE member using the %INCLUDE statement. Data REXX is not supported in the CNMSTYLE member. Instead, you can define tower and subtower conditions in the CNMSTUSR or CxxSTGEN members to control the statements in the CNMSTYLE member.

Note: If you make changes to the included members while the NetView program is running, the changes become effective when you recycle the NetView program. For certain types of changes, you can use the RESTYLE command to activate these changes without recycling the NetView program. For more information on the types of changes that can be activated using the RESTYLE command, see the NetView online help or *IBM Z NetView Command Reference Volume 2 (O-Z)*.

If you want information about...	Refer to...
RESTYLE command	NetView online help or <i>IBM Z NetView Command Reference Volume 2 (O-Z)</i>

Activating NetView Components

NetView components can be activated with TOWER statements. Tower statements are examined earlier in the initialization process than most other variables (for example, common global variables). This is useful, for example, to conditionally control the initialization process.

This is an example of the default TOWER statement:

```
TOWER = *SA *AON *MSM *Graphics NPDA NLDM TCPIP COLLECT
        *AMI *DVIPA *TEMA *IPMG *NVSOA DISCOVERY
```

Usage notes:

1. A tower is enabled if it is not preceded by an asterisk. To enable a tower, remove the asterisk (*) before the tower name.
2. To disable a tower, add a MODIFY.TOWER statement with a minus sign. For example:

```
MODIFY.TOWER.sortkey = -NLDM
```

3. If multiple TOWER statements exist, the last TOWER statement encountered is processed. It is important to remember that modified TOWER statements are not recognized until the NetView program is restarted.

You can use modify statements (MODIFY.TOWER) to enable specific components within a tower. The following are some examples of statements enabling or disabling subtowers:

```
MODIFY.TOWER.sortkey = +AON.SNA +AON.TCP
MODIFY.TOWER.sortkey = -DISCOVERY.INTERFACES -DISCOVERY.TELNET
MODIFY.TOWER.sortkey = -DISCOVERY.INTERFACES.OSA
MODIFY.TOWER.sortkey = +DISCOVERY.INTERFACES.HIPERSOCKETS
MODIFY.TOWER.sortkey = +DVIPA.DVTAD +DVIPA.DVCONN
MODIFY.TOWER.sortkey = +Graphics.SNATM
MODIFY.TOWER.sortkey = +IPMGT.ACTMON +IPMGT.IDS
MODIFY.TOWER.sortkey = +TCPIP.COLLECT.TCPCONN
MODIFY.TOWER.sortkey = +TCPIP.COLLECT.PKTS
MODIFY.TOWER.sortkey = +TEMA.HEALTH +TEMA.OSA
```

Note that the "sortkey" values indicate the order of processing and must all be different. A plus sign indicates an item is being enabled and a minus sign indicates an item is being disabled. For any attempt to enable or disable a subtower to be effective, the tower must have previously been enabled.

Review the subtower statements that are associated with the towers that are supplied with the NetView product that you enable. To enable a subtower, create a MODIFY.TOWER statement in the CNMSTUSR or CxxSTGEN member. To enable or disable any tower or subtower, you must recycle the NetView program, so review these statements carefully. If you plan on implementing any of the tower and subtower components, consider enabling the functions during this step in the installation process.

Table 10 on page 50 lists the towers and subtowers that are provided with the NetView program:

Table 10. CNMSTYLE towers and subtowers			
Tower	Enables	Subtower	Enables
AMI	Application Management Instrumentation	None	
AON	Network automation (AON component)	SNA	SNA automation. Subtower: X25 AON/SNA X.25 support
		TCP	TCP/IP automation. Subtower: IDS Intrusion Detection Services (IDS) support
DISCOVERY	Discovery of sysplexes, z/OS systems, coupling facilities, TCP/IP stacks, TCP/IP subplexes, and NetView applications	INTERFACES	Discovery of IP interface information. Subtowers: OSA Discovery of OSA channels and ports. HIPERSOCKETS Discovery of HiperSockets configuration and status information.
		TELNET	Discovery of Telnet servers and ports.

<i>Table 10. CNMSTYLE towers and subtowers (continued)</i>			
Tower	Enables	Subtower	Enables
DVIPA	Discovery of dynamic virtual IP address (DVIPA) definition and status data	DVCONN	Discovery of DVIPA connections data.
		DVROUT	Discovery of VIPA routes and distributed DVIPA connection routing data.
		DVTAD	Discovery of DVIPA sysplex distributors, distributed DVIPA targets, distributed DVIPA server health statistics, and distributed DVIPA statistics (if enabled).
GRAPHICS	NetView Management console	SNATM	SNA Topology Manager.
IPMGT	IP management	ACTMON	Active monitoring for IP resources without enabling the AON tower.
		IDS	Intrusion Detection automation without enabling the AON tower.
MSM	MultiSystem Manager	OPN	Open function.
NLDM	Session monitor	None	
NPDA	Hardware monitor	None	
NVSOA	Web Services Gateway	None	
SA	IBM Z System Automation	None	
TCIPCOLLECT	Collection of TCP/IP connection and packet trace data	CONNSEC	Collection of encryption-related data for IP connections.
		TCPCONN	Monitoring of IP connections.
		PKTS	Collection of packets.

Table 10. CNMSTYLE towers and subtowers (continued)			
Tower	Enables	Subtower	Enables
TEMA	Communication with the Z NetView Enterprise Management Agent. Usage Note: Do not enable the TEMA tower unless you are installing the Z NetView Enterprise Management Agent. Only enable the TEMA tower on one NetView program for each LPAR.	CONINACT	Collection and display of inactive TCP/IP connections.
		CONNACT	Collection and display of active TCP/IP connections.
		CONNSEC	Collection and display of encryption-related data for active TCP/IP connections.
		DVCONN	Display of DVIPA connections.
		DVDEF	Display of DVIPA definition and status data
		DVROUT	Display of VIPA route and distributed DVIPA connection routing data.
		DVTAD	Display of distributed DVIPA data.
		HEALTH	Collection and display of NetView task data.
		HIPERSOCKETS	Display of HiperSockets configuration and status information. (Requires RODM.)
		OSA	Display of OSA channels and ports information. (Requires RODM.)
		SESSACT	Collection and display of active sessions. The SESSACT subtower is only supported in one NetView program per system.
		SYSPLEX	Display of stack configuration and status data.
		TELNET	Display of Telnet servers and Telnet server port information.

If you want information about...

Refer to...

AON, hardware monitor, session monitor

IBM Z NetView Installation: Configuring Additional Components

MultiSystem Manager subtowers

IBM Z NetView Installation: Configuring Graphical Components

Z NetView Enterprise Management Agent tower and subtowers

IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent

Creating Tower Statements

You can use the TOWER statement to enable or disable towers that you create. You can use TOWER-qualified statements to define subtowers, for example:

- TOWER.twrname = subtwr

- TOWER.twrname.subtwr = subsubtwr

This is an example of TOWER statements that define a tower named PROBMGMT and a subtower named HighPriProbs:

```
TOWER = *SA *AON *MSM *Graphics NPDA NLDM TCIPCOLLECT
      *AMI *DVIPA *TEMA *IPMG *NVSOA DISCOVERY PROBMGMT
TOWER.PROBMGMT = HighPriProbs
```

Note: In this and other examples, the names PROBMGMT, PROBTASK, and HighPriProbs are used as examples; you can use any 1-16 character name for a tower statement.

In the example that follows, the name of an autotask is defined for a user-written problem management application and the name of an additional autotask is defined for a user-written application. The user written application is used to analyze high priority problems that are handled by the problem management application:

```
(PROBMGMT)function.autotask.probmGR = PROBTASK
(PROBMGMT.HighPriProbs)function.autotask.PrimGR = HPRITASK
```

The PROBTASK autotask is only defined when the PROBMGMT tower is enabled. The HPRITASK autotask is only defined when the HighPriProbs subtower is enabled in the TOWER.PROBMGMT statement. Additional information about the function.autotask statement can be found in *IBM Z NetView Administration Reference*.

You can add Data REXX logic to the CNMSTUSR or CxxSTGEN member to conditionally process definition statements based on whether a particular tower is enabled. This is an example:

```
%> IF tower('towername') THEN
%> DO;
    definition statements
%> END;
```

If you use Data REXX logic, ensure that the %> characters begin in column one. Lines beginning with a blank are considered continuation statements from the previous line. Also, Data REXX files must begin with either a /* %DATA */ or /* %LOGIC */ statement.

If you want information about...	Refer to...
Using Data REXX	<i>IBM Z NetView Programming: REXX and the NetView Command List Language</i>

Defining Variables

Use STYLEVAR to define variables that can be used anywhere within the CNMSTYLE member (except for the command phase, described in “Command Phase” on page 59). You can use these variables to simplify the process of entering repetitious data. The STYLEVAR phase is described in “STYLEVAR Phase” on page 59.

Usage notes:

1. System variable names are not valid names for STYLEVAR variable names.
2. STYLEVAR variable values cannot contain another STYLEVAR variable.
3. If you are using a STYLEVAR variable in the CNMSTYLE member, the construction &&varname is not supported, where varname is the STYLEVAR variable.

If you want information about...	Refer to...
STYLEVAR statement	Comments in the CNMSTYLE member and <i>IBM Z NetView Administration Reference</i>

Configuring Allocation Parameters for Canzlog Archive Data Sets

The Canzlog archiving function preserves message data that is stored in the Canzlog log. When enabled, the Canzlog archiving function allocates sequential data sets after NetView initialization. The data set allocation specifications are controlled by the ARCHIVE statements in the CNMSTYLE member. Because large amounts of data can be archived by the NetView program, make sure that the allocation parameters are configured correctly for your environment.

To enable the Canzlog archiving function, add an ARCHIVE.HLQ CNMSTYLE statement with a valid high-level qualifier to your CNMSTUSR or CxxSTGEN member, for example:

```
ARCHIVE.HLQ = NETVIEW
```

If you are not using the default NetView data set naming convention or want to use a different high-level qualifier for the Canzlog archive data sets, replace the value NETVIEW with the high-level qualifier that you want to use.

The Canzlog archiving function allocates several different types of sequential data sets:

- Message data sets are allocated to store archived message data from the Canzlog log.
- Index data sets are allocated to reference the message data sets as they are created.
- A primary index data set is allocated to reference the index data sets as they are created.

The allocation parameters for the message data sets are specified by the ARCHIVE.MESSAGE CNMSTYLE statements. The allocation parameters for the index data sets and the primary index data set are specified by the ARCHIVE.INDEX CNMSTYLE statements.

Default allocation parameters are provided by the NetView program for allocating non-storage management subsystem (non-SMS) sequential data sets that are written to a 3390 DASD device. Depending on your environment, you might want to customize the allocation specifications. See [Table 11 on page 54](#) for a list of statements that contain allocation parameters for the archive Canzlog data sets.

Table 11. ARCHIVE CNMSTYLE Statements		
Volume	Data set type	CNMSTYLE statements
SMS	Message	• ARCHIVE.MESSAGE.DATACLAS • ARCHIVE.MESSAGE.STORCLAS • ARCHIVE.MESSAGE.MGMTCLAS
	Index and primary index	• ARCHIVE.INDEX.DATACLAS • ARCHIVE.INDEX.STORCLAS • ARCHIVE.INDEX.MGMTCLAS
Non-SMS	Message	• ARCHIVE.MESSAGE.VOLUMES • ARCHIVE.MESSAGE.UNIT • ARCHIVE.MESSAGE.BLOCKSIZE • ARCHIVE.MESSAGE.SPACE
	Index and primary index	• ARCHIVE.INDEX.VOLUME • ARCHIVE.INDEX.UNIT • ARCHIVE.INDEX.BLOCKSIZE • ARCHIVE.INDEX.SPACE

Because most production systems generate large amounts of message data, you might want to use dedicated DASD storage for managing your Canzlog log archive data sets. Your data management set-up

must allow the NetView program to dynamically allocate and catalog the sequential data sets that are allocated by the Canzlog archiving function.

Only one NetView program on a given z/OS system can archive Canzlog data regardless of how many NetView programs are configured with archiving enabled. To disable Canzlog archiving for a particular NetView program, set the ARCHIVE.WRITE statement to NO.

You can change the value for any of the archive allocation parameters after the NetView program initializes by modifying the ARCHIVE statements and issuing the RESTYLE ARCHIVE command. When you issue the RESTYLE command, there is no loss of data.

If you want information about...	Refer to...
ARCHIVE statements	<i>IBM Z NetView Administration Reference</i>
Defining Canzlog archive data sets	<i>IBM Z NetView Installation: Configuring Additional Components</i>
DASD requirements for Canzlog archiving	<i>IBM Z NetView Tuning Guide</i>

Setting up Security

You can use the SECOPTS statement to specify levels of security:

- Operator security
- Command authority
- Span of control authority

If you want information about...	Refer to...
Security options	<i>IBM Z NetView Security Reference</i>
SECOPTS keywords	<i>IBM Z NetView Administration Reference</i>

Assigning Unique EMCS Console Names

EMCS consoles are used to send commands from the NetView program to the MVS operating system and to receive messages from MVS. The console name must be unique for each MVS or each sysplex. To avoid console name conflicts, you can use the NetView program to assign a permanent, unique console for each operator by using the following methods:

- Using the ConsMask statement in the CNMSTYLE member. This is the preferred method.

Choose a combination of letters to generate unique names in your environment. When an MVS command, or a GETCONID command without the CONSOLE keyword, is issued before any SETCONID command, then NetView chooses a console name by substituting the letters in the mask from the environment at that time.

- Using the SETCONID command.
- Defining the CONSNAME attribute in the operator profile.

One way to create unique consoles is to use the NetView support for the MVS cloning function. The &SYSCClone system symbolic variable can be used to construct a unique CONSNAME value for each operator on each system within a sysplex.

- Using the GETCONID command with the CONSOLE keyword.

If you do not use the ConsMask statement or the CONSNAME or GETCONID commands, NetView automatically assigns a default console that is the same as the operator task ID.

The console names you specify in CONSOLxx are not available for use as EMCS console names. Therefore, if you specify any console names in CONSOLxx, ensure that they are different from your NetView operator IDs so that they do not conflict with the default EMCS console names.

Use the RELCONID command to release MVS consoles that were previously obtained using an MVS or a GETCONID command.

If you want information about...	Refer to...
Updating CNMSTYLE statements	“Updating the CNMSTYLE Member” on page 46
ConsMask, MVSPARM, CONSNAME	<i>IBM Z NetView Administration Reference</i>
GETCONID, MVS, RELCONID, and SETCONID commands	NetView online help or <i>IBM Z NetView Command Reference Volume 1 (A-N)</i>
z/OS cloning function, system symbolic variables	<i>z/OS library</i>

Disabling XCF Group Participation

To support discovery manager, by default, the NetView program participates in an XCF group that enables communication with other NetView programs in your sysplex environment. If you do not want the NetView program to participate in this XCF group, you can disable this function by adding the following statement in the CNMSTUSR or CxxSTGEN member:

```
XCF . RANK=-1
```

XCF group participation can be enabled after NetView initialization by changing the XCF.RANK value and issuing the RESTYLE XCF command.

Usage note: If you are running the GDPS Continuous Availability solution, do not disable XCF group participation. For information on setting the XCF rank for the solution, see *IBM Z NetView for Continuous Availability Configuring and Using the GDPS Continuous Availability Solution*.

If you want information about...	Refer to...
XCF.RANK statement	<i>IBM Z NetView Administration Reference</i>
Discovery manager	<i>IBM Z NetView IP Management</i>
XCF Services	<i>IBM Z NetView IP Management</i>

Specifying Commands to Run Automatically When the NetView Program Is Started

To define a command or a command list to run automatically when the NetView program is started, use the auxInitCmd statement in CNMSTUSR or CxxSTGEN. You can specify any number of commands or command lists to be run. The EBCDIC value following the auxInitCmd keyword determines the order the commands are run.

An example follows:

```
auxInitCmd.A = MSG SYSOP,Auxiliary commands beginning.  
auxInitCmd.AC = LISTVAR
```

In this case, the MSG SYSOP command (A) runs before the LISTVAR command (AC).

Note: These AuxInitCmd commands run before any commands at any autotask. All commands for autotasks, including both task initial command lists and commands sent by EXCMD, are queued and held up. They run only after all AuxInitCmds have completed. Messages are also queued; they are not submitted to automation nor logged until all AuxInitCmds have completed.

If you want information about...	Refer to...
Creating a command list to run at NetView initialization	<i>IBM Z NetView Programming: REXX and the NetView Command List Language</i>

Starting the NetView Subsystem Interface

You can start the NetView Subsystem Interface (SSI) by using the SSI.ProcString statement in CNMSTUSR or CxxSTGEN:

```
SSI.ProcString = CNMPSSI.SS,SUB=MSTR,ARM='*ARM'
```

Specify the procedure name (for example, CNMPSSI). The NetView program provides a value for the PPIOPT parameter. You can also specify additional start parameters (such as SUB=MSTR) that are required for your installation.

Note: Do not specify the PPIOPT parameter on the SSI.ProcString statement because the NetView program adds this parameter during processing. To update this value, use the SSI.PPI statement.

If you specify *NONE* for SSI.ProcString statement, the CNMCSSIR task does not start the SSI procedure. This is the default.

If you want information about...	Refer to...
Starting the SSI using CNMPSSI	“Modifying the NetView Subsystem Interface Procedure” on page 39
SSI CNMSTYLE statements	Comments in the CNMSTYLE member and <i>IBM Z NetView Administration Reference</i>

Specifying the Subsystem Command Designator

To specify the NetView subsystem command designator, use the MVSPARM.Cmd.Designator statement in the CNMSTYLE member. For more information about the MVSPARM.Cmd.Designator statement, see the *IBM Z NetView Administration Reference*.

Including Japanese Support

The NetView program supports EBCDIC or Kanji character sets. All NetView workstations in a domain must support the character set that you decide to use. Multilingual support is not available. Japanese and English domains can coexist. The NetView program sends messages in English between these domains. To include the Kanji character set, copy the transTbl statement from CNMSTYLE to CNMSTUSR or CxxSTGEN, and specify DSIKANJI.

```
transTbl = DSIKANJI
```

To include Japanese translations of messages, copy the following statement from CNMSTYLE to CNMSTUSR or CxxSTGEN, and remove the asterisk from the beginning of the statement:

```
transMember = CNMTRMSG
```

The CNMTRMSG member contains the Japanese message translations. It is located in the NETVIEW.V6R3M0.SDSIMSG1 data set in the DSIMSG DD concatenation in the NetView procedure (CNMPROC). Edit the CNMTRMSG member and uncomment the %INCLUDE for CNMMSJPN. To preserve the SMP/E-installed copy of the CNMTRMSG member, allocate a user SDSIMSG1 data set, copy the CNMTRMSG member into this user data set, and then edit the CNMTRMSG member in the user data set. To make sure that user data set changes are in effect, concatenate the user data set ahead of the NETVIEW.V6R3M0.SDSIMSG1 data set.

If you want information about...	Refer to...
Enabling Japanese support	<i>IBM Z NetView Installation: Configuring Additional Components</i>
CNMSTYLE statements	Comments in the CNMSTYLE member and <i>IBM Z NetView Administration Reference</i>

CNMSTYLE Processing

Processing for the CNMSTYLE member occurs in five phases:

- [“Early Initialization Phase” on page 58](#)
- [“Command Phase” on page 59](#)
- [“STYLEVAR Phase” on page 59](#)
- [“Common Phase” on page 59](#)
- [“Primary Task Phase” on page 59](#)

During each phase, except for the command phase, the CNMSTYLE member and any members that it includes are read from disk. Statements that define product-supported values and functions are processed. Each time the CNMSTYLE member reads an included member, any Data REXX statements found are run, causing customized data to be processed during NetView initialization. Because values and settings can change from phase to phase, the results of running any Data REXX statements can vary from phase to phase. However, any given CNMSTYLE statement is processed by only one of the processing phases. Ensure that any Data REXX statements used to control generated statements resolve to values that are expected in the phase in which the statements are processed. For example, do not use a condition in Data REXX that tests for a value that is set in the same phase as the generated statement it is controlling. The phase descriptions that follow can be used to determine which Data REXX functions or variables are appropriate to use in Data REXX statements during that phase. Note that statements are evaluated in the order determined by the NetView program and not by the order that they are listed in the CNMSTYLE member.

Early Initialization Phase

During the first initialization phase, the following actions occur:

1. The &NV2I variable returns the two-character value you set in your start procedure (CNMPROC). You can embed this symbol in %INCLUDE member names. The REXX function SUBSYM('&NV2I') provides this value for use in Data REXX.
2. The domain() function returns a value and the &DOMAIN variable has a value only if you specified DOMAIN on your start procedure. Otherwise, null is returned by the function and &DOMAIN resolves to "&DOMAIN".
3. In this phase, the tower() function returns false regardless of the options enabled.
4. Any variables you define in member IEASYMxx in SYS1.PARMLIB can be used with the REXX subsym() function and symbolic variables (&symbolname.).
5. Common global variables CNMSTYLE.STYLE and CNMSTYLE.NV2I are set to the name of the CNMSTYLE member being read and to the value of the &NV2I variable, respectively. All other global variables are null.
6. All other symbols and functions are null or set to their documented default values.

Prior to the end of early initialization, the following actions occur:

- DSIJSTYL variables are resolved. For a list of the DSIJSTYL variables, refer to procedure CNMESTYL.
- If your domain ID was not set in the start procedure, it is set to the value that is specified in the CNMSTYLE member (refer to the DOMAIN keyword in the DSIJSTYL member).
- TOWER statements are evaluated.
- MODIFY.TOWER statements are evaluated.
- The following variables are assigned values based on their associated CNMSTYLE statements. This overrides values found in member IEASYMxx of SYS1.PARMLIB.

Variable	CNMSTYLE Statement
&CNMNETID.	NETID
&CNMRODM.	RODMNAME

Variable	CNMSTYLE Statement
&CNMTCPN.	TCPNAME

The system variables can then be used throughout the NetView program.

Command Phase

During the command phase, the CNMCMD command definitions are evaluated. The values of tower, domain, type, and other variables set in the early initialization phase control how these members are read.

Member DSISCHED is read and other initialization members not associated with a CNMSTYLE statement or optional task are also read.

If you coded a command definition of CNMSTRLY in DISPARM member CNMCMD, the defined REXX procedure is called. Do not call any long running commands from this procedure, except PIPE. Do not call the following commands:

- AUTOTASK
- AUTOTBL
- REFRESH
- START

You can use MVS commands. Do not use POI VTAM commands. If a problem occurs with this procedure, the NetView program might not initialize. Also, if you preset variables designed to be set by CNMSTYLE processing, the NetView program might not initialize properly.

STYLEVAR Phase

During the STYLEVAR phase, the CNMSTYLE member specifications are reevaluated. All &xxx. stylevar variables that are used in CNMSTYLE statements are replaced with the value specified in the corresponding stylevar.xxx entry.

Common Phase

During the common phase, the CNMSTYLE member specifications are re-evaluated. Because of possible changes in the value of tower, domain, type, and other variables, different statements might be included by Data REXX or %INCLUDE processing.

Note: Data REXX can be used only in the members that are included in the CNMSTYLE member using the %INCLUDE statement. Do not use Data REXX in the main CNMSTYLE member (CxxSTYLE).

Because tower statements have already been evaluated, they are used when reading the CNMSTYLE member for this phase. If a CNMSTYLE statement is preceded by a tower name in parentheses, it is included during initialization, if the tower is active. Otherwise, the statement is ignored. Do not use variables or other substitutions within these parentheses.

The CNMSTYLE statements identified by the values in variable earlyCommon (in procedure CNMESTYL) are processed. For more information about the variable earlyCommon and on which CNMSTYLE statements are processed, refer to the earlyCommon in procedure CNMESTYL.

Common global variables are set. This includes interpretation of function.autotask.xxx statements.

Primary Task Phase

The CNMSTYLE member is read again. Because of variables being set during the common phase, different CNMSTYLE statements might be included. All CNMSTYLE statements that were not processed previously are now processed. The following actions then take place:

- Optional tasks are started beginning with netlog (DSILOG).
- NetView tracing is restarted if options different from the default values are specified in the CNMSTYLE member.

- Autotasks are started. However, command and message processing is suspended in these tasks until NetView initialization is complete.
- If VTAM is active, the NetView ACB is opened. When the NetView ACB is opened successfully, operators can log on.

After all other statements are processed, auxInitCmd statements are called. You can use auxInitCmd statements to add your own processing to NetView initialization.

After the last auxInitCmd command returns control, command and message processing is enabled in autotasks and operator station tasks. Initialization is complete.

Note: Some activities related to initialization can run for an extended time (for example, population of the RODM database and resolution of IP domain names).

Using the Report Generator

You can use the CNMSTYLE report generator to analyze the CNMSTYLE member. You can use the report that is created to take the following actions:

- List the %INCLUDE structure.
- Analyze multiple occurrences of statements within the CNMSTYLE member. Use this to determine which value is used during NetView initialization. For statements that are listed multiple times in the report, the last statement that is listed is the one used for initialization.
- List the CNMSTYLE towers that are enabled.
- Analyze initialization statements for a particular function.

Running the report

To run the CNMSTYLE report generator, use the CNMSJCRG sample in the NETVIEW.V6R3USER.INSTALL data set. This INSTALL data set was created during installation by the CNMSJBUP sample job. The CNMSJCRG sample is a job that runs outside of the NetView address space and runs the REXX program CNMECRG under the TSO terminal monitor program.

The CNMSJCRG sample requires the following data sets:

STEPLIB

The NetView CNMLINK data set from the current release, NETVIEW.V6R3M0.CNMLINK.

SYSEXEC

The concatenated data set list of the NetView CNMCLST data sets from the current release.

DSIPARM

The concatenated data set list containing current release versions of the CNMSTYLE member. Ensure that the data set concatenation order is the same as that specified in the NetView start procedure CNMPROC.

CNMPNL1

The NetView CNMPNL1 data set from the current release, NETVIEW.V6R3M0.CNMPNL1.

DSIWRIT

The output partition data set to which the generated report member is written. The output of the report generator is written as a member of a partition data set. If you use the NetView default naming convention, the data set name is NETVIEW.V6R3USER.CNM01.DSILIST.

The CNMSTYLE report is written to member CNMCRG in the DSIWRIT data set. If member CNMCRG already exists, a backup copy of the existing CNMCRG member is created and named CNMCRGBK. If member CNMCRGBK exists, it is overwritten with the existing CNMCRG member.

You can specify keyword parameters in CNMSJCRG. Each keyword parameter and value must be specified on a separate line, just below the CNMECRG command. Do not continue the value onto a second line. All characters typed on a line are interpreted as input to CNMECRG. If a keyword parameter is specified more than once, the first value is used and all subsequent values specified are ignored. Input ends when either a blank line or a /* occurs.

You can specify the following keywords:

TASKS=YES | NO

Specifies whether to include CNMSTASK statements in the report.

YES

Includes statements from the CNMSTASK member. This is the default value.

NO

Does not include CNMSTASK statements.

&NV2I=xx

The default value for xx is NM. If a value that is not valid is specified, an error message is issued and the default value NM is used in the report. If you use alphabetic characters, the characters are converted to uppercase.

&symbolic_name= value

Provides the *value* of a system or NetView symbolic variable (&symbolic_name) that you are using in the CNMSTYLE member. A symbolic parameter must be passed to CNMECRG to be resolved in the report.

The CNMSTYLE Report Generator, when reading a NetView definition member, cannot resolve symbolic references that refer to a substring of a symbolic variable such as

```
%INCLUDE C&DOMAIN(2:2).STGEN
```

Usage notes:

1. Precede the *symbolic_name* with an ampersand (&).
2. The *symbolic_name* can optionally include a trailing period (.).
3. Use single quotation marks (' ') if *value* has leading or trailing blanks.
4. Do not specify a value that contains a symbolic variable, such as

```
&AAAAA=' C&NV2I.01 '
```

or that contains a substring of a symbolic variable, such as

```
&AAAAA=' C&DOMAIN(2:2).01 '
```

The following example shows keyword parameters for CNMECRG within CNMSJCRG:

```
CNMECRG
TASKS=NO
&DOMAIN=CNM01
&NV2I=NM
&CNMTCPN=TCPIP
&CNMRODM=RODMNAME
&CNMNETID=NETA
&MYSYMBL=' A B C '
/*
```

Analyzing the Report

The CNMSTYLE report includes the following sections:

1. General information and CNMSTYLE statements that pertain to all of NetView
2. CNMSTYLE statements that pertain to specific functions of NetView
3. auxInitCmd statements and user-defined statements
4. User-defined and unrecognized statements
5. Data REXX and %INCLUDE statements

Because the TASKS parameter is set to NO, the report in this example does not include CNMSTASK statements.

The first section of the CNMSTYLE report is shown in Figure 4 on page 62. This part of the report contains general information related to the CNMSTYLE member, such as:

- The date and time the report was created
- The &NV2I symbolic variable value being used
- A nested listing of the members included by the CNMSTYLE member
- A list of the CNMSTYLE towers that are enabled when NetView initializes
- A list of CNMSTYLE statements that apply to base NetView

```

                                CNMSTYLE REPORT

DATE: 21 Jul 2019
TIME: 14:03:25

&NV21 value: NM

%INCLUDE structure of: CNMSTYLE

    CNMSTYLE
      CNMSTPWD
      CNMSTASK
      CNMSTIDS
      CNMSTTWR
      CNMSTWBM
      CNMSTUSR
      MYINCLUD
      CNMSTGEN

Enabled Towers:  NPDA  NLDM  TCIPCOLLECT  DISCOVERY

Statements for function: NetView General

Member   Line#  Indicators  Statement
-----
CNMSTYLE  217  Y          DOMAIN = CNM01
CNMSTYLE  304  Y          NetID = NETA
CNMSTYLE  747          TOWER = *SA *AON *MSM *Graphics NPDA NLDM
                        | TCIPCOLLECT *AMI *DVIPA *TEMA *IPMG
                        | *NVSOA DISCOVERY
CNMSTYLE  1589        CNMI = Yes
CNMSTYLE  582        SECOPTS.OPERSEC = NETVPW
CNMSTYLE  596        SECOPTS.SURROGAT = NO
CNMSTYLE  615        SECOPTS.CMDAUTH = TABLE.CNMSCAT2
CNMSTYLE  626        SECOPTS.AUTHCHK = SOURCEID
CNMSTYLE  634        SECOPTS.OPSPAN = NETV
CNMSTYLE  653        SECOPTS.SPANAUT = *NONE*
      .           .
      .           .
      .           .

```

Figure 4. First section of CNMSTYLE report

The format of the CNMSTYLE statements presented in the generated report includes the following fields:

Member

Member name containing the statement

Line#

Line number within the member where the CNMSTYLE statement is located. If a statement is a continuation statement, only the line number where the statement begins is listed.

Indicators

Lists information about the statement. This information is formatted in the following way:

```

R CCCCCC

```

where **R** represents the Resolve indicator and **CCCCCC** represents the Condition indicator:

Resolve

Indicates whether the given CNMSTYLE statement was modified by the report generator. A specification of **Y** indicates that the statement was modified. For example, a symbolic variable was substituted or an autotask statement that uses the question mark (?) feature was resolved.

If the resolve field has no value listed, no modifications were made to the statement.

Condition

Indicates that a condition is required for the listed CNMSTYLE statement to be active, such as a tower that must be enabled. If only one tower is required to be enabled the condition field is set to the required tower name. The first 10 characters of the tower name are listed. If more than one tower must be enabled or if some other condition must be met, the condition field is set to four asterisks (****).

If the condition field has no value listed, no conditions are required for the statement to be active.

Statement

Lists the CNMSTYLE statement and its value. Extra spacing in the statement might be removed, along with any tower conditionals that are found at the beginning of the statement. Statements can be further modified by having values substituted into either the CNMSTYLE keyword or its value.

Values of CNMSTYLE keywords that contain passwords and other values critical to security are identified as a security risk and are listed in the report as four asterisks (****) to prevent unauthorized viewing.

The second section of the report lists CNMSTYLE statements for specific NetView functions. For example, [Figure 5 on page 63](#) lists statements for the hardware monitor (NPDA) component.

If a CNMSTYLE statement applies to multiple NetView functions, that statement is listed for each NetView function to which it applies. For example, the TOWER statement applies to both the hardware monitor and the session monitor, and various other NetView functions.

Within a function, the most critical statements are listed first, followed by less critical statements. NetView functions are presented in the report alphabetically.

Statements for function: Hardware Monitor (NPDA)

Member	Line#	Indicators	Statement
CNMSTYLE	747		TOWER = *SA *AON *MSM *Graphics NPDA NLDM TCPIPCOLLECT *AMI *DVIPA *TEMA *IPMG *NVSOA DISCOVERY
CNMSTYLE	1701		TASK.BNJMNPD.INIT = N
CNMSTYLE	1690		TASK.BNJDSEV.INIT = N
CNMSTYLE	2733		NPDA.ALCACHE = WRAPCNT
MYINCLUD	18	NPDA	NPDA.ALCACHE = 500
CNMSTYLE	2749		NPDA.ALERTFWD = SNA-MDS-LOGONLY
CNMSTYLE	2807		NPDA.ALERTLOG = RANDRANG
CNMSTYLE	2783		NPDA.ALRTINFP.RECORD = Yes
CNMSTYLE	2793		NPDA.ALT_ALERT = DOMAIN
CNMSTYLE	2864		NPDA.AUTORATE = 1
CNMSTYLE	2701		NPDA.DSRBO = 5
CNMSTYLE	2693		NPDA.DSRBU = 5
CNMSTYLE	2901		NPDA.ERR_RATE = 10 50
CNMSTYLE	2708		NPDA.MACRF = LSR
CNMSTYLE	2758		NPDA.MDSIND = Yes
CNMSTYLE	2686		NPDA.PDDNM = BNJLGPR
CNMSTYLE	2714		NPDA.PNA = No
CNMSTYLE	2873		NPDA.PRELOAD_BER = No
CNMSTYLE	2720		NPDA.REPORTS = OFF
CNMSTUSR	14		NPDA.REPORTS = ON
CNMSTYLE	2687		NPDA.SDDNM = BNJLGSE
CNMSTYLE	2728		NPDA.TECROUTE = IHSATEC
MYINCLUD	17	NPDA	NPDA.W.1 = AL 500
CNMSTYLE	1846		function.autotask.HMONdbMaint = DBAUT02

Figure 5. NetView Function Information

The third section of the report lists the auxInitCmd statements as shown in [Figure 6 on page 64](#). The auxInitCmd statements are listed in the order they are encountered in the CNMSTYLE member.

auxInitCmd Statements

Member	Line#	Indicators	Statement
CNMSTYLE	4633		auxInitCmd.A = MSG SYSOP,Auxiliary commands beginning.
CNMSTYLE	4636	NLDM	auxInitCmd.SNLDM = STARTCNM NLDM
CNMSTYLE	4637	NPDA	auxInitCmd.SNPDA = STARTCNM NPDA
CNMSTYLE	4638		auxInitCmd.POLICY = EXCMD ?Policy,EZLEANTL
CNMSTYLE	4639	DISCOVERY	auxInitCmd.ZDISC = EXCMD ?Policy,CNMEERSC
MYINCLUD	16	NPDA	auxInitCmd.BB = MSG SYSOP,NPDA will be activated

Figure 6. auxInitCmd Statements

The fourth section of the report lists CNMSTYLE statements that are not recognized by the CNMSTYLE report generator as belonging to a specific NetView function or to the general NetView information in the first section of the report. A statement might be listed under *User-Defined/Unrecognized CNMSTYLE Statements* for the following reasons:

- A statement might be owned by another product such as IBM Z System Automation.
- A statement might require an additional CNMSTYLE statement to become active.

For example, the AUTOTASK.?APSERV.Console CNMSTYLE statement requires the function.autotask.APSERV statement before it can become active.

- A statement was moved from IBM Z NetView to another product.

For example, the ACTACT.MASTER.DELAY statement was part of the IBM Z NetView product but was moved to the IBM Z NetView for Continuous Availability product. If the ACTACT.MASTER.DELAY statement was modified and placed in the CNMSTUSR member and the IBM Z NetView product but was moved to the IBM Z NetView for Continuous Availability product is not installed, the ACTACT.MASTER.DELAY statement is not recognized by the Report Generator until the IBM Z NetView product but was moved to the IBM Z NetView for Continuous Availability product is installed.

- A statement is created by a user.

For example, you define the OPAAA01 autotask as follows:

```
%> IF TOWER('NPDA') THEN DO;
function.autotask.MyAutoOp = OPAAA01
%> END;
```

The function.autotask.MyAutoOp statement is listed as a user-defined/unrecognized statement in Figure 7 on page 64.

User-Defined/Unrecognized CNMSTYLE Statements

This section identifies statements in your CNMSTYLE member that were either created by the user, require additional CNMSTYLE statements to become active, were moved from Z NetView to another product, or are owned by another product.

Member	Line#	Indicators	Statement
CNMSTYLE	361		AUTOTASK.?Helper.Console = D761CON
CNMSTYLE	1079		AUTOTASK.?APSERV.Console = *NONE*
CNMSTYLE	1080		AUTOTASK.?APSERV.InitCmd = APSERV xyz
CNMSTYLE	1090		AUTOTASK.?SMF30.Console = *NONE*
CNMSTYLE	1091		AUTOTASK.?SMF30.InitCmd = CNMSMF3R
MYINCLUD	15	NPDA	function.autotask.MyAutoOp = OPAAA01

Figure 7. User-Defined/Unrecognized CNMSTYLE Statements

The fifth section of the report lists Data REXX and %INCLUDE statements, as shown in Figure 8 on page 65. During report processing, Data REXX statements are ignored. Data REXX statements are listed in the report in the order that they are encountered in the CNMSTYLE member. Only the first 63 characters of each Data REXX statement are placed in the report. CNMSTYLE statements within a %DATA portion of a Data REXX block that are affected by an IF-THEN statement are also listed to help you understand which CNMSTYLE statements are impacted by your Data REXX statements.

%INCLUDE statements are also listed in section 5 and are listed within the Data REXX statements in the order that they are encountered in the CNMSTYLE member. All %INCLUDE members that are found are included by the CNMSTYLE report generator in the report. Because some %INCLUDE members might be conditionally included, review the conditions surrounding a %INCLUDE statement to ensure that the report reflects your environment.

Member names on %INCLUDE statements that are not found in the DSIPARM concatenation result in a return code of 4 from the report generator. If you feel that a member is missing but should have been included, review your DSIPARM concatenation in CNMSJCRG for any missing data sets. However, be aware that some members might ship with a feature that you have not installed currently.

Data REXX and %INCLUDE Statements

Member	Line#	Statement
CNMSTYLE	242	%INCLUDE CNMSTPWD
CNMSTYLE	1737	%INCLUDE CNMSTASK
CNMSTASK	1	/*%DATA REXX -----
CNMSTASK	17	/*%LOGIC -----
CNMSTASK	18	IF TYPE() = 'ENT' THEN /* Enterprise NetV
CNMSTASK	19	DO;
CNMSTASK	20	polAuto = cglobal('CNMSTYLE.AUTO.POLICY')
CNMSTASK	21	aonAuto = cglobal('CNMSTYLE.AUTO.AON')
CNMSTASK	22	IF aonAuto = '' THEN aonAuto = 'AUTOAON' /* Default name
CNMSTASK	23	IF polAuto = '' THEN polAuto = 'AUTOAON' /* Same default
CNMSTASK	24	IF polAuto ^= aonAuto THEN
.	.	.
.	.	.
.	.	.
CNMSTUSR	11	%INCLUDE MYINCLUD
MYINCLUD	1	/*%DATA REXX -----
MYINCLUD	13	%> IF TOWER('NPDA') THEN DO;
MYINCLUD	15	function.autotask.MyAutoOp = OPAAA01
MYINCLUD	16	auxInitCmd.BB = MSG SYSOP,NPDA will be activated
MYINCLUD	17	NPDA.W.1 = AL 500
MYINCLUD	18	NPDA.ALCACHE = 500
MYINCLUD	20	%> END;
CNMSTYLE	5230	%INCLUDE CNMSTGEN
CNMSTGEN	1	/*%DATA REXX -----
CNMSTGEN	13	%> IF domain() = 'CNM01' THEN /* Never true! Data REXX runs...
CNMSTGEN	14	%> /* earlier than resolution of system-sym
CNMSTGEN	15	%> 'STYLEMSG = Illustration only. You will never see this.'

Figure 8. Data REXX and %INCLUDE Statements

Note: Only the first 63 characters of each Data REXX statement are displayed.

Return codes

The following return codes are set by CNMECRG:

- 0** Successful completion; a file was created in DSIWRIT
- 4** Minor errors encountered; a file was created in DSIWRIT
- 8** Major error encountered; a file was not created in DSIWRIT

For non-zero return codes, error messages can be found in the CNMSJCRG job log.

Starting the NetView Program Before Starting JES

If you plan to start the NetView program and the SSI under the master subsystem before you start JES, the following rules apply:

- Start the PROC with the START command using the parameter SUB=MSTR.
- When you start the NetView program with the SUB=MSTR parameter, ensure that the TASK.DSIRQJOB.INIT statement in the CNMSTYLE member has a value of YES to start the DSIRQJOB task. This is needed for the SUBMIT or ALLOCATE (for INTRDR or SYSOUT files only) commands to complete successfully.

Note: When DSIRQJOB ends, it does not release the job ID. JES releases the job ID.

- Store the procedure in the data set SYS1.PROCLIB, not in a user PROCLIB supported by JES.
- The procedures must contain only a single job step.
- You cannot reference SYSIN, SYSOUT, or VIO data sets. If you are using the sample start procedures, comment out all references to the symbolic variable SOUTA=A in CNMPROC (CNMSJ009).
- JES is coded as the primary subsystem. In the IEFSSN member for JES, code the NOSTART parameter so that MVS does not automatically start JES at initialization.
- You cannot specify AMP=AMORG on any log data set.
- After the DSIRQJOB job receives a job ID from the JES program, if the JES program ends abnormally or ends without notifying the DSIRQJOB job to release the job ID, the DSIRQJOB job and the NetView program cannot be stopped before the JES program becomes active again. If the JES program ends abnormally or is stopped by a user from the command line, you can use the NetView MVS Command Revision function to stop the DSIRQJOB job and the NetView program.

These are the steps that you can use to set up the NetView MVS Command Revision function to stop the DSIRQJOB job when a command is entered to abnormally end the JES program (for example \$PJES2,ABEND or \$PJES2,TERM).

1. Install the NetView MVS Command Revision function. For more information, see *IBM Z NetView Installation: Configuring Additional Components*.
2. Create a REXX procedure for the NetView program and use it to issue two commands:
 - 'STOP TASK=DSIRQJOB'
 - 'REISSUE MVS' arg(1)
3. In a new or existing command revision table, include code similar to the following example:

```
UPON(CMDVERB=' $PJES2 ' )
  SELECT
    WHEN(W2 = 'ABEND')  NETVONLY=rexex_proc
    WHEN(W2 = 'TERM')  NETVONLY=rexex_proc
    OTHERWISE
  END
```

where *rexex_proc* is the name of the REXX procedure that you created in step “2” on page 66.

4. Activate the command revision table from step “3” on page 66 using the REVISE command.

If you want information about...

Refer to...

MVS Command Revision function

IBM Z NetView Automation Guide

Starting the NetView Program Without Restarting the MVS System

The normal process for installing and starting the NetView program for the first time requires restarting the MVS system, in order to load required LPA modules and activate required subsystems. If this is not practical, you can avoid restarting the MVS system by completing the same steps manually.

Before you begin

This procedure assumes that you have completed all installation procedures previously documented in this information, but without restarting the MVS system.

Important: This procedure assumes that you are installing the NetView program for the first time. If you are migrating from a previous version of the NetView program, see the *IBM Z NetView Installation: Migration Guide*.

Procedure

1. Dynamically add the following MVS system symbols:

```
&CNMTCPN=tcpip_name  
&CNMRODM=iocdm_name  
&CNMNETID=network_id
```

If you are running z/OS 2.2, use the **SETLOAD** command to dynamically update your system symbols:

```
SETLOAD xx,IEASYM,DSN=data_set_name
```

where *xx* specifies the LOADxx member in the specified data set (*data_set_name*). The LOADxx member specifies the IEASYMyy member where your system symbols are defined.

2. Dynamically add the required NetView modules to the link pack area (LPA):

- BNJLINTB
- BNJMTERM
- BNJSVC76
- (CNMCNETV,CNMNETV)
- CNMCSRVP
- CNMCSSEI
- CNMCSSPI
- DSICSENF
- DSIELSVC
- DSIRVCEX
- DSISPNOT
- DSI4LCUI
- DSI4LSIT

Use the **SETPROG LPA ADD** command to add each module:

```
SETPROG LPA,ADD,MODNAME=module,DSNAME=NETVIEW.V6R3M0.SCNMLPA1
```

where *module* is the module name or names and NETVIEW.V6R3M0.SCNMLPA1 is the NetView library where the module was installed. Note that CNMCNETV and its alias, CNMNETV, must be specified together, separated by commas, and within parentheses, in the **SETPROG** command.

3. Add the required NetView modules to the list of APF-authorized libraries.

Issue the following MVS command to dynamically add NetView libraries to the list of APF-authorized libraries without requiring a system restart:

```
SETPROG APF,FORMAT=DYNAMIC
```

Issue the following MVS commands to add APF authorizations for the required NetView libraries:

```
SETPROG APF,ADD,DSNAME=NETVIEW.V6R3M0.CNMLINK,VOLUME=vol-ser  
SETPROG APF,ADD,DSNAME=NETVIEW.V6R3M0.SAQNLINK,VOLUME=vol-ser  
SETPROG APF,ADD,DSNAME=NETVIEW.V6R3M0.SCNMLNKN,VOLUME=vol-ser  
SETPROG APF,ADD,DSNAME=NETVIEW.V6R3M0.SCNMLNK1,VOLUME=vol-ser  
SETPROG APF,ADD,DSNAME=NETVIEW.V6R3M0.SCNMLPA1,VOLUME=vol-ser
```

where *vol-ser* is the name of the volume where the NetView V6R3 libraries were allocated.

4. Verify that the subsystem (SSI) procedure and the main NetView procedure both include a STEPLIB DD statement that specifies the data set names for the NetView libraries.

If you plan to start RODM, the RODM Loader, GMFHS, or the Event/Automation Service (E/AS), verify that these procedures also include STEPLIB DD statements that specify the data set names for the NetView libraries.

5. Optional: If you plan to run multiple instances of the NetView program within the same LPAR, decide which is the first instance.

Typically, if only one instance manages system messages, that instance is first; however, you can choose the instance that you prefer. For more information about running multiple NetView programs in the same LPAR refer to *IBM Z NetView Installation: Configuring Additional Components*.

6. Dynamically add the subsystem for the first NetView instance by issuing the **SETSSI ADD** command:

```
SETSSI ADD,SUBNAME=ssiname,INITRTN=DSI4LSIT,INITPARM=parms
```

where *ssiname* is the subsystem name of the first NetView instance, and *parms* is the initialization parameters you want to specify (omit this keyword to use the default initialization parameters). The subsystem name is the four-character name specified either by the IEFSSNxx member or by the **SETSSI ADD** command.

Tip: By convention, the subsystem name for an instance of the NetView program consists of the first four characters of the job name associated with its main address space and the matching SSI procedure. The subsystem name you use for an instance of NetView must not be the same as any other subsystem name in use; multiple NetView address spaces cannot use the same subsystem name.

Including the INITPARM parameter with any value on the **SETSSI** command causes all settings that are specified on previous INITPARM parameters for NetView subsystems to be erased and the active Canzlog data space to be deleted and reallocated, resulting in loss of all messages from the active data space. This means that those messages that are issued during the current IPL will only be available from an archive.

For example, additionally specifying INITPARM= 'TINYDS,STATIC' on the SETSSI command causes the Canzlog data space to be deleted and reallocated with a size of 512 MB and not expandable, and foreign messages are not to be logged in the Canzlog log.

Omitting the INITPARM parameter causes the Canzlog data space with the existing options to be preserved.

Message CNM617 is displayed when the **SETSSI ADD** command completes.

Usage note: The **SETSSI ACTIVATE** command is not supported by the NetView subsystem.

7. Optional: Repeat the **SETSSI ADD** command for any additional instances of the NetView program in the LPAR.

Make sure the CNM617 message is displayed for each instance before you proceed to the next.

8. Start the subsystem (SSI) procedure for each NetView instance as documented in [“Starting the NetView Subsystem Address Space Using Job CNMPSSI”](#) on page 71.

Make sure the CNM541 message is displayed for each procedure before you proceed to the next.

Tip: This step is required before you start the NetView program for the first time. However, if you do not intend to use the NetView Program-to-Program Interface (PPI), the Message Revision Table (MRT), or the Command Revision Table (CRT), you can stop the subsystem after the CNM541 message is displayed.

9. Start the main procedure for each NetView instance as documented in [“Starting the NetView Program Using Job CNMPROC” on page 72](#).

Chapter 6. Activating the NetView Program

You are ready to start the NetView program.

If you have installed the NetView system in a subarea other than 01 and domain CNM01, make the appropriate substitution as you go through the steps in this section. Modify the PROCLIB members to match your system naming conventions before running the steps in this chapter.

Note: Be sure to read each step in its entirety before starting.

Starting VTAM Using Job CNMNET

To start the VTAM program, follow these steps:

1. Before starting the VTAM program, you might have to restart the MVS system or make dynamic updates to the MVS system. For example, you might have to activate APF authorizations, define the NetView subsystem, and load LPALIB modules. Make sure that these tasks are complete before starting the VTAM program.
2. Read the comments in the CNMNET sample VTAM startup procedure to determine if any additional changes are required to your procedure.
3. If your VTAM procedure was not automatically started during a restart of the MVS system, start the procedure by entering the following command at the system console:

```
S CNMNET , , , (LIST=xx)
```

Where xx specifies the VTAM start options list (ATCSTRxx) to use when you start the VTAM procedure. For example, specify LIST=01 when you start the VTAM procedure for NetView domain CNM01 in subarea 01 in network NETA.

Starting the NetView Subsystem Address Space Using Job CNMPSSI

The following components use the NetView subsystem address space:

- Message Revision Table (MRT)

You can use the MRT to examine messages flowing in the system and make changes to certain aspects of the messages. The MRT can remain active as long as the SSI address space is active, even when the NetView program is not active.

- Command Revision Table (CRT)

You can use the CRT to intercept MVS commands before they are processed. You can make changes to the command text, write a message to the command issuer, and then run the command or suppress the command. The CRT can remain active as long as the SSI address space is active, even when the NetView program is not active. However, loading or querying the CRT, or gathering statistics, depends on the NetView address space being active.

- Program-to-Program Interface (PPI)

The PPI enables application programs to communicate with the NetView program and other applications that are running in the same host.

You can start the NetView subsystem address space by using the SSI.ProcString statement in the CNMSTYLE member or by entering the following command at the system console:

```
S CNMPSSI
```

You see messages similar to the following messages:

```
$HASP373 CNMPSSI  STARTED
IEF403I CNMPSSI - STARTED - TIME=12.13.35
CNM226I NetView Program to Program Interface initialization is completed
CNM541I NetView subsystem CNMP is fully functional
```

If you want information about...

Refer to...

SSI.ProcString statement

IBM Z NetView Administration Reference

Starting the NetView Program Using Job CNMPROC

You can start the NetView program using the CNMPROC job. To do this, enter the following command at the system console:

```
S CNMPROC
```

You see messages similar to the following messages:

```
$HASP373 CNMPROC  STARTED
IEF403I CNMPROC - STARTED - TIME=12.14.27
BNJ080I BNJLINTB - BUFFER SIZE=24K,SLOT SIZE=200
DSI530I 'DSIMONIT' : 'DSIMONIT' IS READY AND WAITING FOR WORK
DSI530I 'DSIDCBMT' : 'DSIDCBMT' IS READY AND WAITING FOR WORK
DSI530I 'DSITIMMT' : 'DSITIMMT' IS READY AND WAITING FOR WORK
BNH350I NETVIEW HIGH-LEVEL LANGUAGE SUPPORT HAS BEEN SUCCESSFULLY
INITIALIZED WITH LE/370 LIBRARIES
DSI530I 'DSIHLLMT' : 'DSIHLLMT' IS READY AND WAITING FOR WORK
DSI530I 'DSISTMMT' : 'DSISTMMT' IS READY AND WAITING FOR WORK
*0003 DSI802A CNM01  REPLY WITH VALID NCCF SYSTEM OPERATOR COMMAND
DSI530I 'DSILOGMT' : 'DSILOGMT' IS READY AND WAITING FOR WORK
```

Usage notes:

1. Notice the DSI530I messages for the following tasks:

- DSIDCBMT
- DSIHLLMT
- DSILOGMT
- DSIMONIT
- DSISTMMT
- DSITIMMT

These tasks are part of the main task of the NetView program. They are automatically started when the NetView program is started and are not defined by TASK statements externally. They cannot be started or stopped by the STARTCNM and STOPCNM commands. They have no user exits, and cannot be stopped with the STOP TASK command either. Although these tasks can be forced to ABEND with the STOP FORCE command, after the ABEND, NetView automatically restarts the task. Therefore, do not try to stop these tasks during NetView initialization.

2. If you did not configure the allocation parameters for the Canzlog archive data sets for your environment, you might receive the following BNH803I message:

```
BNH803I VALUE *NONE* FOR KEYWORD ARCHIVE.HLQ CONFLICTS WITH VALUE YES
FOR KEYWORD ARCHIVE.WRITE
```

To configure the Canzlog archiving function for this NetView program, review the ARCHIVE statements in the CNMSTYLE member. For information about modifying the ARCHIVE statements, see [“Configuring Allocation Parameters for Canzlog Archive Data Sets” on page 54](#).

3. VSAM might return IEC161I messages. Do not be concerned with these messages if all the tasks start properly.
4. Do not respond to the DSI802A message at this time.

Chapter 7. Verifying the Installation

This section leads you through a series of commands to test the NetView program that you just started. Run the commands in exactly the order presented. If you make errors, or issue the commands out of sequence, the contents of the network logs and the results of the operator tests are different from the examples you see in this chapter.

To verify the NetView installation, issue instructions that test the installation. If you cannot start the tests, ensure that you correctly performed all the installation procedures in the NetView program directory and all the installation procedures in this book.

Usage notes:

1. Your data can differ from the panels in the examples because the display format depends on the type of terminal that you are using.
2. If you customized your system to suppress messages, remember to check the system and Canzlog logs for critical information that might be suppressed during the verification of your NetView installation.
3. For a list of tasks and automated tasks that are initialized by default, see [Table 14 on page 95](#) and [Table 15 on page 97](#).

Testing the Command Facility

To start the test of the NetView program, enter the following command:

```
LOGON APPLID(CNM01)
```

A panel similar to [Figure 9 on page 73](#) is displayed.

```

NN  NN      VV      VV
NNN NN      VV      VV  II  EEEEE WW      WW  TM
NNNN NN      EE      TT  VV      VV  II  EE      WW      W  WW
NN NN NN      EEEE      TT      VV      VV  II  EEEE      WW  WWW  WW
NN  NNNN      EE      TT      VV      VV  II  EE      WWW  WWW
NN  NNN      EEEEE      TT      VVV      II  EEEEE      WW  WW
NN  NN      V
5697-NV6 © Copyright IBM Corp.      1986, 2019 - All Rights Reserved
U.S. Government users restricted rights - Use, duplication, or disclosure
restricted by GSA ADP schedule contract with IBM corporation.
Licensed materials - Property of IBM Corporation
Domain = CNM01      NetView V6R3 - NM

OPERATOR ID ==>      or LOGOFF
PASSWORD ==>
PROFILE ==>      Profile name, blan=default
HARDCOPY LOG ==>      device name, or NO, default=NO
RUN INITIAL COMMAND ==>      YES or NO, default=YES
Takeover session ==>      YES, NO, or FORCE, default=NO

Enter logon information or PF3/PF15 to logoff
```

Figure 9. IBM Z NetView Logon Panel

If the value of the SECOPTS.OPERSEC statement in the CNMSTYLE member is SAFDEF, or if the OPERSEC operand was specified as SAFDEF on the REFRESH command, no PROFILE field is shown on the Logon panel and the HARDCOPY LOG field does not have a default value.

In the PROFILE field, system symbolic substitution is performed on records read from the DSIOPF member in the DSIPARM data set and the specified profile member in the DSIPRF data set. The &DOMAIN symbolic variable that is supplied with the NetView product is also included in the substitution

process. The substitution is performed after comment removal but before record processing. After substitution, comments are also removed. Substitution is always performed on the &DOMAIN symbolic variable, unless substitution was disabled when NetView was started.

Logging on

To log on, enter the following text in both the OPERATOR ID field and in the PASSWORD field:

```
NETOP1
```

A panel similar to [Figure 10 on page 74](#) is displayed.

```
NetView V6R3 - NM          NetView  CNM01 NETOP1  06/22/19 15:52:25
- CNM01      DSI020I OPERATOR NETOP1 LOGGED ON FROM TERMINAL NT68L703 USING
              PROFILE (DSIPROFB ), HCL ( )
- CNM01      DSI083I AUTOWRAP STOPPED
C CNM01      CNM357I PFKDEF : PF KEY SETTINGS NOW ESTABLISHED. 'DISPFK' TO SEE
              YOUR PF KEY SETTINGS
| CNM01

      Enter LOG or LOGOFF to terminate session.
      Enter HELP to obtain help.
      Lead operator has been notified of your logon.
      To obtain help from the network control center, enter

      MSG PPT, your question here

| CNM01
News for 30 Nov 2019

The following information lists the enhancements for IBM Z NetView
V6R3, formerly known as IBM Tivoli NetView for z/OS.
This information also lists the enhancements for IBM Z NetView
for Continuous Availability V6R3, formerly known as IBM Tivoli NetView
Monitoring for GDPS if it is running in the NetView program.

For additional information, see the following website:
https://ibm.biz/Bdz9uw
```

Figure 10. NetView Initial Message Display

The information that is displayed on this panel is contained in member CNMNEWS in the NETVIEW.V6R3M0.SDSIOPEN data set. The date and time at the top of your panel will differ from the examples.

Usage Note: When verifying your installation, be sure to clear the console between each step. In [Figure 10 on page 74](#), the 3 asterisks at the bottom of a panel indicate a held screen.

Clearing the screen

To clear the screen and go to the NetView main menu, press the Enter key and continue to press the Enter key until the main menu is displayed ([Figure 11 on page 75](#)).

```

CNM1NETV                      IBM Z NetView V6R3                      Main Menu

                          Operator ID = NETOP1      Application = CNM0100E

Enter a command (shown highlighted or in white) and press Enter.

      IP Management Menu      NETVIP command
      Browse Facility         BROWSE command
      Command Facility        NCCF command
      News                    NEWS command
      PF Key Settings         DISPFK command
      Help Facility           HELP command
      Index of help topics    INDEX command
      Hardware Monitor        NPDA command
      Session Monitor         NLDM command
      Automated Operations Network AON command

      To log off or disconnect LOGOFF command or DISC command

TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
Action====>

```

Figure 11. NetView Main Menu Panel

Displaying help information

To go to the NetView help facility, enter the following command:

```
HELP NETVIEW
```

A panel similar to [Figure 12 on page 75](#) is displayed.

```

CNMKNEEW                      NETVIEW HELP FACILITY MAIN MENU

Select  To get information about

      1  Operator's overview of the NetView Program
      2  Using the NetView Help Desk for operators
      3  Using NetView online message help
      4  Using command and command list help
      5  Finding help on VTAM in NetView
      6  Finding help on RODM (Resource Object Data Manager)
      7  Finding help on GMFHS (Graphic Monitor Facility host subsystem)
      8  Help for the NETVIEW stage (NetView Pipelines)
      A  All NetView commands
      I  Finding help in the Index
      P  Help for PIPE syntax

Type a value (1 to 8, A, I, or P) and press ENTER.

TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
Action====>

```

Figure 12. NetView Help Facility Main Menu

Press the F2 or F3 key twice to return to the command facility.

For help information on the command facility, enter the following command:

```
HELP
```

A panel similar to [Figure 13 on page 76](#) is displayed.

```

CNMKNCCF                      COMMAND FACILITY HELP MENU

Select  To get information about

    1    Operator's overview of the command facility
    2    Using the terminal access facility (TAF)

    3    The command facility screen
    4    Command facility commands and command lists

    5    Field level help

Type a number (1 through 5) and press ENTER.

                      HELP NETVIEW ---> NetView Help Menu

TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
Action====>

```

Figure 13. Command Facility Help Menu Panel

Select option 4 to display the command facility commands. A panel similar to [Figure 14 on page 76](#) is displayed.

```

CNMPNL1.EUYCLIST              HELP NCCF COMMANDS                      LINE 0 OF 284
*----- Top of Data -----*
ACQ      Acquire a resource . . . . . A2
ACT      Activate VTAM resources . . . . . A3
ADDCMD   Add or replace user-written commands . . . . . A6
AFTER    Schedule a command to run later . . . . . A7
AGTBRW   Browse the network log from the NetView EMA . . . A11
AGTSESMG Display SNA sessions from the NetView EMA . . . A12
AGTTCP   Display TCP/IP data from the NetView EMA . . . A13
AINQ     Display translation tables . . . . . A14
ALLOCATE Allocate an MVS data set . . . . . A19
APPLS    Display minor nodes . . . . . A41
APPLSPEN Display sessions state . . . . . A42
APSERV   Accept commands/messages from secure programs A44
ASSIGN   Define operators and groups . . . . . A45
AT       Schedule a command to run . . . . . A48
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
CMD==>

```

Figure 14. Using NetView Command and Command List Help Panel

Press the F3 key twice to return to the command facility.

Enter the **CLEAR** command to refresh the command facility screen.

Verifying that your automation table is active

To display the name of the active NetView automation table, enter:

```
AUTOTBL STATUS
```

If the default NetView automation table (DSITBL01) was loaded during NetView initialization, a panel similar to [Figure 15 on page 76](#) is displayed.

```

NetView V6R3 - NM           NetView   CNM01 NETOP1   06/22/19 15:54:10
* CNM01   AUTOTBL STATUS
' CNM01
BNH361I THE AUTOMATION TABLE CONSISTS OF THE FOLLOWING LIST OF MEMBERS:
CNM01PPT COMPLETED INSERT  FOR TABLE #1: DSITBL01 AT 06/22/19 15:51:29 (FIRST)
CNM01PPT COMPLETED INSERT  FOR TABLE #2: DSIAMIAT AT 06/22/19 15:51:29

-----
???
```

Figure 15. Displaying the active automation table

Enter the **CLEAR** command to refresh the command facility screen.

Displaying session information

To display the status of all operator terminals, NetView-NetView task (NNT) sessions requested by other NetView systems, and information about your session, enter the following command:

```
WHO
```

A panel similar to [Figure 16 on page 77](#) is displayed.

```
NetView V6R3 - NM          NetView  CNM01 NETOP1  06/22/19 15:54:39
* CNM01      WHO
C CNM01      LIST STATUS=OPS
- CNM01      OPERATOR: NETOP1    TERM: NT68L703 STATUS: ACTIVE
- CNM01      OPERATOR: AUTOAON   TERM: AUTOAON  STATUS: ACTIVE
- CNM01      OPERATOR: AUTOCT7   TERM: AUTOCT7  STATUS: ACTIVE
- CNM01      OPERATOR: AUTOEDAT  TERM: AUTOEDAT STATUS: ACTIVE
- CNM01      OPERATOR: AUTOPSAV  TERM: AUTOPSAV STATUS: ACTIVE
- CNM01      OPERATOR: AUTOXCF   TERM: AUTOXCF  STATUS: ACTIVE
- CNM01      OPERATOR: AUTOXDSC  TERM: AUTOXDSC STATUS: ACTIVE
- CNM01      OPERATOR: AUTO1     TERM: AUTO1    STATUS: ACTIVE
- CNM01      OPERATOR: AUTO2     TERM: AUTO2    STATUS: ACTIVE
- CNM01      OPERATOR: DBAUTO1   TERM: DBAUTO1  STATUS: ACTIVE
- CNM01      OPERATOR: DBAUTO2   TERM: DBAUTO2  STATUS: ACTIVE
- CNM01      OPERATOR: DSIIPCHK  TERM: DSIIPCHK STATUS: ACTIVE
- CNM01      OPERATOR: MVSCMDS   TERM: MVSCMDS  STATUS: ACTIVE
- CNM01      OPERATOR: DSILCOPR  TERM: DSILCOPR STATUS: ACTIVE
- CNM01      OPERATOR: AUTOTCPS  TERM: AUTOTCPS STATUS: ACTIVE
- CNM01      END OF STATUS DISPLAY
C CNM01      LIST STATUS=NNT
- CNM01      MAX SESS: 00032767
- CNM01      NO ACTIVE NCCF TO NCCF SESSIONS FOUND
??? ***
```

Figure 16. Information about Your NetView Session Panel

Press the ENTER key to release any messages being held by the command facility. The three asterisks (***) at the bottom of the screen indicate that there are more messages to display.

Enter the CLEAR command to refresh the command facility screen.

Displaying PF key settings

To display the PF key settings, enter the following command:

```
DISPFK
```

The PF key settings are displayed for the component from which you entered the command. A panel similar to [Figure 17 on page 78](#) is displayed.

```

CNMKWIND OUTPUT FROM  DISPFK                                LINE 0 OF 32
*----- Top of Data -----*
DISPLAY OF PF/PA KEY SETTINGS FOR NCCF
KEY  ----TYPE-----COMMAND-----SET-APPL
PA1  IMMED,IGNORE    RESET              NETVIEW
PA2  IMMED,IGNORE    AUTOWRAP TOGGLE     NETVIEW
PA3  IMMED,IGNORE    RETRIEVE AND EXECUTE NETVIEW
PF1  IMMED,APPEND    HELP                NETVIEW
PF2  IMMED,APPEND    GO                  NCCF
PF3  IMMED,IGNORE    RETURN              NETVIEW
PF4  IMMED,APPEND    DISPFK              NETVIEW
PF5  IMMED,APPEND    BROWSE LOG           NETVIEW
PF6  IMMED,IGNORE    ROLL                NETVIEW
PF7  IMMED,APPEND    BACK                NETVIEW
PF8  IMMED,APPEND    FORWARD             NETVIEW
PF9  DELAY,IGNORE    PIPE HELDMSG | CONSOLE DELETE NCCF
PF10 IMMED,APPEND    WINDOW              NETVIEW
PF11 IMMED,IGNORE    HOLD                NCCF
PF12 IMMED,IGNORE    RETRIEVE            NETVIEW
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
CMD==>

```

Figure 17. PF Key Settings for the Command Facility

Use the F8 key to page forward through the contents.

Press the F3 key to return to the command facility.

Viewing operator identifiers and passwords

To view NetView operator identifiers and passwords, enter the following command:

```
BROWSE DSIOPF
```

A panel that is similar to [Figure 18 on page 78](#) is displayed.

```

NETVIEW.BRWS ----- BROWSE DSIOPF (DSIPARM) --- LINE 00000 TO 00017 OF 0117
                                           SCROLL ==> CSR
-----1-----2-----3-----4-----5-----6-----7-----
***** TOP OF DATA ***** DATASET: 4
*****
* LICENSED MATERIALS - PROPERTY OF IBM *
* 5697-NV6 © Copyright IBM Corp. 1986, 2019 *
* ALL RIGHTS RESERVED. *
* US GOVERNMENT USERS RESTRICTED RIGHTS *
* - USE, DUPLICATION OR DISCLOSURE RESTRICTED BY *
* GSA ADP SCHEDULE CONTRACT WITH IBM CORPORATION. *
*****
* NAME(DSIOPF) SAMPLE(DSIOPF) RELATED-TO( ) *
* DESCRIPTION: SAMPLE DSIPARM - OPERATOR DEFINITIONS/PASSWORDS *
* *
*-----*
* Include customer operator definitions. *
*-----*
***** START OF MEMBER DSIOPFU FROM DSIOPF DATASET: 2
*****
* 5697-NV6 © Copyright IBM Corp. 1986, 2019 *
*
CMD==>
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'

```

Figure 18. Display of Operator Identifiers and Passwords

Use the F8 key to page forward through the contents.

Press the F3 key to return to the command facility panel.

Using the terminal access facility (TAF)

You also need to verify your installation of the NetView system by entering the NetView program through the terminal access facility (TAF). Use the BFSESS command list to start a full-screen session with another subsystem. If you are using subarea 01 for your installation, enter:

```
BFSESS CNM01
```

Press ENTER again to display the NetView logon panel, if needed.

To log on to the NetView system, enter:

```
OPER4
```

in the OPERATOR ID field, and

```
OPER4
```

in the PASSWORD field. A panel similar to [Figure 19 on page 79](#) is displayed.

```
NetView V6R3 - NM          NetView  CNM01 OPER4    06/22/19 15:56:32
- CNM01      DSI020I OPERATOR OPER4 LOGGED ON FROM TERMINAL TF01#000 USING
              PROFILE (DSIPROFA ), HCL ( )
- CNM01      DSI083I AUTOWRAP STOPPED
C CNM01      CNM357I PFKDEF : PF KEY SETTINGS NOW ESTABLISHED. 'DISPFK' TO SEE
              YOUR PF KEY SETTINGS
| CNM01

      Enter LOG or LOGOFF to terminate session.
      Enter HELP to obtain help.
      Lead operator has been notified of your logon.
      To obtain help from the network control center, enter

          MSG PPT, your question here

| CNM01
News for 30 Nov 2019

The following information lists the enhancements for IBM Z NetView
V6R3, formerly known as IBM Tivoli NetView for z/OS.
This information also lists the enhancements for IBM Z NetView
for Continuous Availability V6R3, formerly know as IBM Tivoli NetView
Monitoring for GDPS if it is running in the NetView program.

For additional information, see the following website:
https://ibm.biz/Bdz9uw
```

Figure 19. Entering the Command Facility after Logging On

To go to the NetView main menu, press the Enter key and continue to press the Enter key until the NetView main menu is displayed.

Press the F3 key to go to the command facility.

Enter the CLEAR command to refresh the screen.

To view the operator identifiers and passwords, enter the following command:

```
BR DSIOPF
```

BR is a synonym for BROWSE. A panel similar to [Figure 20 on page 80](#) is displayed.

```

NetView V6R3 - NM          NetView  CNM01 OPER4    06/22/19 15:57:07
- CNM01    BNH236E 'OPER4' IS NOT AUTHORIZED TO USE THE KEYWORD 'DSIOPF' AND
            VALUE 'DSIOPF' COMBINATION
C CNM01    DSI238I BR COMMAND FAILED, INVALID PARAMETER: DSIOPF
- CNM01    BNH237E THE KEYWORD 'DSIOPF' AND VALUE 'DSIOPF' ARE PROTECTED BY
            COMMAND IDENTIFIER '*.*.READSEC.DSIOPF.*' IN 'TBLNAME=CNMSCAT2'
C CNM01    CNM337I VIEW : CODE 12 - YOU ARE NOT AUTHORIZED TO BROWSE 'DSIOPF'.
            THE DSIKVS MACRO RESULTED IN A NONZERO RETURN CODE.
-----

```

???

Figure 20. Command Facility Message Panel

Usage notes:

1. OPER4 does not have the authority to display the DSIOPF member.
2. Command authorization is being performed by using the NetView command authorization table (CNMSCAT2). If a SAF product (such as RACF) is being used for security, the information in [Figure 20](#) on page 80 will be different.

To end your OPER4 operator session, enter:

```
LOGOFF
```

at the cursor.

You return to a panel similar to [Figure 21](#) on page 80 (the command facility for the NETOP1 operator).

```

NetView V6R3 - NM          NetView  CNM01 NETOP1    06/22/19 15:57:42
* CNM01    BFSESS CNM01
- CNM01    DSI498I SRCLU HAS BEEN DEFAULTED TO SRCLU = TF01#000 FOR FLSCN
            SESSION TO APPLID = CNM01
- CNM01    DSI496I FLSCN SESSION BETWEEN APPLID = CNM01    AND SRCLU =
            TF01#000 ENDED
-----
???
```

Figure 21. Command Facility Panel for Operator NETOP1

Enter the CLEAR command to refresh the command facility screen.

Displaying common global variables

To list the common global variable names that begin with CNMSTYLE.NV, enter the following command:

```
QRYGLOBL COMMON VARS=CNMSTYLE.NV*
```

A panel similar to [Figure 22](#) on page 80 is displayed.

```

NetView V6R3 - NM          NetView  CNM01 NETOP1    06/22/19 15:59:00
* CNM01    QRYGLOBL COMMON VARS=CNMSTYLE.NV*
' CNM01
BNH031I NETVIEW GLOBAL VARIABLE INFORMATION
BNH103I COMMAND ISSUED AT: 06/22/19 15:59:00
BNH061I
BNH032I COMMON GLOBAL VARIABLES
BNH036I GLOBAL VARIABLE NAME:          GLOBAL VARIABLE VALUE:
BNH061I -----
BNH039I CNMSTYLE.NVROLE                NETWORK
BNH039I CNMSTYLE.NV2I                  NM
BNH035I NUMBER OF VARIABLES FOUND: 2
BNH061I
BNH037I NETVIEW GLOBAL VARIABLE INFORMATION COMPLETE

```

Figure 22. Command Facility Panel to Display Common Global Variables

Browsing Canzlog

The consolidated audit, NetView program, and z/OS log (Canzlog) provides a comprehensive means of displaying the following kinds of information:

- MVS messages
- NetView messages
- Broadcast messages
- DOMS
- Command echoes
- Trace and audit messages

To browse the Canzlog log, enter the following command:

```
BROWSE CANZLOG
```

A panel similar to [Figure 23 on page 81](#) is displayed.

```
Canzlog TAG=(NVMSG,MVSMMSG,DOM) 06/22/19 15:57:07 -- 15:59:20

15:57:07 DSI238I BR COMMAND FAILED, INVALID PARAMETER: DSIOPF
15:57:07 BNH237E THE KEYWORD 'DSIPARM' AND VALUE 'DSIOPF' ARE PROTECTED BY COMM

15:57:07 CNM337I VIEW : CODE 12 - YOU ARE NOT AUTHORIZED TO BROWSE 'DSIOPF'. TH

15:57:28 LOGOFF
15:57:28 CNM493I CNMSDCA : #0000023 : CNME8251
15:57:28 DSI081I OPERATOR OPER4, LOGOFF PROCEEDING: TERMINAL = TF01#000
15:57:30 DSI496I FLSCN SESSION BETWEEN APPLID = CNM01 AND SRCLU = TF01#000 E

15:58:38 CLEAR
15:59:00 QRYGLOBL COMMON VARS=CNMSTYLE.NV*
15:59:00 BNH031I NETVIEW GLOBAL VARIABLE INFORMATION
          BNH103I COMMAND ISSUED AT: 06/22/19 15:59:00
          BNH061I
          BNH032I COMMON GLOBAL VARIABLES
          BNH036I GLOBAL VARIABLE NAME: GLOBAL VARIABLE VALUE:
          BNH061I -----
          BNH039I CNMSTYLE.NVROLE NETWORK
          BNH039I CNMSTYLE.NV2I NM
          BNH035I NUMBER OF VARIABLES FOUND: 2
          BNH061I
          BNH037I NETVIEW GLOBAL VARIABLE INFORMATION COMPLETE
15:59:20 BROWSE CANZLOG
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
CMD==>
```

Figure 23. View Canzlog data

Usage notes:

1. You can use the BR LOG command to display messages that match a filter set by the DEFAULTS or OVERRIDE commands for CANZLOG.
2. You can use the BR *filtername* command to display messages that match the specified filter name. You can display a list of defined filters by using the LIST CZFILTER command.

Press the F3 key to return to the command facility. Enter the CLEAR command to refresh the command facility screen.

Testing Basic IP Management Functions

You can issue a PING command to an IP host from the command facility. For example, to ping IP address 1.2.3.4 with a host name of my.host.name.com enter the following command from the command facility:

```
PING 1.2.3.4
```

A panel similar to [Figure 24 on page 82](#) is displayed.

Usage note: For the **PING** command to resolve IP host names, the resolver configuration must be correct in TCP/IP or the SYSTCPD DD statement in CNMPROC (CNMSJ009) must identify the location of your TCP/IP configuration data.

```
NetView V6R3 - NM          NetView  CNM01 NETOP1  06/22/19 16:00:04
* CNM01  PING 1.2.3.4
C CNM01  BNH765I Pinging my.host.name.com at 1.2.3.4 with 3 packets of
length 16 bytes
C CNM01  BNH767I 16 bytes received from 1.2.3.4: seq=1 in 2ms
C CNM01  BNH767I 16 bytes received from 1.2.3.4: seq=2 in 2ms
C CNM01  BNH767I 16 bytes received from 1.2.3.4: seq=3 in 1ms
C CNM01  BNH769I 3 packets sent, 3 packets received, 0.00% packet loss
C CNM01  BNH770I Round trip times from 1 to 2 ms, averaging 1ms
-----
???      Pings of 1.2.3.4 averaging 1ms
```

Figure 24. PING command output

To display a menu of available IP Management functions, enter the following command:

```
NETVIP
```

A panel similar to [Figure 25 on page 82](#) is displayed.

```
CNM4NVIP          NetView IP Management Functions Menu

Type the number or move the cursor to a function and press Enter

  1. Ping a device (PING)
  2. Trace the route to a device (TRACERTE)
  3. Check TCP connection status (IPSTAT)
  4. Work with IP traces (IPTRACE)
     for SP: -----
  5. Manage IP Active Monitoring (IPMAN)
  6. Issue SNMP commands (NVSNMPP)
  7. Manage Sysplex
  8. Manage DVIPA
  9. Check the status of an IP port (TESTPORT)
 10. Show EE information for a VTAM resource (DIS PATH)

Command ===>
F1=Help          F3=Return          F6=Roll
                  F12=Cancel
```

Figure 25. IP Management functions menu

Usage note: To select a function, you can enter the number associated with the function and press Enter, or move the cursor to the function description and press Enter.

To display a list of the NetView sysplex discovery commands, move the cursor to option 7 and press Enter. A panel similar to [Figure 26 on page 83](#) is displayed.

```

CNM4NVSP          NetView Sysplex Management Functions Menu

Type the number or move the cursor to a function and press Enter

  1. Stack configuration and status (CNMSSTAC)
  2. IP stack interfaces (CNMSIFST)
  3. NetView configuration and status (CNMSNVST)
  4. OSA channel and ports (CNMSOSAP)
  5. HiperSockets adapters (CNMSHIPR)
  6. Telnet servers (CNMSTNST)
  7. Telnet server ports (CNMSTPST)

Command ==>
F1=Help          F3=Return          F6=Roll
                  F12=Cancel

```

Figure 26. NetView Sysplex Management Functions Menu Panel

You can view configuration and status information about the NetView domains known to this NetView program. To do this, move the cursor to option 3 and press enter. A panel similar to [Figure 27 on page 83](#) is displayed.

```

CNMKWIND OUTPUT FROM  CNMSNVST                                LINE 0 OF 3
*----- Top of Data -----*
BNH495I NUMBER OF NETVIEW APPLICATIONS: 1
# Domain   Status   Total-CPU % Total-Stor Role           Sysplex-Role Syspl
1  CNM01    ACTIVE    0.02         11157    NETWORK        MCAP          1
*----- Bottom of Data -----*

TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
CMD==>

```

Figure 27. Configuration and status information

Usage note: Not all IP management functions are enabled by default. To fully enable IP management, see *IBM Z NetView Installation: Configuring Additional Components*.

Press the F3 key until you return to the command facility panel.

Enter the **CLEAR** command to refresh the command facility screen.

Testing the Hardware Monitor

To generate an alert and record the alert to the hardware monitor database, enter the following command:

```
GENALERT N TYPE=PERM MAJ=1 MIN=1 ACT=1 HIER=ATCHM003,COMC PRID=3725
```

A panel similar to [Figure 28 on page 83](#) is displayed.

```

NetView V6R3 - NM          NetView  CNM01 NETOP1   06/22/19 16:01:54
* CNM01    GENALERT N TYPE=PERM MAJ=1 MIN=1 ACT=1 HIER=ATCHM003,COMC PRID=3725
-----
???
```

Figure 28. GENALERT command output

To go to the hardware monitor component, enter the following command:

```
NPDA
```

A panel similar to [Figure 29 on page 84](#) is displayed.

```
N E T V I E W          SESSION DOMAIN: CNM01    NETOP1    06/22/19 16:02:08
NPDA-01A                * MENU *                HOST DOMAIN: CNM01

SEL#  PRODUCES:
( 1)  ALERTS-DYNAMIC DISPLAY
( 2)  TOTAL EVENTS DISPLAY
( 3)  TOTAL STATISTICAL DATA DISPLAY
( 4)  HELP MENU DISPLAY

      REQUEST DATA FROM NETWORK RESOURCES:
( 5)  SNA CONTROLLERS (CTRL)

      DATA TYPES INITIALIZED/PURGED
AL.. (06/22/19)  EV.. (06/22/19)  ST.. (06/22/19)  GMFALERT.. (06/22/19)

ENTER SEL#

???
CMD==>
```

Figure 29. Hardware Monitor Main Menu

Note: If the sample job CNMSJ005 is used to initialize a newly allocated hardware monitor (NPDA) database cluster, the database initialization dates shown on the hardware monitor main menu **NPDA-01A** will be 05/30/96.

Examine the dates in the AL, EV, and ST fields. The dates cannot be asterisks. Your dates might not match the dates in [Figure 29 on page 84](#).

Use the ALERTSH (ALH) command to display the alerts recorded in the hardware monitor database. To see a panel similar to [Figure 30 on page 84](#), enter:

```
ALH

N E T V I E W          SESSION DOMAIN: CNM01    NETOP1    06/22/19 16:02:16
NPDA-31A                * ALERTS-HISTORY *                PAGE    1 OF    1

SEL# DOMAIN RESNAME TYPE TIME  ALERT DESCRIPTION:PROBABLE CAUSE
( 1) CNM01 ATCHM003 COMC 16:01  INITIALIZATION FAILURE:RING INTERFACE COUPLER

ENTER SEL# (ACTION),OR SEL# PLUS M (MOST RECENT), P (PROBLEM), DEL (DELETE)

???
CMD==>
```

Figure 30. Alerts History Panel

If no alerts were recorded, you receive message BNJ925I:

```
BNJ925I NO DATA EXISTS FOR COMMAND SPECIFIED
```

Press the F3 key twice to return to the command facility panel.

Testing the Session Monitor

To go to the session monitor component, enter the following command:

```
NLDM
```

A panel similar to [Figure 31 on page 85](#) is displayed.

```
NLDM.MENU                                PAGE    1

                                NETVIEW

                                DOMAIN  CNM01

SEL#                                DESCRIPTION

( 1)  LUNAME LIST    LIST OF ALL ACTIVE LOGICAL UNIT NAMES
( 2)  SLUNAME LIST   LIST OF ACTIVE SECONDARY LOGICAL UNIT NAMES
( 3)  PLUNAME LIST   LIST OF ACTIVE PRIMARY LOGICAL UNIT NAMES
( 4)  PUNAME LIST    LIST OF ACTIVE PHYSICAL UNIT NAMES
( 5)  CPNAME LIST    LIST OF ACTIVE CP AND SSCP NAMES
( 6)  DOMAIN LIST    LIST OF NLDM DOMAINS
( 7)  ER LIST        LIST OF ACTIVE EXPLICIT ROUTES
( 8)  VR LIST        LIST OF ACTIVE VIRTUAL ROUTES

ENTER: H OR HELP FOR INFORMATION ON THE USE OF NLDM
      HELP NLDM COMMANDS FOR NLDM COMMAND LIST

      NLDM FILE LAST INITIALIZED 08/22/19

ENTER SEL# OR COMMAND
CMD==>
```

Figure 31. Session Monitor Main Menu

Note: If the sample job CNMSJ005 is used to initialize a newly allocated session monitor (NLDM) database cluster, the database initialization date shown on the session monitor main menu **NLDM.MENU** will be 07/17/87.

To display a list of all inactive NetView sessions enter:

```
SESS * * INACT
```

A panel similar to [Figure 32 on page 85](#) is displayed.

```
NLDM.SESS                                PAGE    1

                                SESSION LIST

NAME: *                                DOMAIN: CNM01

-----
***** PRIMARY ***** ***** SECONDARY *****
SEL#  NAME  TYPE  DOM  NAME  TYPE  DOM  START TIME  END TIME
( 1)  CNM0100F LU  CNM01  TF01#000 LU  CNM01  08/22 15:56:13 08/22 15:57:29
( 2)  NT68MVS SSCP CNM01  CNM0100F LU  CNM01  08/22 15:56:13 08/22 15:57:29
( 3)  CNM01   LU  CNM01  TF01#000 LU  CNM01  08/22 15:56:13 08/22 15:56:13
( 4)  NT68MVS SSCP CNM01  TF01#000 LU  CNM01  08/22 15:56:13 08/22 15:57:29
( 5)  CNM01   LU  CNM01  NT68L703 LU  CNM01  08/22 15:52:07 08/22 15:52:07

END OF DATA
ENTER SEL# (CONFIG), SEL# AND CT (CONN. TEST), SEL# AND STR (TERM REASON)
CMD==>
```

Figure 32. Session List Panel

Press the F3 key twice to return to the command facility panel.

Logging Off

To log off from the NetView program, enter:

```
logoff
```

Extending the Base Functions

You completed the basic installation and configuration of the NetView program.

You might want to consider additional configuration to extend the base NetView functions.

<i>Table 12. Additional Configuration and Customization Options</i>	
Task	Refer to...
Configuring the NetView program for your environment	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Configuring additional NetView functions For each function that you implement, test to make sure that it works correctly.	“NetView Functions” on page 89
Installation, configuration, and setup for NetView REST Server	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Installing the NetView program on the same LPAR as an existing NetView program	<i>Running Multiple NetView Programs in the Same LPAR in IBM Z NetView Installation: Configuring Additional Components</i>
Using the NetView program for IP management	<i>IBM Z NetView Installation: Configuring Additional Components</i> <i>IBM Z NetView IP Management</i>
Using the NetView program for sysplex management	<i>IBM Z NetView IP Management</i>
Implementing graphics support	<i>IBM Z NetView Installation: Configuring Graphical Components</i>
Extending automation Make sure that your automation tables are updated and correctly specified using AUTOCMD statements in the CNMSTYLE member. The DSITBL01 automation table is required for the NetView program and provides basic automation statements to respond to messages and management services units (MSUs). Use %INCLUDE members to define changes for your environment and add these changes before any automation statements that are provided with the NetView program. Code any local customization with CONTINUE(YES).	<i>IBM Z NetView Automation Guide</i>
Using installation exits	<i>IBM Z NetView Programming: Assembler or IBM Z NetView Programming: PL/I and C</i>
Writing command processors	<i>IBM Z NetView Programming: Assembler or IBM Z NetView Programming: PL/I and C</i>
Installing the IBM Z NetView for Continuous Availability product	<i>IBM Z NetView for Continuous Availability Configuring and Using the GDPS Continuous Availability Solution</i>

Preparing for Production

To run the NetView program in production, consider the following steps:

- Make sure that the V6.3.0 modules are active in the production system.

- If you are using the NetView status monitor or AMI function, make sure that the V6.3.0 SCNMLNK1 members are in use by VTAM.
- Allocate the VSAM data sets for the production LPAR.
- Complete any tuning and customization tasks your system requires. See [“Extending the Base Functions” on page 85](#) for more information.
- If you are using the NetView program for system automation, review your system automation planning and verify that any new operating procedures are ready for implementation.

For each administration task that you prepared, test to ensure that it was done correctly. When you are satisfied, the NetView program is ready for full production.

Note: If you resume production under a previous release of the NetView program, cancel the NetView subsystem job and close the V6.3.0 application.

Appendix A. NetView Functions and Tasks

This section lists NetView functions and tasks.

NetView Functions

The following table contains some of the NetView functions with information on how they benefit you and where more information can be found.

<i>Table 13. Benefits of Individual NetView Functions</i>		
NetView Function	Benefit to You	Where Documented in NetView Library
Application programming	Tailor or supplement NetView to satisfy unique requirements or operating procedures. Command procedures can be written in PL/I, C, assembler, REXX, or NetView CLIST languages. Installation exits can be written in PL/I, C, or assembler.	• <i>IBM Z NetView Application Programmer's Guide</i> • <i>IBM Z NetView Programming: Assembler</i> • <i>IBM Z NetView Programming: PL/I and C</i> • <i>IBM Z NetView Programming: REXX and the NetView Command List Language</i> • <i>IBM Z NetView Programming: Pipes</i> • <i>IBM Z NetView User's Guide: Automated Operations Network</i>
Automated operation for networks (AON)	Provides automation for TCP/IP, VTAM, and SNA using NetView AON automation policy	<i>IBM Z NetView User's Guide: Automated Operations Network</i>
Browse facility ¹	View logs, data sets, and files on either a local or remote NetView	<i>IBM Z NetView User's Guide: NetView</i>
Canzlog log	Provides consolidated access to all IBM Z NetView system messages (including the netlog and syslog). The Canzlog log includes expanded access to message attributes and robust filtering to retrieve specific log records. By using the Canzlog function, you can significantly reduce problem determination time. You can archive the Canzlog log for later recall, as needed for audit or further problem analysis.	• <i>IBM Z NetView Installation: Configuring Additional Components</i> • <i>IBM Z NetView User's Guide: NetView</i>

Table 13. Benefits of Individual NetView Functions (continued)

NetView Function	Benefit to You	Where Documented in NetView Library
Canzlog Print	By issuing the PRINT command, you can print the specified Canzlog messages to a sequential data set or a member of a partitioned or partitioned extended data set. It can be issued from one of the following ways: • NETVIEW operator's console • BROWSE window • Canzlog Panel The NVINFO command is enhanced to support Canzlog log printing as well.	• <i>IBM Z NetView Administration Reference</i> • <i>IBM Z NetView Installation: Configuring Additional Components</i>
CNMSTYLE migration tool	Converts previous releases of CNME1034 command list and DSIPARM initialization members to the CNMSTYLE member format. This tool also converts your DSICMD definitions to the CNMCMD member format.	<i>IBM Z NetView Installation: Migration Guide</i>
CNMSTYLE report generator	Creates a report that provides information on the CNMSTYLE member. The report includes the following information: • Listing of the %INCLUDE structure of the CNMSTYLE member • Grouping of statements by NetView function • Grouping of multiple occurrences of statements for quick analysis • Listing of enabled towers	“Using the Report Generator” on page 60
Command facility ¹	Provides basic command support for the NetView product; lets you issue VTAM, MVS, and NetView commands from the NetView command line.	<i>IBM Z NetView User's Guide: NetView</i>
Command Statistics	Provides usage statistics for each invocation of a command that runs in the NetView address space.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Discovery manager	Collects sysplex, z/OS image, TCP/IP stack, IP interface, OSA, HiperSockets, NetView program, and Telnet server data for display in the NetView management console and the NetView agent. Provides real-time 3270 TCP/IP stack, IP interface, Telnet server, OSA, HiperSockets, and NetView application commands.	<i>IBM Z NetView Installation: Configuring Additional Components</i>

<i>Table 13. Benefits of Individual NetView Functions (continued)</i>		
NetView Function	Benefit to You	Where Documented in NetView Library
DVIPA support	Collects dynamic virtual IP address (DVIPA) data for display in the NetView agent. Provides real-time 3270 DVIPA commands.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Event/Automation Service (E/AS)	Serves as a gateway for event data between the IBM Z NetView management environment, managers and agents that handle Event Integration Facility (EIF) events, and SNMP trap managers and agents. With this gateway function, you can manage all network events from the management platform of your choice.	• <i>IBM Z NetView Installation: Configuring Additional Components</i> • <i>IBM Z NetView User's Guide: NetView</i>
Event correlation engine	Maps NetView messages and alerts to events that consist of name and value pairs. These events are made available to correlation automation.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Extended multiple console support (EMCS) consoles	Provides a consistent MVS operation interface, an alternative to the subsystem interface (SSI)	<i>IBM Z NetView Automation Guide</i>
Graphic monitor facility host subsystem (GMFHS)	Interfaces with RODM for the various components of NetView (SNA topology manager, NetView management console, MultiSystem Manager, and NetView resource manager) to manage the resources in your environment	<i>IBM Z NetView Installation: Configuring Graphical Components</i>
Hardware monitor	Collects and stores data about failed resources in networks	• <i>IBM Z NetView Installation: Configuring Additional Components</i> • <i>IBM Z NetView User's Guide: NetView</i>
Help facility ¹	Provides online help for messages and commands	<i>IBM Z NetView User's Guide: NetView</i>
Help desk facility ¹	Provides online help for diagnosing problems	<i>IBM Z NetView User's Guide: NetView</i>
Intrusion detection services	Defines automated responses to the following items: • Scans • Attacks • Traffic regulation for TCP connections and UDP receive queues Using notification and inform policies, you can send an e-mail to a security administrator, issue a message, generate an alert or Event Integration Facility (EIF) event, issue commands, or generate a report in response to an intrusion.	<i>IBM Z NetView Installation: Configuring Additional Components</i>

<i>Table 13. Benefits of Individual NetView Functions (continued)</i>		
NetView Function	Benefit to You	Where Documented in NetView Library
IP management	Supports management of heterogeneous networks that use both SNA and TCP/IP. NetView IP management capabilities include discovery, connection monitoring, packet collection and analysis, and DVIPA management.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Message revision table	The message revision table (MRT) enables you to intercept MVS messages before they are displayed, logged, automated, or routed through your sysplex. You can make decisions about the message based on its message ID, job name, and many other properties.	<i>IBM Z NetView Automation Guide</i>
MultiSystem Manager	Provides an integrated, centralized network management facility that you can use to manage your non-SNA network resources from a single workstation	• <i>IBM Z NetView Installation: Configuring Graphical Components</i> • <i>IBM Z NetView User's Guide: NetView Management Console</i>
NetView automation	Perform repetitive or routine tasks without operator action	<i>IBM Z NetView Automation Guide</i>
NetView management console	Displays topology and status information about network and system resources in a graphical format, from any platform enabled by Java™	<i>IBM Z NetView Installation: Configuring Graphical Components</i>
NetView MVS command revision table	The NetView MVS command revision table (CRT) enables you to intercept MVS commands before they are processed. Command sources include the MVS console and the NetView MVS command.	<i>IBM Z NetView Automation Guide</i>
NetView Resource Manager	Used to graphically monitor and manage NetView task resource utilization and status using the NetView management console. You can monitor all NetView programs in your enterprise using one NetView management console.	<i>IBM Z NetView Installation: Configuring Graphical Components</i>
REST Server	Externalize NetView functions through Restful APIs. Provides users the capabilities to access NetView from anywhere that can make REST request. This helps facilitates the creation of web applications.	• <i>IBM Z NetView Application Programmer's Guide</i> • <i>IBM Z NetView Installation: Configuring Additional Components</i> • <i>IBM Z NetView User's Guide: NetView</i> • <i>IBM Z NetView Security Reference</i>

<i>Table 13. Benefits of Individual NetView Functions (continued)</i>		
NetView Function	Benefit to You	Where Documented in NetView Library
Z NetView Enterprise Management Agent	Used to manage both TCP/IP availability and performance data from a single user interface. The NetView program provides TCP/IP availability data and OMEGAMON for Networks provides TCP/IP performance data. You can also monitor the health of your NetView system including processor utilization, storage, message queue count, input and output message rates, and input and output rates.	<i>IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent</i>
Packet trace data	The NetView program provides real-time capture and formatting of IP packet and OSA packet trace data, including both headers and payloads.	<i>IBM Z NetView IP Management</i>
Program-to-Program Interface (PPI)	Enables user programs to send or receive data buffers from other user programs; also allows system and application programs to send alerts to the hardware monitor	<i>IBM Z NetView Application Programmer's Guide</i>
Remote operations	Interact with remote SNA systems without the overhead of cross-domain logons. (The receiver can be an unattended system.)	RMTCMD in the <i>IBM Z NetView Command Reference Volume 2 (O-Z)</i>
Resource Object Data Manager (RODM)	Provides a central location for storing, retrieving, and managing operational resource information	• <i>IBM Z NetView Installation: Configuring Graphical Components</i> • <i>IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide</i>
Save and restore function	Saves timers, global variables, and focal point information to VSAM and then restores this data when the NetView program is restarted	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Session monitor	Collects and stores data about SNA resources in subarea, Advanced Peer-to-Peer Networking, and mixed networks.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
SNA topology manager	Obtains the status and topology information for SNA subarea and SNA Advanced Peer-to-Peer Networking resources dynamically, for graphical display using the NetView management console	<i>IBM Z NetView SNA Topology Manager Implementation Guide</i>
Status monitor	Collects status information about SNA resources in the network	<i>IBM Z NetView Installation: Configuring Additional Components</i>

Table 13. Benefits of Individual NetView Functions (continued)

NetView Function	Benefit to You	Where Documented in NetView Library
Subsystem interface (SSI)	Used by MVS operators to send commands to the NetView program and by NetView operators to receive output from commands sent to MVS through the SSI; also can be used to monitor MVS operations, because unsolicited messages are sent to NetView program through the SSI	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Sysplex management	The NetView program can be used to manage a sysplex from a single point of control.	<i>IBM Z NetView IP Management</i>
TCP/IP connection data collection	Collects TCP/IP connection data to be displayed in NetView workspaces in the Tivoli Enterprise Portal, or on the 3270 console. Performance data collected by IBM OMEGAMON for Networks can also be displayed.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
TESTPORT function	Provides monitoring capability for critical ports using the TESTPORT command. Monitors a port that refuses a connection but appears to be normal when the NETSTAT command is issued.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
TSO command server	Supports operations and procedures that issue commands through TSO.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
UNIX command server	Enables UNIX commands to be entered from the NetView command line and returns output to the NetView console.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Web Services Gateway	Provides an open interface into the NetView program to issue commands and receive responses. SOAP is used for communications and uses HTTP or HTTPS as the transport mechanism.	• <i>IBM Z NetView Installation: Configuring Additional Components</i> • <i>IBM Z NetView Application Programmer's Guide</i>
XCF services	In sysplex management, z/OS XCF services are used to help initiate resource discovery and facilitate forwarding of management data in the sysplex, and to implement a master NetView program. The master NetView program has the complete view of the sysplex.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
zERT data collection	Collects TCP/IP connection security data to be displayed in NetView workspaces in the Tivoli Enterprise Portal, or on the 3270 console.	<i>IBM Z NetView Installation: Configuring Additional Components</i>
Note: No additional steps are required before using this function.		

Optional NetView Tasks

Table 14 on page 95 lists the optional NetView tasks and how they are started.

Table 14. Optional NetView Tasks		
Task	Description	Started by Default?
AAUTCNMI	Passes data to and from the communication network management interface (CNMI). This task is used by the session monitor.	Y
AAUTSKLP	Writes data to the session monitor database	Y
ALIASAPL	Creates aliases for VTAM resources	N
BNJDSERV	Processes events and alerts. This task is used by the hardware monitor.	Y
BNJMNPD	Receives certain types of locally generated event and statistical records. This task is used by the hardware monitor.	Y
CNMCALRT	Receives alerts through the PPI	Y
CNMCSSIR	Routes commands and messages to appropriate NetView tasks	Y
CNMTAMEL	Receives status changes for resources and forwards them to the NetView management console. This task is needed for LU 6.2 and IP connections to the NetView management console server using the NetConv command.	Y
<i>domainid</i> BRW	Provides the log-browse facility. The <i>domainid</i> is the NetView domain identifier.	Y
<i>domainid</i> LUC	Provides communications between the functions of NetView and their counterparts in other NetView domains. The <i>domainid</i> is the NetView domain identifier. This task is used by the hardware monitor, session monitor, and NetView management console.	Y
<i>domainid</i> VMT	Provides communication between NetView and VTAM. The <i>domainid</i> is the NetView domain identifier. This task is used by the status monitor and NetView management console.	N
DSIACBMT	Opens and processes session requests for the NetView domain	Y
DSIAL2WS	Provides alert dynamic function for the SNA/6000 workstation	N
DSIAMLUT	Receives session awareness and other data from VTAM. This task is used by the session monitor.	Y
DSIATOPT	Performs disk write services for the AUTOTEST command	N
DSICORSV	Manages the TCP/IP socket interface with the event correlation engine	N
DSICRTR	Routes data from the CNMI to the appropriate NetView component	Y
DSIDB2MT	Provides a DB/2 environment	N

Table 14. Optional NetView Tasks (continued)		
Task	Description	Started by Default?
DSIELTSK	Provides external SMF logging support	N
DSIGDS	Provides a CNM interface for communication with resources in the network such as service points using the RUNCMD command.	N
DSIGKPMT	Manages start up and shut down for global KEEP message repositories.	Y
DSIHPDST	Provides high performance LU 6.2 communications for sending and receiving large amounts of data	Y
DSIIPLOG	Provides system logging for remote users	N
DSILOG	Writes data to the network log	Y
DSIQTSK	Provides facilities for communicating with the Resource Object Data Manager (RODM)	N
DSIRQJOB	Ensures that the NetView program is known to JES	N
DSIRSH	Enables the RSH server	N
DSIRTTR	Allows the NetView program to receive alerts over a TCP/IP connection	N
DSIRXEXC	Enables the REXEC server	N
DSISVRT	Enables the save and restore function for timers, global variables, and focal point information to VSAM	Y
DSITCONT	Enables the collection and querying of connection data using the TCPCONN START, TCPCONN STOP, and TCPCONN QUERY commands	Y
DSITCPIP	Provides a TCP/IP environment for NetView	N
DSITRACE	Writes trace data to the trace log	N
DSIUDDST	Sends RMTCMD data to another NetView program and receives responses	Y
DSIWBTSK	Interfaces with the NetView Web server task	N
DSIWTOMT	Processes NetView commands using the WTOR function	Y
DSIXCFMT	START XCFGROUP and STOP XCFGROUP command processing. Also XCF group exit and message exit event processing	Y
DSI6DST	Provides management services (MS) transport function for sending and receiving management data	Y
DUIDGHB	Gets the host name from the IP address NetView management console	Y
DUIFSSCO	Provides command and span authorization checking for NetView management console commands and provides other NetView management console-to-NetView or NetView-to-NetView management console communication services	N
EZLTCFG	Stores the NetView policy file definitions	Y

<i>Table 14. Optional NetView Tasks (continued)</i>		
Task	Description	Started by Default?
EZLTDDF	Provides DDF functions for AON	N
EZLTLOG	Provides automation logging functions for AON	N
EZLTSTS	Provides automation status functions for AON	N
SQLOGTSK	Writes data to a sequential log data set	N
VPDTASK	Solicits and logs vital product data from network inventory	N

Automated NetView Tasks

Table 15 on page 97 lists the automated NetView tasks and whether they are started by default.

<i>Table 15. Automated NetView Tasks</i>		
Task	Description	Started by Default?
AUTIPMGT	Autotask used to initialize the IP management ACTMON and IDS functions.	N
AUTOAMI	Autotask for Application Management Instrumentation	N
AUTOAON	Autotask to read policy definitions and initialize AON	Y
AUTOCMON	Autotask for processing data of Command Statistics function	Y
AUTOSEC	Autotask for connection security information	N
AUTOCSMF	Autotask for writing data of Command Statistics function to SMF	Y
AUTOCT1	Autotask for DVIPA definition and status	N
AUTOCT2	Autotask for DVIPA sysplex distributors, distributed targets, and server health.	N
AUTOCT3	Autotask for DVIPA connections	N
AUTOCT4	Autotask for VIPA routes and distributed DVIPA connection routing	N
AUTOCT5	Autotask for OSA channels and ports.	N
AUTOCT6	Autotask for Telnet servers and Telnet server ports.	N
AUTOCT7	Autotask for NetView applications.	Y
AUTODCn	Autotask for the NetView agent to collect data	N
AUTOEDAT	Routes commands and messages to and from the enterprise master NetView program.	Y
AUTOIPA	Autotask for MultiSystem Manager IBM Tivoli Network Manager agent	N
AUTOMSM	Autotask for the MultiSystem Manager that is used for alert processing.	N
AUTOMSMD	Autotask for the MultiSystem Manager and is used during GETTOPO command processing.	N

Table 15. Automated NetView Tasks (continued)

Task	Description	Started by Default?
AUTONALC	Autotask for the NetView agent	N
AUTONRM	Autotask for NetView Resource Manager	N
AUTONVSP	Autotask for NetView Web Services Gateway	N
AUTOOPKT	Autotask to collect OSA packet trace data for a given TCP/IP stack	N
AUTOOPNA	Autotask for MultiSystem Manager Open agent	N
AUTOPKTS	Autotask to collect IP packet trace data for a given TCP/IP stack	N
AUTOPORT	Autotask for TESTPORT function	N
AUTOPSAV	Autotask used with IPTRACE function to store packet trace data as part of the packet trace analysis function. By default the autotask used for this function is AUTOPSAV, but can be changed to another valid autotask to fit your environment.	Y
AUTOREST	Autotask for REST SERVER CMDSERV RECEIVER	N
AUTOSMF3	Autotask for the CNMSMF3R sample (PPI receiver for SMF records)	N
AUTOTCPC	Autotask to collect TCP/IP connection data for a given TCP/IP stack	N
AUTOTCPS	Autotask to collect DVIPA configuration changes for a given TCP/IP stack	N
AUTOTMSI	Autotask for Tivoli Enterprise Monitoring Services	N
AUTTRAN	Autotasks for concurrent packet trace instances	N
AUTOVTAC	Autotask for VTAM ACB monitor	N
AUTOVTDB	Autotask for VTAM ACB monitor	N
AUTOXCF	Autotask that processes XCF-related commands	Y
AUTOXDSC	Routes messages to the sysplex master NetView program	Y
AUTO1	Autotask for NetView initialization	Y
AUTO2	Autotask for NetView initialization	Y
BRIGOPER	Autotask for NetView Bridge	N
DBAUTO1	Autotask for NetView VSAM DB automation	Y
DBAUTO2	Autotask for NetView VSAM DB automation	Y
DSIIPCHK	Autotask used to perform security checks for DSTs and OPTs that accept connections from TCP/IP	Y
DSILCOPR	Autotask for focal point autodial	Y
DSINVGR	Autotask for NetView generic receiver	N
DSIWEB	Autotask for Web server interface functions	N

<i>Table 15. Automated NetView Tasks (continued)</i>		
Task	Description	Started by Default?
DUIFCSGW	Autotask for common operations services (COS) gateway functions for NetView management console.	N
DUIFEAUT	Autotask for GMFHS	N
DUIFPOLI	Autotask for GMFHS to process NMCSTATUS policy definitions	N
DVIPAUTO	Autotask on which DVIPA automation runs to handle event processing, forwarding data to the master NetView program, and requesting data from the master NetView program.	N
DVIPSTAT	Autotask that processes the distributed DVIPA statistics and logs them to sequential data sets	N
FLBTOPO	Autotask to provide SNA topology manager support for subarea and Advanced Peer-to-Peer Networking resources.	N
MVSCMDS	Autotask for MVS command revision function	Y
REMOPER	Autotask for NetView Bridge	N
Notes: 1. For AON autotasks associated with the AON tower, see the EZLOPF member that is included in the DSIOPF member. 2. For AON autotasks associated with the AON.SNA subtower, see the FKVOPF member that is included in the DSIOPF member. 3. For AON autotasks associated with the AON.TCP subtower, see the FKXOPF member that is included in the DSIOPF member. 4. For autotasks associated with the IPMGIT tower, see the FKXOPFIP member that is included in the DSIOPF member. 5. For MSM autotasks associated with the MSM tower, see the FLCSOPF member that is included in the DSIOPF member.		

Changing an operator ID

If you change an operator ID, you might also have to change the corresponding operator ID in other DSIPARM members. For example, the following DSIPARM members might require modifications:

NetView style sheet

CNMSTYLE and its included members

NetView automation table

DSITBL01 and its included members

NetView command authorization table

CNMSCAT2 or CNMSCATU member, if the following statement is specified:
SECOPTS.CMDAUTH=TABLE

Security product definitions for operators

CNMSAF2 member (CONNECT statements), if the following statement is specified:
SECOPTS.CMDAUTH=SAF.option

Important: Do not change the following automated task IDs that are defined in DSIOPF and CNMSAF2 or their included members:

AUTOAMI

AUTOVTAC
AUTOVTDB
BRIGOPER
DSIIPCHK
DSILCOPR
DSINVGR
DSIWEB
DUIFCSGW
DUIFEAUT
FLBTOPO
REMOPER
SYSADMIN

Internal NetView Tasks

Table 16 on page 100 lists internal tasks which perform vital NetView services. These tasks automatically start when NetView starts. Do not routinely start or stop them.

Table 16. Internal NetView Tasks	
Task	Description
DSIACBMT	NetView main VTAM ACB processing
DSIDCBMT	NetView data set management
DSIGKPMT	NetView global keep processing
DSIHLLMT	NetView high-level language, pre-initialized environments
DSILOGMT	NetView log browse synchronization
DSIMONIT	NetView task resource monitoring
DSISTMMT	NetView STOP and MODIFY processing
DSITIMMT	NetView main task auxiliary processing
DSIXCFMT	NetView XCF processing

Appendix B. NetView Samples Overview

All of the sample definitions listed in this book (and some not listed) are contained on the NetView distribution media. These definitions are distributed as members of the following data sets:

- NETVIEW.V6R3M0.CNMSAMP
- NETVIEW.V6R3M0.DSIPARM
- NETVIEW.V6R3M0.DSIPRF
- NETVIEW.V6R3M0.SAQNPARM
- NETVIEW.V6R3M0.SAQNSAMP
- NETVIEW.V6R3M0.SDSIMSG1
- NETVIEW.V6R3M0.SDSIOPEN

Each samples name begins with AAU, BNJ, CNMB, CNMCM, CNMS, DSI, DUIF, EKG, EZL, FKV, FKX, or FLC, and can have a maximum length of 8 characters. The last four or five characters of all samples vary. [Table 17 on page 101](#) summarizes the samples distributed with the NetView program.

Table 17. Summary of Sample Members

Name	Description
AAUxxxxx	Session monitor samples
application.yml	Sample setup for server configuration, provided in the NetView ZFS.
BNJxxxxx	Hardware monitor samples
CNMBnnn	Sense code descriptions
CNMCMxxx	Command definitions
CNMSAAnn	SAA DM members
CNMSCSxx	Command Statistics samples
CNMSInnn	MVS IDCAMS statement members
CNMSJHnn	GMFHS installation samples
CNMSJIIn	MVS IEBCOPY procedures and control statements
CNMSJMnn	Miscellaneous MVS samples
CNMSJSNF	A new PROCEDURE JCL needed to create Sniffer trace format data sets.
CNMSJVnn	IDCAMS verify statements
CNMSJnnn	MVS installation samples
CNMSnVSM	Sample VSAM load data
CNMS0nnn	VTAM and NCP definitions for sample network
CNMS1Ann	External logging support samples
CNMS42nn / CNMS43nn	High-level language (HLL) PL/I samples
CNMS42nn / CNMS44nn	High-level language (HLL) C samples
CNMS42nn / CNMS45nn	Assembler samples
CNMS6nnn	Automated operations samples

Table 17. Summary of Sample Members (continued)

Name	Description
CNMS71nn	IBM Network Configuration Application/MVS samples
CNMS80nn	REXX samples
DSIxxxxx	Command facility samples
DUIFxxxx	GMFHS samples
EJNSSRST	Startup procedure for the NetView REST Server, provided so the server can be started as a data space within NetView.
EKGxxxxx	RODM samples
EZLxxxxx	AON samples
FKVxxxxx	AON/SNA samples
FKXxxxxx	AON/TCP samples
FLCxxxxx	MSM samples
NetViewRestServer.yml	Sample yml file for integration with Zowe™ Mediation Layer.
NetViewSample	Sample application for Zowe™ that provides some guidance on how to use the new NetView APIs.

Appendix C, “NetView Samples Reference,” on page 119 lists all the samples on the distribution media and a brief description of each sample. Use CNMSJ003 to copy sample members into the correct libraries where they can be used by the NetView and VTAM programs. Sometimes, you must rename the samples to meet NetView or VTAM program requirements. For example, VTAM start options are distributed on the samples media in CNMS0010. However, the VTAM program requires that you name this member ATCSTR00. CNMSJ003 copies CNMS0010 into the VTAM library as ATCSTR00.

If you want information about...	Refer to...
HLL samples	<i>IBM Z NetView Programming: PL/I and C</i>
Assembler samples	<i>IBM Z NetView Programming: Assembler</i>
REXX samples	<i>IBM Z NetView Programming: REXX and the NetView Command List Language</i>
Automation samples	<i>IBM Z NetView Automation Guide</i>

The sample network uses a standard naming convention for the NCP and VTAM definitions.

VTAM Major Node Names

Table 18 on page 102 lists the names of the VTAM major nodes in the sample network (NETA).

Table 18. Summary of Sample VTAM Major Nodes at A01MPU in Network NETA

Name	Distributed As	Description
ATCCON01	CNMS0003	Defines configuration start list
ATCSTR00 ATCSTR01	CNMS0010 CNMS0007	Defines VTAM start options
A01ADJ	CNMS0012	Defines adjacent SSCPs for VTAM to query

Table 18. Summary of Sample VTAM Major Nodes at A01MPU in Network NETA (continued)

Name	Distributed As	Description
A01APPLS	CNMS0013	Defines the NetView program application major node
A01CDRM	CNMS0014	Defines cross-domain resource managers in NETA
A01CDRM1	CNMS0088	Defines cross-domain resource managers in NETB
A01CDRSC	CNMS0015	Defines cross-domain resources
A01CDRS1	CNMS0089	Defines independent LU cross-domain resources
A01LOCAL	CNMS0016	Defines local non-SNA terminals
A01MVS	CNMS0047	Defines MVS applications
A01NVAS	CNMS0087	Defines the NetView access services applications
A01PATH	CNMS0018	Defines path table for host A01MPU
A01SNA	CNMS0073	Defines local SNA terminals
A01USER	CNMS0041	Defines various application programs
A04NTRI	CNMS0061	Defines the dial deck coded for NTRI lines
CNMCON01	CNMS0084	Defines major nodes not activated during VTAM initialization
CTCA0102	CNMS0038	Defines a channel-to-channel attachment
CTNA0104	CNMS0081	Defines a channel-to-NCP channel attachment

Table 19 on page 103 lists the names of the VTAM major nodes in the sample network (NETB). The actual samples are not shown in this book, however, their NETA equivalents are listed in the table for easy reference.

Table 19. Summary of Sample VTAM Major Nodes at B01MPU in Network NETB

Name	Distributed As	Description	NETA Equivalent
ATCCONB1	CNMS0102	Defines configuration start list	ATCCON01
ATCSTRB1	CNMS0101	Defines VTAM start options	ATCSTR01
B01ADJ	CNMS0104	Defines adjacent SSCPs for VTAM to query	A01ADJ
B01APPLS	CNMS0113	Defines the NetView program application major node	A01APPLS
B01CDRM	CNMS0106	Defines cross-domain resource managers in NETA	A01CDRM
B01CDRSC	CNMS0107	Defines cross-domain resource managers in NETB	A01CDRSC
B01LOCAL	CNMS0108	Defines local non-SNA terminals	A01LOCAL
B01MVS	CNMS0111	Defines MVS applications	A01MVS
B01PATH	CNMS0105	Defines path table for host B01MPU	A01PATH
B30NTRI	CNMS0110	Defines dial deck coded for NTRI lines	A04NTRI

Table 19. Summary of Sample VTAM Major Nodes at B01MPU in Network NETB (continued)

Name	Distributed As	Description	NETA Equivalent
CNMCONB1	CNMS0103	Defines major nodes not activated during VTAM initialization	CNMCON01
CTNB0130	CNMS0115	Defines a channel-to-NCP attachment	CTNA0104

VTAM Definitions

You can modify the sample network definitions to fit your network. This process includes modifying VTAM definition statements and VTAM tables.

VTAM Definition Statements

After installing the NetView program, begin to define the network, starting with VTAM definitions. Define the programs, devices, and paths to other nodes in the network. Define these by coding major nodes.

A **major node** is a representation of a set of resources. Specifically, a major node is a set of VTAM definition statements. For MVS, a VTAM major node is a member of the partitioned data set identified on the VTAMLST DD statement in the VTAM start procedure.

When you have coded the necessary major nodes, the VTAM operator can activate or deactivate resources in the network by activating and deactivating major or minor nodes. (A **minor node** is one resource in a major node.) For example, at A01MPU, local terminals are defined in the major node named A01LOCAL. When the VTAM program is active, the VTAM operator can use the following VARY ACT command:

```
v net,act,id=a01local
```

Conversely, the operator can deactivate these terminals by entering the following VARY INACT command:

```
v net,inact,id=a01local
```

Configuration Lists

The major nodes you have defined are not processed by the VTAM program until you activate them. Use one of the following two ways to activate these nodes:

1. Have the VTAM operator enter a VARY ACT command for each major node.
2. Include the major node names in a configuration list. When you start the VTAM program, it automatically activates every node that you have listed if CONFIG=xx is coded in your start list, where xx is the last two characters of your configuration list name (ATCCONxx).

VTAM Start Options

VTAM start options define such things as buffer pools, major nodes to be activated when the VTAM program is started, the network name, and the identifier given to the VTAM program for the host subarea. The only VTAM start options that you need to code are SSCPID=*n*, NETID=*n*, and SSCPNAME=*n*. However, most users need to specify some options, especially the host subarea (HOSTSA=*n*). All subareas within a network must be unique.

Code VTAM start options in a major node named ATCSTRyy, where yy is specified by the user. The default value is 00. You can code start options in more than one file. The VTAM program always processes the file named ATCSTR00. You can add additional start options in another file as was done in the sample network.

Start options that apply to all hosts in the network are coded in major node ATCSTR00. The VTAM program processes default values first. Default values are overridden by ATCSTRyy. Start options specific to host A01MPU are coded in major node ATCSTR01. Both nodes are processed when the VTAM program

is started. ATCSTR00, the default value, is always processed. ATCSTR01 is processed when the operator starts the VTAM program and identifies 01 as the start option list by entering:

```
s cnmnet.net,,, (list=01)
```

If the same option is coded in both ATCSTR00 and ATCSTR01, the option coded in ATCSTR01 takes precedence.

The VTAM start options for B01MPU in NETB are coded in ATCSTRB1. To start the VTAM program in host B01MPU, enter:

```
s cnmnet.net,,, (list=B1)
```

ATCSTR00 (CNMS0010) (VTAM Startup Options)

ATCSTR00, the default start option file, is on the distribution media. If you want to use these definitions in your network:

1. Check the PPOLOG start option. This option is used to specify whether messages are sent to the network log. The sample network uses the default value of PPOLOG=NO. If you are running a release of the VTAM program before V3R3, change this value to PPOLOG=YES to keep the status monitor panels in the NetView program accurate.
2. Review the default values for each start option. Default values are coded for all start options except TRACE and IOBUF.
3. Check the TRACE start option. This option indicates which functions are traced when the VTAM program is started. OPT=ALL was coded here, meaning all functions are traced when the VTAM program is started. To conserve storage, delete the OPT=ALL operand or specify a specific function that you want to trace.
4. Check the IOBUF start option. This option is the message pool in fixed storage and is used for input/output data. For the sample network, this option is set to 256 bytes. This represents the value of the UNITSZ operand on the NETA definition statements in A04A54C (CNMS0065) and A31A54C (CNMS0066). The value of the UNITSZ operand on the NETB definition statement in B30A54C (CNMS0109) is set to 384 bytes.

ATCSTR01 (CNMS0007) (Start Options for Host Subarea)

ATCSTR01 is the second start option file coded for A01MPU in NETA. The start option file coded for B01MPU in NETB is ATCSTRB1 (CNMS0101). If you want to use these definitions in your network:

1. Change the HOSTPU keyword parameter to the physical unit name that you want assigned to the VTAM program. In this example, the HOSTPU keyword parameter is the SSCP name followed by the string PU (for example: A01MPU). The HOSTPU name is displayed on some NetView panels.
2. Change the HOSTSA keyword parameter to match your host subarea.
3. Change the NETID keyword parameter to match your network name. The NETID keyword is a required parameter. If you code the NETID keyword parameter using the NLDN.NETID statement in the CNMSTYLE member, the NETID keyword parameter must be the same as the start parameter that you used for the VTAM program.
4. Change the SSCPID keyword parameter to a number that is unique within your network and across all interconnected networks. In this example, the host subarea is also used for the SSCPID keyword parameter.
5. Change the SSCPNAME keyword parameter to the name you want to use for your SSCP. You can use the name A01M.
6. You might want to add the CSA24 start option. This start option is used to specify the maximum amount of 24-bit common storage area that the VTAM program can access.

Configuration Start List

The configuration list must be named ATCCONxx, where xx is defined by the user. You identify the file name to the VTAM program with a start option. For example, if you have coded **config=01**, the VTAM program expects to find a configuration list named ATCCON01. ATCCON01 is coded for host A01MPU. It contains all the major nodes that are activated when the VTAM program is initialized, except the NCPs (which can also be initialized but are not initialized in this example). If you intend to use this list in your own network, change the major node names to match the major nodes your host activates.

ATCCON01 (CNMS0003) (Configuration Start List)

The configuration list coded for host A01MPU in NETA is included on the distribution media. Sample configuration lists for A02MPU and A99MPU are also included on the distribution media. They are ATCCON02 (CNMS0004) and ATCCON99 (CNMS0005), respectively. The configuration list coded for B01MPU in NETB is ATCCONB1 (CNMS0102).

CNMCON01 (CNMS0084) (Major Nodes Not Activated During VTAM Initialization)

The STATMON preprocessor CNMNDEF (CNMSJ007) uses this major node to inform the status monitor which resources at this VTAM host (A01MPU) were not activated during VTAM initialization. The STATMON preprocessor also informs the status monitor that these resources need to be monitored. To use this sample in your network, change the node names to match the major nodes (in your host) that must not be activated during VTAM initialization.

Also included on the distribution media are equivalent samples for A02MPU and A99MPU. They are CNMCON02 (CNMS0085) and CNMCON99 (CNMS0086) respectively. The major node coded for B01MPU in NETB is CNMCONB1 (CNMS0103).

Application Programs

Each VTAM application program must be defined with an APPL definition statement. The statement is coded in an application major node defined with a VBUILD statement (TYPE=APPL).

In the sample network, the following major nodes are available for host A01MPU:

A01APPLS

Defines application programs common to all operating systems. APPL statements for the NetView program are in this major node.

A01MVS

Defines MVS-based application programs, such as TSO and NetView Performance Monitor (NPM).

A01NVAS

Defines NetView access services application major node.

A01USER

Defines additional application programs, such as IMS and CICS

A detailed description of each application major node follows.

A01APPLS (CNMS0013) (NetView APPL Definitions for Subarea)

A01APPLS is the first APPL major node coded for A01MPU. The application major node coded for B01MPU in NETB is B01APPLS (CNMS0113). If you want to use these definitions in your network, perform these steps:

1. For each APPL statement, check the names of the logon mode table and default logon mode. The sample network uses a logon mode table called AMODETAB and various default logon modes.

2. Check the description in each STATOPT operand. The lines beginning with STATOPT are operands used by the status monitor component of the NetView program. The VTAM program interprets these lines as comments, so they have no effect if you do not use the NetView program. The STATOPT operands control the functions of the status monitor and are coded in the major and minor node members in VTAMLST. The STATOPT operands are displayed on the status monitor panel.
3. If you plan to have more than 10 concurrent NetView users, add additional &CNMDOMN.*nnn* and TAF01*nnn* APPL statements.

Note: If the status monitor preprocessor is not used, define one APPL statement for NetView users and one APPL statement for TAF. For more information, see [“Defining Multiple Concurrent NetView Operators”](#) on page 31 and [“Allowing for Additional TAF LU1 Operators”](#) on page 32.

If you want information

Refer to...

How to code the STATOPT operand

IBM Z NetView Administration Reference

A01MVS (CNMS0047) (MVS Unique APPL Definitions)

A01MVS is an APPL major node coded for host A01MPU. The APPL major node coded for NETB is B01MVS (CNMS0111). If you want to use these definitions in your network, complete the following steps:

1. If you plan to have more than 10 concurrent TSO users, add additional TSOA01*nn* APPL statements.
2. This major node contains APPL definitions for NPM. The sample network does not contain NPM. This definition is included only as an example. If you do not plan to run NPM in your network, you can remove the NPM APPL definition.
3. Add APPL statements for any other VTAM applications you plan to run.

A01NVAS (CNMS0087) (Defines NetView APPL Statements)

This major node contains APPL statements for the NetView access services (NVAS) application. If you want to use these definitions in your network, complete the following steps:

1. If you are not using NVAS, remove this sample. EMSMODE is a logon mode table shipped with the NVAS product and is not a part of the samples shipped with the NetView program.
2. If you plan to have more than 10 concurrent users for NVAS, add additional PSA01*nnn* APPL statements.

A01USER (CNMS0041) (Defines APPL Samples)

This major node contains definitions for other VTAM application programs, such as IMS and CICS.

Local Non-SNA Terminals

Host A01MPU in the sample network has six local non-SNA 3277 terminals that you must define to the VTAM program. The definition is done with a local non-SNA major node. This major node consists of an LBUILD statement followed by a LOCAL statement for each terminal. The local non-SNA major node for A01MPU is named A01LOCAL (CNMS0016). The local non-SNA terminal major node coded for NETB is B01LOCAL (CNMS0108). The terminals defined by A01LOCAL are illustrated by [Figure 33 on page 109](#).

If you want to use these definitions in your network, complete the following steps:

1. Verify that the terminal types match the non-SNA terminals in your network. Check the model numbers and the use of the extended data stream feature. (The sample network uses 3277 Model 2 terminals, which use the extended data stream feature.)
2. Check the control unit address for each terminal. The sample network uses addresses 701 through 706. You can choose these addresses or change them to those available on your system.
3. Check the names of the default logon mode and logon mode table. The sample network uses a default logon mode called M2BSCQ in a logon mode table called AMODETAB.
4. Check the name of the USS table. The sample network uses a USS table called AUSSTAB.

5. Check the description in the NetView STATOPT operand to make sure that it is appropriate for the device. This description is displayed in status monitor panels.

Local SNA Devices

Besides local non-SNA terminals, host A01MPU has a channel-attached SNA control unit with six terminals and two printers attached to it. These resources must be defined to the VTAM program with a local SNA major node. This major node consists of a VBUILD TYPE=LOCAL statement, followed by a PU statement for the control unit and an LU statement for each device.

The major node name is A01SNA (CNMS0073). The SNA devices defined by A01SNA are illustrated by [Figure 33 on page 109](#).

If you want to use these definitions in your network, complete the following steps:

- For the PU statement:
 1. Check the control unit address (CUADDR). For the sample network, the address is 7A0.
 2. Check the names of the default logon mode (DLOGMOD) and logon mode tables (MODETAB). In this example, the default logon mode is M23278I and the logon mode table is AMODETAB.
 3. Check the name of the USS table. A01MPU uses a USS table called AUSSTAB.
 4. Check the description in the NetView STATOPT operand. What you code on this operand is displayed on status monitor panels.
- For each LU statement:
 1. Make sure the local address (LOCADDR) is appropriate.
 2. If necessary, change the default logon mode. In this example, the default logon mode for the LUs at addresses 5 and 9 (printers), was changed to the correct mode. These devices use a logon mode named M3287SCS, which is for 3287 sequential character stream (SCS) printers. The other LUs use the default value defined in the preceding PU statement.
 3. Check the description in the NetView STATOPT operand.
 4. Check the SSCPFM operand. If the device supports character-coded messages in its communication with the VTAM program, code SSCPFM=USSCS. Otherwise the default value is SSCPFM=FSS, which is for formatted commands only.

Example of Devices Defined by A01LOCAL and A01SNA

[Figure 33 on page 109](#) illustrates the local SNA and non-SNA devices defined by A01LOCAL and A01SNA.

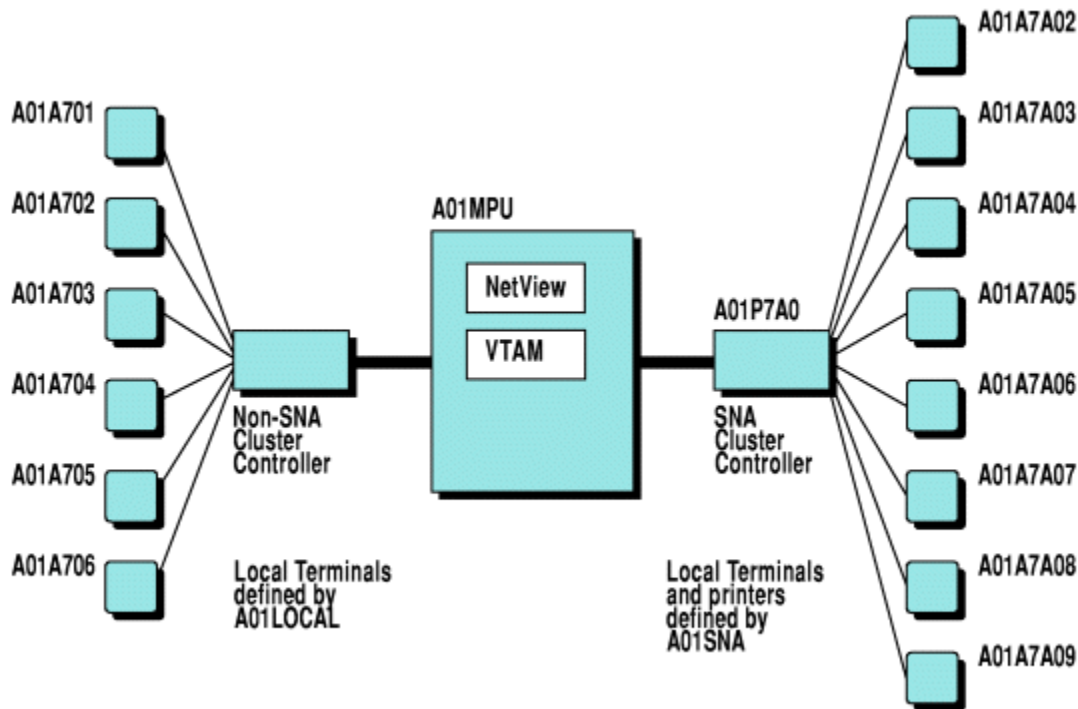


Figure 33. Local Devices Defined at A01MPU

Cross-Domain Resources

The logical units (LUs) in other domains that the VTAM program accesses are called **cross-domain resources** (CDRSCs). Some cross-domain resources are applications that reside in other hosts, such as CICS and TSO. A terminal in another domain can also be a CDRSC.

CDRSC definitions are not required. CDRSCs can dynamically define the owner of a CDRSC by polling other hosts in the network.

You can define CDRSCs in one or more CDRSC major nodes. The major node consists of a VBUILD statement (TYPE=CDRSC), followed by a CDRSC statement for each cross-domain resource.

The following sections define the cross-domain resources for A01MPU. The NETB equivalent cross-domain resource major node is B01CDRSC (CNMS0107).

A01CDRSC (CNMS0015) (Cross-Domain Resource Definitions)

In this example, only the NetView programs at subareas 2 (CNM02) and 99 (CNM99) are defined as CDRSCs. In this example, the owning CDRM for the NetView program is not defined. The VTAM program uses an adjacent SSCP table to locate the owning CDRM. See [“Adjacent SSCP Table”](#) on page 110 for additional information.

A01CDRS1 (CNMS0089) (Independent LU CDRSC Major Node)

The cross-domain resource sample A01CDRS1 (CNMS0089) contains the independent LUs defined in the sample network for host A01MPU.

Cross-Domain Resource Managers

The owner of a CDRSC is called a **cross-domain resource manager** (CDRM). A CDRM is the VTAM program in another host.

Identify the CDRMs in all the other hosts with which your host needs to communicate. For A01MPU, these are A02M, A99M, and B01M. Also define the CDRM component of the VTAM program in your local host. Therefore, a CDRM statement is required for A01M, A02M, A99M, and B01M for A01MPU.

Define each CDRM with a CDRM definition statement. The statement is coded in a CDRM major node that begins with a VBUILD statement (TYPE=CDRM).

It is not necessary to define B01M to A02MPU and A99MPU because they communicate through A01MPU.

To use the sample definitions, complete the following steps:

1. Change the NETID in the NETWORK statement to the name of your network. You can omit NETWORK statements if you do not use SNA Network Interconnection (SNI).
2. In each CDRM statement, change the label to match the name of the host being defined. In this sample and in most circumstances, use a CDRM name to match the SSCP name of the host. (Hosts define their SSCP names as a start option.)
3. Also, in each CDRM statement, ensure that the SUBAREA number is correct.
4. Change the ISTATUS statements for the external CDRMs (A02M, A99M, and B01M in the sample) from ACTIVE to INACTIVE if you do not want SSCP-to-SSCP sessions to be automatically established when this major node is activated.
5. Adjust the VPACING operand if it does not meet the needs of your network.
6. The CDRDYN=YES and CDRSC=OPT are coded for each CDRM. Code these operands to enable dynamic definition of cross-domain resources.
7. If necessary, add more CDRM statements. Include a CDRM statement for every host with which to communicate (inside or outside your network).

A01CDRM (CNMS0014) (Cross-Domain Resource Manager)

A01CDRM defines the cross-domain resource managers in NETA to A01MPU. The CDRM major node coded for B01MPU in NETB is B01CDRM (CNMS0106).

A01CDRM1 (CNMS0088) (Cross-Domain Resource Manager)

This major node defines the cross domain-resource manager in NETB to A01MPU. A01CDRM1 is an example of an interconnected network CDRM.

When these major nodes and the CDRM minor nodes are activated, A01M sets up sessions with the remote VTAMs—A02M, A99M, and B01M.

If you want information about...	Refer to...
To see examples of CDRM definitions in interconnected networks	z/OS Communications Server library

Adjacent SSCP Table

An adjacent SSCP table is a list of SSCPs that you want the VTAM program to query when the resource is unknown. If you code an adjacent SSCP table, you can omit all or selected CDRSC statements. You can also code a CDRSC statement without defining the CDRM that owns the resource. The CDRSC is useful in networks where applications are frequently moved from one host to another.

The adjacent SSCP table filed at B01MPU in NETB is B01ADJ (CNMS0104). If you want to use this table in your own network, modify it in the following way:

1. As with CDRMs, you can delete the NETWORK statement if you do not intend to use SNA Network Interconnection (SNI). If you do use SNI, you can add another adjacent SSCP table for each network. The sample network, however, uses one adjacent SSCP table for all CDRMs in NETA and NETB.
2. For each ADJCDRM, change the name to match the name of the hosts that are next to your host.

When the VTAM program searches for a CDRSC whose location is unknown, the VTAM program queries every SSCP in the list, in sequence, until it locates the resource or until it reaches the end of the list. If the VTAM program does not have a current session with an SSCP in this list, that SSCP is not queried.

Figure 34 on page 111 illustrates how default SSCP routing works. This figure illustrates what occurs when a user at host A01MPU enters:

cnm99

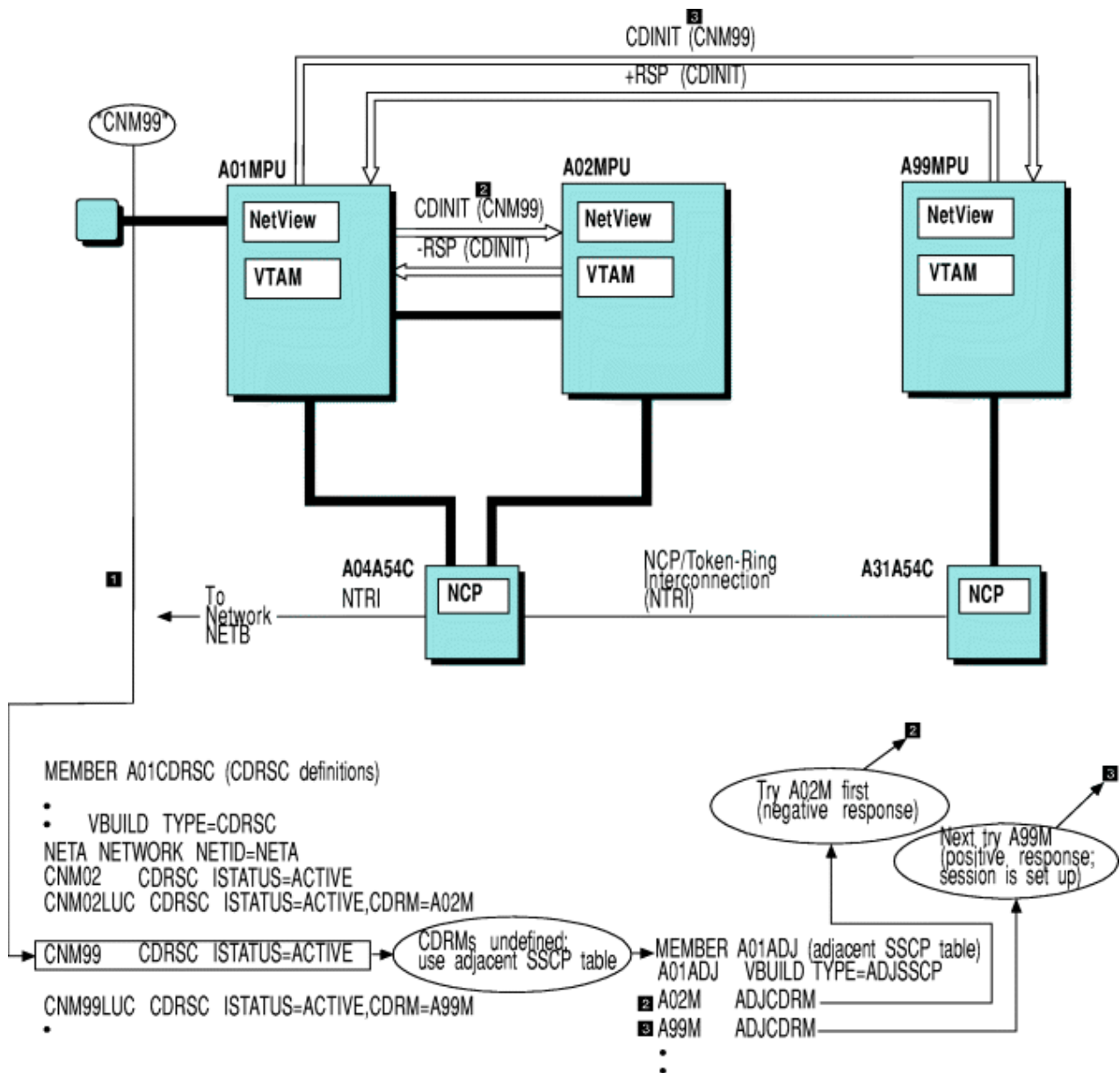


Figure 34. Use of Adjacent SSCP Table

When the VTAM program searches its list of cross-domain resources, it tries to locate an entry for CNM99. Because there is no indication where this resource resides, the VTAM program checks an adjacent SSCP table to see if adjacent SSCPs are defined. The VTAM program then uses the table to try to locate CNM99.

The first SSCP in the list is A02M. The VTAM program sends a session initiation request (CDINIT) to A02M, seeking to establish a session with CNM99. A01M has already established an SSCP-to-SSCP session with A02M. Therefore, the VTAM program knows which route to use.

A02M reacts to the initiation request by searching its resources for CNM99. A02M does not find CNM99 and returns a negative response. After receiving the negative response, the VTAM program tries the next SSCP in the list, which is A99M. Here, A99M owns CNM99 and accepts the initiation request.

Dynamic CDRSC definition increased the overhead required to set up this session. Explicit definition of the cross-domain resource allows the VTAM program to route the session request to its owner immediately.

This increase in overhead is minor and occurs only during initial session setup. For many sites, the increased overhead might be offset by the advantages of dynamic CDRSC definition. With fewer CDRSCs, less coding is required. Also, when an application moves to a different host you do not have to change CDRSC statements.

If you want information about...

Refer to...

Adjacent SSCP tables

z/OS Communications Server library

NCP/Token-Ring Interconnection (NTRI)

The NCP/Token Ring Interconnection (NTRI) allows a communication controller to attach to an IBM token-ring network using one physical medium, one attachment scheme, and one communication protocol.

The NCP at subarea 04 (A04A54C) has NTRI line links that allow an NCP to connect to a token ring. To the VTAM program, NTRI links are switched links and require a switched major node.

A04NTRI defines the dial deck coded for A01MPU in NETA for A04A54C NTRI lines. The dial deck coded for B01MPU in NETB for B30A54C NTRI lines is B30NTRI (CNMS0110).

Channel-to-Channel Attachment

As the illustration of the network in Figure 35 on page 112 shows, A01MPU has a channel-to-channel attachment to A02MPU. This attachment must be defined to the VTAM program in a channel attachment major node. The channel attachment major node consists of a VBUILD statement (TYPE=CA) followed by GROUP, LINE, and PU statements.

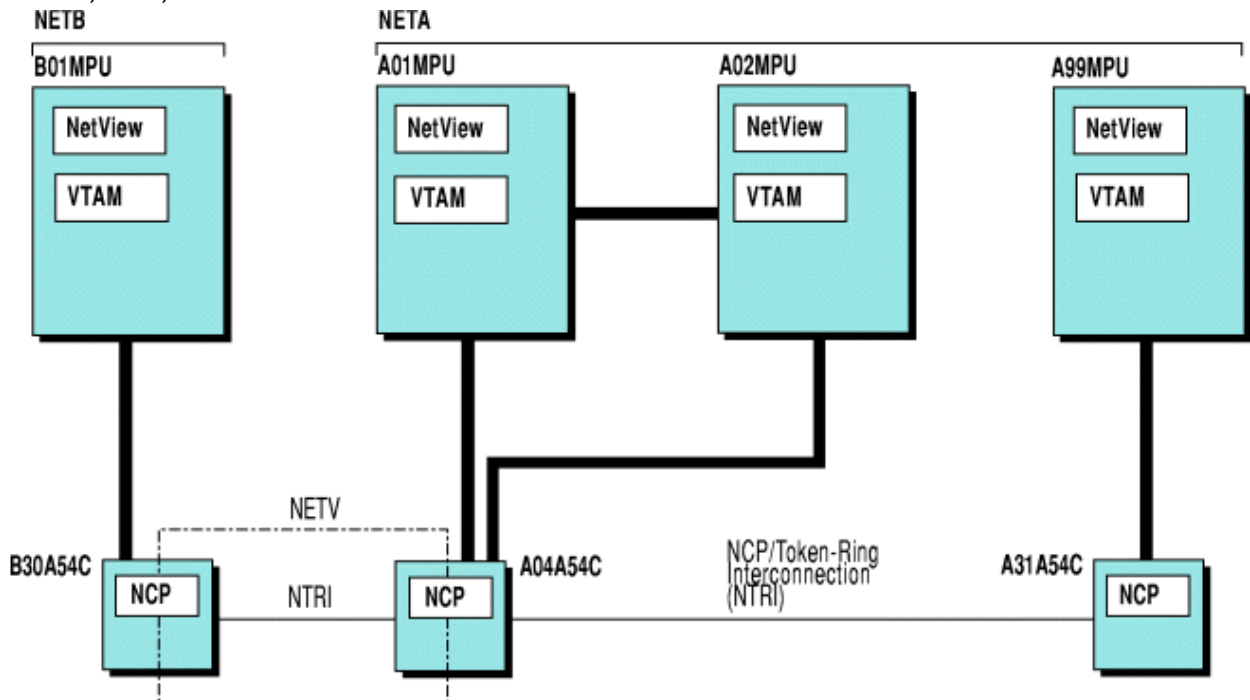


Figure 35. Network (NETA and NETB)

The major nodes named CTCA0102 (CNMS0038) and CTCA0201 (CNMS0039) define the channel attachment between A01MPU and A02MPU. If you want to use these definitions in your network, complete the following steps:

1. Check the DELAY and REPLYTO operands. Default values are coded for each. The value specified in the DELAY operand is the time in seconds that the VTAM program in A01MPU must wait before sending

low-priority data to the VTAM program in A02MPU. The value specified in the REPLYTO operand is the time the VTAM program must wait after completing a channel program.

2. Change the ADDRESS operand on the LINE statement to match the correct value for your network.
3. Verify that the CTC address is defined to the other host.

Channel-to-NCP Attachment

As the illustration of the sample network shows (see [Figure 35 on page 112](#)), A01MPU has a channel-to-NCP attachment to A04A54C as coded in CTNA0104 (CNMS0081). To use these definitions in your network, change the ADDRESS operand on the LINE statement to match the correct value for your network.

The NCP channel attachment between A02MPU and A04A54C in NETA is coded in CTNA0204 (CNMS0082) and the NCP channel attachment between A99MPU and A31A54C in NETA is coded in CTNA9931 (CNMS0083). The channel-to-NCP definition coded for B01MPU in NETB is CTNB0130 (CNMS0115).

Paths

When you know the physical structure of your network, remember to define the paths between the different nodes with PATH definition statements.

[Figure 36 on page 113](#) illustrates the meaning of the PATH statement.

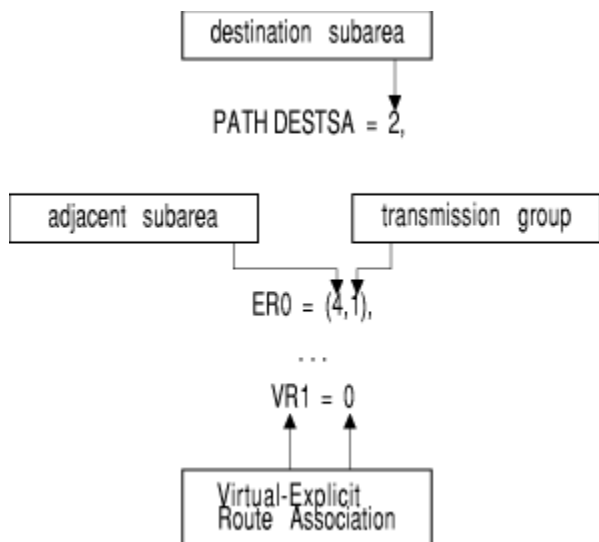


Figure 36. Meaning of PATH Statement

Code a PATH statement for each subarea (host or NCP) with which the VTAM program needs to communicate. Code a PATH statement for every subarea except your own.

The statement first defines the destination subarea. You can use the same statement for more than one subarea. Next, the statement must define the explicit routes and virtual routes (ERs and VRs) that make up that path.

An ER defines the physical elements that connect two subareas. A VR is the logical connection between two subareas. A VR uses the physical route defined by an ER. Associate each ER with a VR. ERs have the additional characteristic of a transmission priority.

The VTAM program needs both ERs and VRs to set up a session. The VTAM program begins with a VR and then uses the PATH table to determine which ER to which the VR maps. A class-of-service table tells the VTAM program which VR to use. In [Figure 36 on page 113](#), if the class of service says that VR1 is to be used, then the VTAM program uses the PATH statement and map that statement into ER0.

2. The VTAM program locates the LOCAL definition statement for the terminal at address 701. As the figure illustrates, the definition for the terminal defines the logon mode table entry as M2BSCQ (DLOGMOD=M2BSCQ). The name of the logon mode table that contains this entry is also given (MODETAB=AMODETAB).
3. The VTAM program searches AMODETAB for the logon mode table entry named M2BSCQ. In [Figure 38 on page 115](#), the class of service specified for that entry is INTERACT.

The VTAM program at A02MPU carries out these steps in selecting the route:

1. The VTAM program searches the class-of-service table for the class named INTERACT. The class-of-service entry defines the virtual route and transmission priority for the session. In [Figure 38 on page 115](#), the entry for INTERACT specifies that virtual route 0 with a transmission priority of 1 is used.
2. The VTAM program uses its path table, specifically the definition of the path to subarea 2. The COS entry specifies that virtual route 0 is used for this session. The PATH statement maps virtual route 0 into explicit route 10 (VR0=10). Therefore the VTAM program uses explicit route 10 for this session. The PATH statement specifies the adjacent subarea and transmission group number for this ER. As the figure shows, for this ER, the VTAM program uses transmission group 1 (a channel) to subarea 2. The VTAM program activates the VR if required, completing the routing from A01MPU to A02MPU.

A02MPU accepts the session initiation request, and the user logs on to the NetView program.

VTAM Tables

Besides VTAM major nodes, you need to code the CNM routing table. You have the option of also coding the following tables:

- Unformatted system services (USS)
- Logon mode
- Class of service (COS)

The VTAM and NetView programs provide default values or samples for each table. You can use the tables as they exist or customize them to fit the requirements of your network.

If you plan to modify one or more of these tables, copy the IBM sample and change it to meet your requirements. Then assemble and link edit the table.

The following sections provide an overview of each table and include a description of how to assemble and link edit a VTAM table.

If you want information about...	Refer to...
Coding these tables	z/OS Communications Server library

USS Table

You can use the USS table to define user (and operator) commands and messages. As an MVS user at subarea 01, remember to remove the VM and the VM01 commands. If hosts A02MPU and A99MPU are MVS hosts, remove the VM02 and VM99 commands. AUSSTAB (CNMS0011) is the USS table that is used for the sample network.

A default USS table is also shipped with the VTAM program.

If you want information about...	Refer to...
A listing of the default USS table that is shipped with the VTAM program	z/OS Communications Server library

Logon Mode Table

A logon mode table defines the session protocols for the different devices and applications in the network. The table also defines the terminal screen size and the class of service used for the session. Two logon mode tables were coded for the sample network:

AMODETAB (CNMS0001)

Supports various devices for the sample network

INTERCOS (CNMS0050)

Supports various applications

To use these tables, assemble and link-edit them into NETVIEW.V6R3USER.VTAMLIB.

Note: Do not change RU sizes for a bisynchronous session.

Class-of-Service Table

Use a class-of-service (COS) table to define the routes that the VTAM program uses in establishing cross-domain sessions. High-priority sessions might be assigned a class of service that uses high-speed routes. Conversely, low-priority sessions might be assigned a class of service that uses slower routes.

You define a class of service by creating an entry for it in a class-of-service table. You then specify the name of the entry in the logon mode table entry to create the routes used for the session.

The VTAM program does not require a class-of-service table. However, if you do not code one, the only classes of service you can use are the unnamed class of service and ISTVTCOS, which is the name the VTAM program uses to set up its own sessions.

For the sample network, the COS table has five classes of service:

- Unnamed class of service
- ISTVTCOS
- NETOPER
- INTERACT
- BATCH

All the classes of service have the same selection order but different transmission priorities. ISTVTCOS is always the class of service with the highest transmission priority. The following list shows the selection order for the unnamed class of service:

1. Virtual Route 0, Transmission Priority 0
2. Virtual Route 1, Transmission Priority 0
3. Virtual Route 2, Transmission Priority 0
4. Virtual Route 3, Transmission Priority 0
5. Virtual Route 4, Transmission Priority 0
6. Virtual Route 5, Transmission Priority 0
7. Virtual Route 6, Transmission Priority 0
8. Virtual Route 7, Transmission Priority 0

Assembling and Link-Editing the Tables

The CNMSJ006 sample that is shipped with the NetView program, assembles, and link-edits the USS, logon mode, and class-of-service tables.

Modifying the VTAM Start Procedure

A VTAM start procedure (CNMSJ008) is included in the samples. You can use the procedure as it is shipped, or you can modify it to meet the needs of your installation. The comments in the start procedure

list some considerations for modifying the start procedure. Symbolic parameters are used for the region size, data set name qualifier, and the SYSABEND output class. For example, the data set name on the VTAMLST statement results in a data set name of NETVIEW.V6R3USER.CNM01.VTAMLST. For more information about modifying the VTAM start procedure, see [“Modifying the VTAM Start Procedure” on page 36](#).

Appendix C. NetView Samples Reference

Table 20 on page 119 lists the samples that can be found in the following libraries:

- NETVIEW.V6R3M0.CNMSAMP
- NETVIEW.V6R3M0.DSIPARM
- NETVIEW.V6R3M0.DSIPRF
- NETVIEW.V6R3M0.SDSIMSG1
- NETVIEW.V6R3M0.SDSIOPEN

The table includes a brief description of each sample and the data set name where each member resides when installation is complete.

Usage notes:

1. Samples that reside in VTAMLST and PROCLIB data sets are initially distributed in NETVIEW.V6R3M0.CNMSAMP.
2. The National Language Support feature contains CNMMSJPN, the NetView sample for the Kanji feature program definition file for the sample network.
3. Although the samples were tested and are supported by the NetView program, you are responsible for making the proper modifications to the samples for your operating system.
4. For information about RODM, GMFHS, SNA Topology Manager, and MultiSystem Manager samples, refer to *IBM Z NetView Installation: Configuring Graphical Components*.

Table 20. List of Samples

Distributed As	Name	Description	Data Set Name
AAUCNMTD	same	Initialization values for the task AAUTCNMI of the session monitor. AAUTCNMI collects data from other network components such as the VTAM and NCP programs.	DSIPARM
AAUKEEP1	same	KEEP classes that control the amount of data kept by the session monitor component of the NetView program. Also included are the MAPSESS statements that map sessions into KEEP classes.	DSIPARM
AAUPRMLP	same	Initialization values for the task AAUTSKLP of the session monitor	DSIPARM
AAURTM1	same	Response time performance classes for the response time monitor (RTM). The performance class sets a standard for performing a certain percentage of the transactions in less than a specified response time.	DSIPARM
BNJMBDST	same	Initialization values for the BNJDSERV hardware monitor task. Initialization values for this task are specified in the CNMSTYLE member. Do not modify the BNJMBDST member.	DSIPARM
CNMB000	same	Sample sense code	DSIPARM
CNMB001	same	Sample sense code	DSIPARM
CNMB002	same	Sample sense code	DSIPARM

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMB080	same	Sample sense code	DSIPARM
CNMB081	same	Sample sense code	DSIPARM
CNMB082	same	Sample sense code	DSIPARM
CNMB083	same	Sample sense code	DSIPARM
CNMB084	same	Sample sense code	DSIPARM
CNMB085	same	Sample sense code	DSIPARM
CNMB086	same	Sample sense code	DSIPARM
CNMB087	same	Sample sense code	DSIPARM
CNMB088	same	Sample sense code	DSIPARM
CNMB089	same	Sample sense code	DSIPARM
CNMB08A	same	Sample sense code	DSIPARM
CNMB08B	same	Sample sense code	DSIPARM
CNMB100	same	Sample sense code	DSIPARM
CNMB101	same	Sample sense code	DSIPARM
CNMB200	same	Sample sense code	DSIPARM
CNMB201	same	Sample sense code	DSIPARM
CNMB400	same	Sample sense code	DSIPARM
CNMB401	same	Sample sense code	DSIPARM
CNMB402	same	Sample sense code	DSIPARM
CNMB800	same	Sample sense code	DSIPARM
CNMB801	same	Sample sense code	DSIPARM
CNMB802	same	Sample sense code	DSIPARM
CNMBA00	same	Sample sense code	DSIPARM
CNMBA01	same	Sample sense code	DSIPARM
CNMBFF0	same	Sample sense code	DSIPARM
CNMBFF1	same	Sample sense code	DSIPARM
CNMBFF2	same	Sample sense code	DSIPARM
CNMBFF3	same	Sample sense code	DSIPARM
CNMBFF5	same	Sample sense code	DSIPARM
CNMBFF6	same	Sample sense code	DSIPARM
CNMBFF7	same	Sample sense code	DSIPARM
CNMBFF8	same	Sample sense code	DSIPARM
CNMBFF9	same	Sample sense code	DSIPARM
CNMBFFC	same	Sample sense code	DSIPARM

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMBINDF	same	Sample sense code	DSIPARM
CNMBUNBD	same	Sample sense code	DSIPARM
CNMCAU00	same	MVS command management exclusion/inclusion table	CNMSAMP
CNMCMD	same	NetView command definitions for NetView commands The CMDDEF statements provide the definitions for the commands. Some of the CMDDEF statements have command synonyms (CMD SYN). These statements provide a synonym for the command. This definition also provides the cross-domain logon definitions and the CMDDEF statements (and synonyms) for the terminal access facility (TAF) and the VTAM program. The following files are included in NETVIEW.V6R3M0.CNMSAMP: • CNMS6206 • CNMS6401 - CNMS6404 Include these files in CNMCMD so that you can use the automation command lists that are also included on the distribution media.	DSIPARM
CNMCMDU	same	Command definitions for user-defined commands	DSIPARM
CNMCMDO	same	Command definitions for product-specific defined commands	DSIPARM
CNMCMENT	same	NetView command definitions	DSIPARM
CNMCMSYS	same	NetView command definitions	DSIPARM
CNMCRC1	same	Input data for the CNMSTYLE report generator	DSIPARM
CNMCRCR1	same	Input data for the CNMSTYLE report generator	DSIPARM
CNMETSO	same	TSO command server CLIST	CNMSAMP
CNMGNETV	same	JPEG NetView logo	SDSIOPEN
CNMGSPCR	same	JPEG - graphic spacer	SDSIOPEN
CNMGTVL	same	JPEG - Tivoli logo	SDSIOPEN
CNMHELPF	same	Map for NetView help panels	DSIPARM
CNMIPMG	same	Automation policy for NetView IP management	DSIPARM
CNMJSCL	same	Java language "count lines" example	CNMSAMP
CNMJSHW	same	Java language "Hello World" example	CNMSAMP
CNMKEYS2	same	Alternate programmable key definitions for NetView applications	SDSIOPEN
CNMKEYS	same	Programmable key definitions for NetView applications	SDSIOPEN

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMMIGKA	same	Programmable key definitions for NetView applications	SDSIOPEN
CNMMIGKV	same	Programmable key definitions for View applications	SDSIOPEN
CNMMSENU	same	NetView date/time message translation sample	SDSIMSG1
CNMMSJPN	same	Japanese only: Defines Japanese message translations to the NetView command facility.	SDSIMSG1
CNMNEWS	same	Customizable sample containing text that can be displayed to operators during log on	SDSIOPEN
CNMPOLCY	same	Automation policy for NetView	DSIPARM
CNMS0001	AMODETAB	Logon mode table for the sample network	VTAMLST
CNMS0003	ATCCON01	Configuration start list for host subarea 01	VTAMLST
CNMS0004	ATCCON02	Configuration start list for host subarea 02	VTAMLST
CNMS0005	ATCCON99	Configuration start list for host subarea 99	VTAMLST
CNMS0006	ATCCON00	Default configuration start list	VTAMLST
CNMS0007	ATCSTR01	Start options for host subarea 01	VTAMLST
CNMS0008	ATCSTR02	Start options for host subarea 02	VTAMLST
CNMS0009	ATCSTR99	Start options for host subarea 99	VTAMLST
CNMS0010	ATCSTR00	VTAM start options	VTAMLST
CNMS0011	AUSSTAB	VTAM USS table	VTAMLST
CNMS0012	A01ADJ	Adjacent SSCP table	VTAMLST
CNMS0013	A01APPLS	NetView APPL definitions for subarea 01	VTAMLST
CNMS0014	A01CDRM	Cross domain resource manager	VTAMLST
CNMS0015	A01CDRSC	Cross domain resource definitions	VTAMLST
CNMS0016	A01LOCAL	Non-SNA channel attached terminals	VTAMLST
CNMS0018	A01PATH	Path definitions for host A01MPU	VTAMLST
CNMS0019	A02ADJ	Adjacent SSCP table	VTAMLST
CNMS0020	A02APPL	NetView APPL definitions for subarea 02	VTAMLST
CNMS0021	A02CDRM	Cross domain resource manager	VTAMLST
CNMS0022	A02CDRSC	Cross domain resource definitions	VTAMLST
CNMS0023	A02LOCAL	Non-SNA channel attached terminals	VTAMLST
CNMS0025	A02PATH	Sample VTAM list - path definition for A02MPU	VTAMLST
CNMS0030	A99ADJ	Adjacent SSCP table	VTAMLST
CNMS0031	A99APPL	NetView APPL definitions for subarea 99	VTAMLST
CNMS0032	A99CDRM	Cross domain resource manager	VTAMLST
CNMS0033	A99CDRSC	Cross domain resource definitions	VTAMLST

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMS0034	A99LOCAL	Non-SNA channel attached terminals	VTAMLST
CNMS0036	A99PATH	Sample VTAM list - path definition for A99MPU	VTAMLST
CNMS0038	CTCA0102	Sample VTAM list - CTC definition for A01MPU	VTAMLST
CNMS0039	CTCA0201	Sample VTAM list - CTC definition for A02MPU	VTAMLST
CNMS0040	ISTSDCOS	Sample VTAM list - class of service for all networks	VTAMLST
CNMS0041	A01USER	APPL definitions for host A01MPU	VTAMLST
CNMS0042	A02USER	APPL definitions for host A02MPU	VTAMLST
CNMS0043	A99USER	APPL definitions for host A99MPU	VTAMLST
CNMS0047	A01MVS	MVS unique APPL definitions	VTAMLST
CNMS0048	A02MVS	MVS unique APPL definitions	VTAMLST
CNMS0049	A99MVS	MVS unique APPL definitions	VTAMLST
CNMS0050	INTERCOS	Sample VTAM list - log on mode table for NETA	VTAMLST
CNMS0055	same	Assembles and link-edits the NetView constants module (DSICTMOD)	CNMSAMP
CNMS0061	A04NTRI	Dial deck for NTRI lines	VTAMLST
CNMS0065	A04A54C	NCP generation definition	VTAMLST
CNMS0066	A31A54C	NCP generation definition	VTAMLST
CNMS0073	A01SNA	SNA channel attached terminals	VTAMLST
CNMS0074	A02SNA	SNA channel attached terminals	VTAMLST
CNMS0075	A99SNA	SNA channel attached terminals	VTAMLST
CNMS0081	CTNA0104	NCP channel attachment major node for A01MPU	VTAMLST
CNMS0082	CTNA0204	NCP channel attachment major node for A02MPU	VTAMLST
CNMS0083	CTNA9931	NCP channel attachment major node for A99MPU	VTAMLST
CNMS0084	CNMCON01	Lists major nodes and major nodes listed in ATCCON01.	VTAMLST
CNMS0085	CNMCON02	Lists major nodes and major nodes listed in ATCCON02.	VTAMLST
CNMS0086	CNMCON99	Lists major nodes and major nodes listed in ATCCON99.	VTAMLST
CNMS0087	A01NVAS	NetView Access Services APPL statements	VTAMLST
CNMS0088	A01CDRM1	Cross domain resource manager	VTAMLST
CNMS0089	A01CDRS1	Independent LU CDRSC major node for A01MPU	VTAMLST
CNMS0090	A99CDRS1	Independent LU CDRSC major node for A99MPU	VTAMLST
CNMS0091	A02NVAS	NetView Access Services APPL statements	VTAMLST
CNMS0092	A99NVAS	NetView Access Services APPL statements	VTAMLST
CNMS0101	ATCSTRB1	VTAM start options for host B01MPU	VTAMLST

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMS0102	ATCCONB1	Configuration start list for host B01MPU	VTAMLST
CNMS0103	CNMCONB1	Lists major nodes in B01MPU and major nodes listed in ATCCON01.	VTAMLST
CNMS0104	B01ADJ	Adjacent SSCP table for host B01MPU	VTAMLST
CNMS0105	B01PATH	Path definitions for host B01MPU	VTAMLST
CNMS0106	B01CDRM	CDRM for host B01MPU	VTAMLST
CNMS0107	B01CDRSC	CDRSC major node for host B01MPU	VTAMLST
CNMS0108	B01LOCAL	Non-SNA channel attached terminals for B01MPU	VTAMLST
CNMS0109	B30A54C	NCP generation definitions for host B01MPU	VTAMLST
CNMS0110	B30NTRI	Dial deck for B30A54C NTRI lines	VTAMLST
CNMS0111	B01MVS	MVS APPL definitions	VTAMLST
CNMS0113	B01APPLS	NetView APPL definitions for host B01MPU	VTAMLST
CNMS0115	CTNB0130	NCP channel attachment major node for host B01MPU	VTAMLST
CNMS1048	HELPMAP	Correlates arguments from the help command to the primary help panel	DSIPARM
CNMS1084	HELPMAPU	Correlates arguments from the help command to the primary help panel for user-defined commands	DSIPARM
CNMS1097	TSTCSCF	Full-screen automation sample that captures event log and response time log for hardware group 1	CNMSAMP
CNMS1098	TSOSDA	Full-screen automation example of a TAF session	CNMSAMP
CNMS1101	same	Contains PIPE samples included in <i>IBM Z NetView Programming: Pipes</i> .	CNMSAMP
CNMS1A01	DSIELFCB	Assembler language program that you can use for external logging. DSIELFCB returns information about the output file being used for logging. Use this sample with DSIELLR, DSIELMEM, and DSIELXIT.	CNMSAMP
CNMS1A02	DSIELLR	Assembler language program that you can use for external logging. DSIELLR contains a sample logoff routine. Use this sample with DSIELFCB, DSIELMEM, and DSIELXIT.	CNMSAMP
CNMS1A03	DSIELXIT	Sample installation exit that you can use to log data in to a file. You can use this program whenever the system management facilities (SMF) are not being used.	CNMSAMP
CNMS1A04	DSISDMA	Message string definitions	CNMSAMP
CNMS2VSM	same	Creates sample database for the session monitor	CNMSAMP
CNMS3VSM	same	Creates sample database for the hardware monitor	CNMSAMP
CNMS4200	PTMPPLT	HLL PL/I template	CNMSAMP

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMS4201	CTMPPLT	HLL C template	CNMSAMP
CNMS4202	ATMPCMDP	ATMPCMDP control section	CNMSAMP
CNMS4210	PEXIT3	HLL PL/I DSIEX03 example	CNMSAMP
CNMS4211	PSNDDAT	HLL PL/I send data example	CNMSAMP
CNMS4212	PWATDAT	HLL PL/I wait for data example	CNMSAMP
CNMS4213	PEXIT2A	HLL PL/I DSIEX02A example	CNMSAMP
CNMS4214	PCNMI	HLL PL/I CNMI example	CNMSAMP
CNMS4215	PKEYIO	HLL PL/I VSAM example	CNMSAMP
CNMS4216	PSCOPCK	HLL PL/I example of command authorization	CNMSAMP
CNMS4217	PFLVIEW	HLL PL/I view example	CNMSAMP
CNMS4218	PACTLU	HLL PL/I wait/trap example	CNMSAMP
CNMS4219	PSEQLOG	HLL PL/I sequential logging example	CNMSAMP
CNMS4220	PXITDI	DST initialization exit for USERVSAM sample DST	CNMSAMP
CNMS4221	PXITVN	DST empty VSAM data exit for sample USERVSAM DST	CNMSAMP
CNMS4222	PSNDDST	Sends VSAM requests to sample USERVSAM DST	CNMSAMP
CNMS4223	PDOVSAM	Processes VSAM requests under sample USERVSAM DST sent by OPRSAM command processor	CNMSAMP
CNMS4224	PPRIME	HLL PL/I DSIXITVN example	CNMSAMP
CNMS4226	PHSNDMU	PL/I sample that shows how to use CNMHSEND in a PL/I program	CNMSAMP
CNMS4227	same	PL/I sample that sends generic alert through program-to-program interface	CNMSAMP
CNMS4228	same	PL/I sample that sends data buffer through program-to-program interface	CNMSAMP
CNMS4229	same	PL/I sample that receives data buffer through program-to-program interface	CNMSAMP
CNMS4230	PRODMCON	Begins the HLL PL/I CNMQAPI member connection to RODM	CNMSAMP
CNMS4231	PAUTOTB	HLL PL/I example	CNMSAMP
CNMS4232	PREGISTR	PL/I sample for CNMRGS	CNMSAMP
CNMS4233	PSENDMU	PL/I sample for CNMSMU	CNMSAMP
CNMS4236	PHREGSTR	PL/I sample for CNMHRGS	CNMSAMP
CNMS4239	PPRSMDB	PL/I sample for CNMPMDB	CNMSAMP
CNMS4240	CEXIT3	HLL C DSIEX03 example	CNMSAMP
CNMS4241	CSNDDAT	HLL C send data example	CNMSAMP

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMS4242	CWATDAT	HLL C wait for data example	CNMSAMP
CNMS4243	CEXIT2A	HLL C DSIEX02A example	CNMSAMP
CNMS4244	CCNMI	HLL C CNMI example	CNMSAMP
CNMS4245	CKEYIO	HLL C VSAM example	CNMSAMP
CNMS4246	CSCOPCK	HLL C example of command authorization	CNMSAMP
CNMS4247	CFLVIEW	HLL C view example	CNMSAMP
CNMS4248	CACTLU	HLL C wait/trap example	CNMSAMP
CNMS4249	CSEQLOG	HLL C sequential logging example	CNMSAMP
CNMS4250	CXITDI	DST initialization exit for USERVSAM sample DST	CNMSAMP
CNMS4251	CXITVN	DST empty VSAM data exit for sample USERVSAM DST	CNMSAMP
CNMS4252	CSNDDST	Sends VSAM requests to sample USERVSAM DST	CNMSAMP
CNMS4253	CDOVSAM	Processes VSAM requests under the USERVSAM DST sent by the CSNDDST command processor	CNMSAMP
CNMS4254	CPRIME	HLL C DSIXITVN sample	CNMSAMP
CNMS4256	CHSNDMU	C sample that shows how to use CNMHSEND in a C program	CNMSAMP
CNMS4257	same	HLL C sample that sends generic alert through program-to-program interface	CNMSAMP
CNMS4260	CRODMCON	Begins the HLL C CNMQAPI member connection to RODM	CNMSAMP
CNMS4261	CAUTOTB	HLL C sample	CNMSAMP
CNMS4262	CREGISTR	C sample for CNMRGS	CNMSAMP
CNMS4263	CSENDMU	C sample for CNMSMU	CNMSAMP
CNMS4266	CHREGSTR	C sample for CNMHRGS	CNMSAMP
CNMS4269	CPRSMDBR	C sample for CNMPMDB	CNMSAMP
CNMS4270	AXITVN	Provides initial record for empty VSAM database	CNMSAMP
CNMS4271	AMSGMOD	Issues DSIMDS to build a user-defined message module	CNMSAMP
CNMS4272	AWRTLOG	Shows how to write a message to the NetView log using DSIWLS	CNMSAMP
CNMS4273	AMLWTO	Demonstrates title line output using DSIPSS type=output	CNMSAMP
CNMS4274	ADATTIM	Gets current data and time and displays message	CNMSAMP
CNMS4275	ASEQLOG	Logs text to a sequential log	CNMSAMP
CNMS4276	ALISTMEM	Reads and displays a member from NetView DSIPARM data set	CNMSAMP

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMS4277	AOPTTSK	User-defined optional subtask	CNMSAMP
CNMS4278	ABLDMSG	Shows how DSIMBS can be used to build user-defined messages	CNMSAMP
CNMS4279	APSSFULL	Uses DSIPSS TYPE=ASYPANEL to display full screen panel	CNMSAMP
CNMS4280	ACALLCMD	Calls another command processor (or command list) directly	CNMSAMP
CNMS4281	DSIUSR00	Sample user-defined message member	CNMSAMP
CNMS4282	ATMPUXIT	ATMPUXIT control section	CNMSAMP
CNMS4283	DSIEX02A	Installation exit called for standard output to the operator terminal	CNMSAMP
CNMS4284	ALERTMSG	Generates automation message for NMVT (Network Management Vector Transport) Alerts	CNMSAMP
CNMS4286	AHSNDMU	Assembler sample for DSIHSNDS	CNMSAMP
CNMS4287	CNMSGENA	Assembler sample that sends generic alert through program-to-program interface	CNMSAMP
CNMS4288	CNMSEND	Assembler sample that sends the data buffer through program-to-program interface	CNMSAMP
CNMS4289	CNMRECV	Assembler sample that receives the data buffer through program-to-program interface	CNMSAMP
CNMS4290	ARODMCON	Activates the assembler DSINOR that connects to RODM	CNMSAMP
CNMS4291	AAUTOTB	Tests automation table statements by sending an MSU directly to the automation table	CNMSAMP
CNMS4292	AREGISTR	Assembler sample for DSI6REGS	CNMSAMP
CNMS4293	ASENDMU	Assembler sample for DSI6SNDS	CNMSAMP
CNMS4294	AGETDS	Assembler sample for DSIGETDS	CNMSAMP
CNMS4295	OPERID	Example of an Automation Table Function (ATF)	CNMSAMP
CNMS4296	AHREGSTR	Assembler sample for DSIHREGS	CNMSAMP
CNMS4297	DSIEX17	Installation exit called for MVS messages and delete operator messages (DOM)	CNMSAMP
CNMS4298	DSIEX18	Installation exit that can be used as a template	CNMSAMP
CNMS4299	APRSMDB	Assembler sample which builds a message data block (MDB) and a source object and calls the process MDB service	CNMSAMP
CNMS4305	PACTPIP	HLL PL/I PIPE command example	CNMSAMP
CNMS4307	DSIEX19	Installation exit that provides command authority checking for the RUNCMD command	CNMSAMP
CNMS4308	DSIEX20	Installation exit that allows filtering of session awareness (SAW) data	CNMSAMP

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMS4402	CNMSNIFF	GMFHS automation example	CNMSAMP
CNMS4403	EKGSNIFF	GMFHS automation example	CNMSAMP
CNMS4405	CACTPIP	HLL C PIPE command example	CNMSAMP
CNMS4406	same	C sample that sends an INIT or DOWN alert to complete the DOMP010/DOMS010 protocols for service points that do not send an adequate alert themselves.	CNMSAMP
CNMS4501	same	Assembler IPCS installation exit for PPI trace records.	CNMSAMP
CNMS4508	TESTQRSR	Command procedure that returns a list of active spans and resources that can be controlled	CNMSAMP
CNMS6201	MPFLSTAC	Conservative MVS MPF message suppression	CNMSAMP
CNMS6202	MPFLSTAA	Aggressive MVS MPF message suppression	CNMSAMP
CNMS6205	ACOTABLE	NetView automation table entries	CNMSAMP
CNMS6206	same	Command definition statements for system automation command lists	CNMSAMP
CNMS6207	same	JES2 and JES3 system log analysis program	CNMSAMP
CNMS6211	CLRLOG	Clears SYS1.LOGREC for future recording	CNMSAMP
CNMS6212	CLRSMF	Clears SYS1.MANX for future recording	CNMSAMP
CNMS6213	LGPRNT	Prints SYS1.LOGREC	CNMSAMP
CNMS6214	DSIPRT	Prints the primary and secondary NetView logs	CNMSAMP
CNMS6221	\$CLRSMF	Input to the CLRSMF procedure	CNMSAMP
CNMS6222	\$SOFT	Input to the LGPRNT procedure, step name SOFT	CNMSAMP
CNMS6223	\$SYSEXN	Input to the LGPRNT procedure, step name SYSEXN	CNMSAMP
CNMS6224	\$SYSUM	Input to the LGPRNT procedure, step name SYSUM	CNMSAMP
CNMS62J1	same	Rename JCL	CNMSAMP
CNMS62J2	RDSIPARM	SYSLOG analysis program	CNMSAMP
CNMS6401	same	Command definition statements for MVS commands	CNMSAMP
CNMS6402	same	Command definition statements for JES2 commands	CNMSAMP
CNMS6403	same	Command definition statements for JES3 commands	CNMSAMP
CNMS6404	same	Command definition statements for advanced starter set command lists	CNMSAMP
CNMS6405	DSITBL11	NetView automation table	CNMSAMP
CNMS6406	AOPUMCMT	TSO command list to copy command lists without comments	CNMSAMP

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMS6408	same	Automated operator definitions	CNMSAMP
CNMS6409	DSIPROFM	AUTOMGR AUTOTASK profile	CNMSAMP
CNMS6410	DSIPROFG	Generic AUTOTASK profile	CNMSAMP
CNMS64P0	same	Panel to display automation status of all products	CNMSAMP
CNMS64P1	same	Panel to display automation information for a specific product	CNMSAMP
CNMS64P2	same	Panel to display message response variable values	CNMSAMP
CNMS64P3	same	Help panel for panel CNMS64P0	CNMSAMP
CNMS64P4	same	Help panel for panel CNMS64P1	CNMSAMP
CNMS64P5	same	Help panel for panel CNMS64P2	CNMSAMP
CNMS7030	same	Retrieves focal point definitions for primary and backup names and stores in global variables	CNMSAMP
CNMS7101	same	Alias table definitions for default information problem records	CNMSAMP
CNMS7102	same	Alias table definitions for default information configuration records	CNMSAMP
CNMS7103	same	Alias table definitions for Network Configuration Application (NCA) configuration records	CNMSAMP
CNMS7104	same	Program interface data table (PIDT) definitions for NCA inquiry transactions	CNMSAMP
CNMS7105	same	PIDT definitions for retrieve transactions for NCA SNA records	CNMSAMP
CNMS7106	same	PIDT definitions for retrieve transactions for NCA circuit records	CNMSAMP
CNMS7107	same	PIDT definitions for retrieve transactions for NCA equipment records	CNMSAMP
CNMS7108	same	PIDT definitions for retrieve transactions for NCA software records	CNMSAMP
CNMS7109	same	Alias table definitions supporting SNA topology manager problem records	CNMSAMP
CNMS7110	same	Alias table definitions supporting SNA topology manager configuration records	CNMSAMP
CNMS7111	same	Alias table definitions supporting SNA topology manager configuration records	CNMSAMP
CNMS7112	same	Alias table definitions supporting SNA topology manager configuration records	CNMSAMP
CNMS8002	RXUFUNC	Demonstrates how to return a value to the REXX caller	CNMSAMP
CNMS8003	AUTODROP	Conditionally drops preloaded NetView command lists	CNMSAMP

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMS8013	DELVSAM	Sample REXX procedure that deletes a VSAM data set	CNMSAMP
CNMS8014	OPENVSAM	Sample REXX procedure that creates or opens a VSAM data set	CNMSAMP
CNMS8015	INITREC	Sample REXX procedure that writes the first record in a VSAM data set	CNMSAMP
CNMS8016	PUTREC	Sample REXX procedure (DSIVSMX) that writes records to VSAM data sets	CNMSAMP
CNMS8017	VSAMGLOB	Sample REXX procedure that uses VSAM for permanent task global	CNMSAMP
CNMS8018	BULKPUT	Sample REXX procedure (DSIVSMX) that writes multiple records to VSAM data sets	CNMSAMP
CNMS8019	STARVSAM	Sample REXX procedure that starts sample VSAM tasks	CNMSAMP
CNMS8020	DEFVSAMS	Sample REXX procedure (IDCAMS) that define VSAM data sets used by VSAM tasks.	CNMSAMP
CNMS8021	SAMPBULK	Sample REXX procedure (DSIVSAM) that writes multiple records to VSAM data sets	CNMSAMP
CNMS8022	same	NetView sample - SAMPVSAM task initialization member	CNMSAMP
CNMS8023	SENDTSO	Sends TSO commands using a TAF LU1 session and produces correlated output	CNMSAMP
CNMS8024	same	Processes TSO commands and returns output that can be processed by NetView pipelines	CNMSAMP
CNMS8027	CSCFIML	NetView sample to IML a 3174 controller	CNMSAMP
CNMS8029	same	Sample to interact with NetView through PPI to issue console commands	CNMSAMP
CNMS8050	ZAIGET	Connects to IBM System z® Advanced Workload Analysis Reporter (IBM zAware) and queries the IBM zAware server for data	CNMSAMP
CNMS8051	ZAIPROC	Defines the IBM zAware query and calls the ZAIGET sample	CNMSAMP
CNMS8052	ZAITIMER	Issues the ZAIPROC sample periodically	CNMSAMP
CNMS8053	same	Saves long global variables	CNMSAMP
CNMS8054	same	Restores long global variables	CNMSAMP
CNMSAF2	same	Sets RACF definitions for NetView operators and commands	CNMSAMP
CNMSALRT	same	Converts SNMP trap data to an alert and sends the alert to a NetView alert receiver	CNMSAMP
CNMSBAK1	same	Backup command authorization table	DSIPARM
CNMSCAT2	same	Sample command authorization table	DSIPARM

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMSCATU	same	User-defined command authorization table	DSIPARM
CNMSCSDP	same	The Command Statistics Data Processor. It will process command statistics data that is sent to it by the CMDMON CAPTURE command, or periodically by the NetView program.	CNMSAMP
CNMSCSFM	same	The Command Statistics Formatter. It will format command statistics data that is sent to it by the CMDMON DISPLAY command.	CNMSAMP
CNMSCSIE	same	The Command Statistics Include/Exclude list. Use this sample to specify which commands will be monitored for command statistics.	CNMSAMP
CNMSCSSU	same	The Command Statistics Summarizer. It will collect command statistics data from CMDMON DISPLAY and aggregate the data based on each command.	CNMSAMP
CNMSCM	same	SNMP community names for TCP/IP stacks	DSIPARM
CNMSCMSG	same	Sample REXX procedure that displays a count of all Canzlog messages matching a specified Basic Filter Syntax (BFS) filter.	CNMSAMP
CNMSCNFT	same	Sample that defines screen colors, prefix data, and prefix display order for message formatting	DSIPARM
CNMSCRT1	same	Command revision table	DSIPARM
CNMSDCA	same	Provides automation statements that help control the data collection autotasks that are managed by the COLLECTL command.	DSIPARM
CNMSDDCR	same	Displays distributed DVIPA connection routing information. It formats the DVIPDDCR command output (BNH815I message) into a readable format.	CNMSAMP
CNMSDVCG	same	Automation table member for z/OS Communications Server VIPADYNAMIC TCP/IP profile updates that is included when the DVIPA tower is enabled. The VIPADYNAMIC TCP/IP profile statements that are changed trigger rediscovery for DVIPA functions on this z/OS system.	DSIPARM
CNMSDVDS	same	Automation table member that is used for DVIPA data discovery (including rediscovery) and distributed DVIPA statistics on the master NetView program. It processes the DSIIIF003I (DVIPA rediscovery), DSIIIF004I (DVIPA data forwarding), DSIIIF006I (DVIPA data request), and BNH867I (distributed DVIPA statistical records) messages.	DSIPARM
CNMSDVEV	same	Automation table member for DVIPA SMF runtime updates that is included when the DVIPA tower is enabled. The automation detects that a z/OS Communications Server SMF DVIPA update was received, which then triggers rediscovery for DVIPA functions on this z/OS system.	DSIPARM

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMSDVIP	same	This sample displays the DVIPA definition and status data. It formats the DVIPSTAT command output (BNH846I message) into a readable format.	CNMSAMP
CNMSDVPC	same	Displays DVIPA connections. It formats the DVIPCONN command output (BNH849I message) into a readable format.	CNMSAMP
CNMSDVPH	same	Displays distributed DVIPA server health information. It formats the DVIPHLTH command output (BNH814I message) into a readable format.	CNMSAMP
CNMSDVST	same	Displays the data in the distributed DVIPA statistical data sets.	CNMSAMP
CNMSDVTP	same	Automation table member for DVIPA SNMP trap automation that is included when the DVIPA tower is enabled. The automation detects that a z/OS Communications Server DVIPA trap was received, which then triggers rediscovery for DVIPA functions on this z/OS system.	DSIPARM
CNMSEMAA	same	Automation table member for the NetView agent	DSIPARM
CNMSEPTL	same	Automation table sample for resource discovery	DSIPARM
CNMSFMSG	same	Sample REXX procedure that finds and displays a single Canzlog message that matches a specified Basic Filter Syntax (BFS) filter.	CNMSAMP
CNMSHIPR	same	Displays HiperSockets configuration and status information. It formats the HIPERSOC command output (BNH598I message) into a readable format.	CNMSAMP
CNMSHMAT	same	Automation sample for hardware monitor instrumentation	DSIPARM
CNMSI101	same	Allocates VSAM databases for the hardware monitor, network logs, save/restore function, session monitor, and TCP/IP connection management	CNMSAMP
CNMSI501	same	Allocates CSCF VSAM databases of sample network	CNMSAMP
CNMSID01	same	Deletes VSAM databases	CNMSAMP
CNMSIFST	same	Displays IP interfaces. It formats the IFSTAT command output (BNH498I message) into a readable format.	CNMSAMP
CNMSIHSA	same	Provides examples of automation table statements for forwarding alerts and messages from the NetView program to alert and message adapters in the Event/Automation Service. The adapters in the Event/Automation Service can create Event Integration Facility (EIF) events or SNMP traps.	DSIPARM
CNMSJ000	same	Changes samples to reference user-defined subarea and domain	CNMSAMP

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMSJ001	same	Defines user ICF catalog and alias	CNMSAMP
CNMSJ002	same	Allocates partitioned data sets	CNMSAMP
CNMSJ003	same	Compresses and copies partitioned data sets	CNMSAMP
CNMSJ004	same	Allocates logs and databases	CNMSAMP
CNMSJ005	same	REPRO sample databases	CNMSAMP
CNMSJ006	same	Assembles and link-edits VTAM tables	CNMSAMP
CNMSJ007	CNMNDEF	Status monitor preprocessor procedure that starts the status monitor preprocessor and creates the DSINDEF member	PROCLIB
CNMSJ008	CNMNET	VTAM start procedure	PROCLIB
CNMSJ009	CNMPROC	NetView application address space start procedure	PROCLIB
CNMSJ010	CNMPSSI	NetView subsystem address space start procedure	PROCLIB
CNMSJ023	same	Reproduces VSAM file for verification	CNMSAMP
CNMSJ032	same	Creates /etc/netview and /tmp/netview directories and copies /usr/lpp/tcpip/samples mibs to /etc/netview/mibs	CNMSAMP
CNMSJ033	same	Loads the initialization record into the Saved Packet Trace database	CNMSAMP
CNMSJBUP	same	Creates the installation JCL samples library	CNMSAMP
CNMSJCRG	same	CNMSTYLE report generator sample job	CNMSAMP
CNMSJH12	same	Loads the RODM data cache for GMFHS	CNMSAMP
CNMSJH13	same	Relink-edits the assembler table, DUIFSMT	CNMSAMP
CNMSJI01	same	Copy statement	CNMSAMP
CNMSJI02	same	Copy statement	CNMSAMP
CNMSJI03	same	IEBCOPY control statements to copy VTAM samples to VTAMLST	CNMSAMP
CNMSJI10	same	IEBCOPY control statements to copy NetView sample procedures to PROCLIB	CNMSAMP
CNMSJM01	same	VSAM LSR buffer definition	CNMSAMP
CNMSJM04	CNMPRT	Prints network logs	PROCLIB
CNMSJM10	same	Generates a list of sense codes and frequency of occurrence	CNMSAMP
CNMSJM11	same	Assembles and link-edits the NetView REXX parameters module (DSIRXPRM)	CNMSAMP
CNMSJM12	same	Symbol substitution utility	CNMSAMP
CNMSJM13	same	DATA REXX initialization parameters	CNMSAMP
CNMSJM14	same	Reorganizes the Saved Packet Trace database	CNMSAMP

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMSJM15	same	Migrates the save/restore (DSISVRT) VSAM cluster	CNMSAMP
CNMSJMIG	same	CNMSTYLE migration tool sample job	CNMSAMP
CNMSJSQL	same	SQL plan installation sample job	CNMSAMP
CNMSJTLS	same	Enables and configures AT-TLS for secure communications between the NetView program and either the NetView management console, NetView Web Services Gateway or both.	CNMSAMP
CNMSJTZO	same	TSO command server sample job	CNMSAMP
CNMSJUNX	same	UNIX command server sample job	CNMSAMP
CNMSJV03	same	Reorganizes NLDM CNM.SA01 database	CNMSAMP
CNMSJV04	same	Reorganizes NPDA CNM.SA01 database	CNMSAMP
CNMSJZCE	same	Sample start job for the event correlation engine	CNMSAMP
CNMSMF3A	same	Sample command list that is called by the automation table when the BNH874I message is issued. This sample is used for SMF record type 30 automation.	CNMSAMP
CNMSMF3E	same	Sample IEFACTRT SMF exit to process type 30 SMF records and send them across the program-to-program interface (PPI) to the NetView program for automation	CNMSAMP
CNMSMF3F	same	Sample that formats the BNH874I message. This sample is used for SMF record type 30 automation.	CNMSAMP
CNMSMF3R	same	Sample PPI receiver that formats the BNH874I message. This sample is used for SMF record type 30 automation.	CNMSAMP
CNMSMRT1	same	Message revision table	DSIPARM
CNMSMSGT	same	Builds and sends an SNMPv1 or SNMPv2c trap	CNMSAMP
CNMSMSIP	same	Sends a CP-MSU over IP	CNMSAMP
CNMSNVST	same	Displays NetView application information. It formats the NVSTAT command output (BNH495I message) into a readable format.	CNMSAMP
CNMSOSAP	same	Displays the OSA channel and port information. It formats the OSAPORT command output (BNH597I message) into a readable format.	CNMSAMP
CNMSPAN2	same	Sample NetView span table	DSIPARM
CNMSPLEX	same	This sample displays the DVIPA sysplex distributor data. It formats the DVIPPLEX command output (BNH847I message) into a readable format.	CNMSAMP
CNMSRPLY	same	Multiple reply support for a single MVS command	CNMSAMP
CNMSRVAR	same	This sample shows how to manipulate the revision variable table that exists in the NetView SSI and is loaded or queried using the SETRVAR command.	CNMSAMP

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMSRVMC	same	This sample provides an example of the NETVONLY action command that is used in the command revision table.	CNMSAMP
CNMSSMON	same	This sample provides z/OS Communications Server sysplex monitoring message automation and is included when the DVIPA tower is enabled.	DSIPARM
CNMSSTAC	same	This sample displays the TCP/IP stack configuration and status data. It formats the STACSTAT command output (BNH845I message) into a readable format.	CNMSAMP
CNMSSTSO	same	Sample JCL to run the NetView TSO command server as an MVS started job	CNMSAMP
CNMSSUNX	same	Sample JCL to run the NetView UNIX command server as an MVS started job	CNMSAMP
CNMSTARG	same	This sample displays the distributed DVIPA target data. It formats the DVIPTARG command output (BNH813I message) into a readable format.	CNMSAMP
CNMSTASK	same	Task statements that are provided with the NetView program	DSIPARM
CNMSTCPC	same	Formats TCPCONN output. This sample calls the WINDOW command. Active connection information displayed by this sample (for example, from the TCPCONN QUERY command) can be missing fields that are available for inactive connections.	CNMSAMP
CNMSTDAT	same	Topology Display Instrumentation automation table sample	DSIPARM
CNMSTGEN	same	You can include additional or modified system-specific CNMSTYLE definition statements, including DATA REXX logic.	DSIPARM
CNMSTIDS	same	Includes Intrusion Detection Services (IDS) initialization statements.	DSIPARM
CNMSTMSG	same	Sample REXX procedure that finds and displays a single Canzlog message that matches a specified Basic Filter Syntax (BFS), overriding the timeout defined by the DEFAULTS CZBRWAIT command.	CNMSAMP
CNMSTNST	same	This sample displays the configuration and status information about Telnet servers. It formats the TELNSTAT command output (BNH496I message) into a readable format.	CNMSAMP
CNMSTNXT	same	Includes modifiable CNMSTYLE statements by release. CNMSTNXT is commented out in the CNMSTYLE member. It is provided for documentation purposes only.	DSIPARM

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
CNMSTPST	same	This sample displays the configuration and status information about Telnet server ports. It formats the TNPTSTAT command output (BNH497I message) into a readable format.	CNMSAMP
CNMSTPWD	same	Includes VSAM and ACB passwords.	DSIPARM
CNMSTSOS	same	MVS start command sample to start the NetView TSO command server as a started task	DSIPARM
CNMSTTWR	same	Includes CNMSTYLE statements from non - NetView towers.	DSIPARM
CNMSTUSR	same	Customization member for the CNMSTYLE member; include additional or modified global (enterprise) definition statements that override statements in the CNMSTYLE member.	DSIPARM
CNMSTWBM	same	Includes webmenu statements	DSIPARM
CNMSTYLE	same	Defines some of the NetView initialization parameters.	DSIPARM
CNMSUNXS	same	MVS start command sample to start the NetView UNIX command server as a started task	DSIPARM
CNMSVPRT	same	Displays status information about VIPA routes. It formats the VIPAROUT command output (BNH824I message) into a readable format.	CNMSAMP
CNMSVTET	same	VTAM monitor auto-table: message suppression	DSIPARM
CNMSVTFT	same	VTAM monitor auto-table entries	DSIPARM
CNMSXCFA	same	Automation sample for XCF sysplex support	DSIPARM
CNMSXENT	same	Sample used to show that NetView is installed	CNMSAMP
CNMSXSYS	same	Sample used to show that NetView System Services is installed	CNMSAMP
CNMSZERT	same	Formats TCPCONN output. This sample calls the WINDOW command and displays Connection encryption information from the CONNSEC QUERY command.	CNMSAMP
CNMTRAPI	same	Contains the data services task initialization parameters required for an SNMP trap automation task	DSIPARM
CNMTRMSG	same	NetView message translation sample	SDSIMSG1
CNMTRUSR	same	Message translation sample for user-defined message translations	SDSIMSG1
CNMTRXMP	same	Message translation sample for examples of message translations	SDSIMSG1
DSI6INIT	same	Prototype of the LU 6.2 transport initialization member	DSIPARM

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
DSI6SCF	same	Defines to the sphere of control manager which entry points to acquire into a focal point sphere of control.	DSIPARM
DSIALATD	same	Includes initialization parameters for alias name translation. Use alias name translation to translate the names of logical units, logon modes, and classes of service. You need this function only if you have naming conflicts between interconnected networks.	DSIPARM
DSIALTAB	same	Includes alias translation definitions. You can use this sample to define new names for a logical unit (LU), a class of service (COS), or a logon mode (MODE).	DSIPARM
DSIAMAT	same	Automation table for Application Management Instrumentation	DSIPARM
DSIAMIE	same	Application Management Instrumentation for the NetView program on a host with an Event/Automation Service message adapter	DSIPARM
DSIAMII	same	Application Management Instrumentation configuration file for initializing and stopping autotasks	DSIPARM
DSIAMIN	same	Application Management Instrumentation for the NetView program on a host not running the Event/Automation Service	DSIPARM
DSIAMIR	same	Application Management Instrumentation for NetView on a host without an Event/Automation Service message adapter	DSIPARM
DSIAMIT	same	Application Management Instrumentation for NetView on a host with an Event/Automation Service message adapter	DSIPARM
DSIAMLTD	same	Includes initialization parameters for the access method LU function	DSIPARM
DSIAPMI	same	APM autotask configuration file	DSIPARM
DSIAPML	same	APM instrumentation	DSIPARM
DSIAPMR	same	APM instrumentation	DSIPARM
DSIAUTB	same	Part list for usage of the AUTOBYPAS REXX or CLIST function	DSIPARM
DSIAUTBU	same	User-defined part list for AUTOBYPAS REXX or CLIST function	DSIPARM
DSIBKMEM	same	Defines the initialization values for the task DSIAL2WS. The DSIAL2WS task sends alerts to workstations running AIX® SNA Manager/6000.	DSIPARM
DSICCDEF	same	Defines conditions for stopping commands and is used with the CCDEF command.	DSIPARM

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
DSICCSYS	same	Defines conditions for stopping commands and is used with the CCDEF command for NetView System Services.	DSIPARM
DSICNM	same	Contains initialization parameters for the status monitor.	DSIPARM
DSICPINT	same	Contains definitions for the network product support communications network management interface (CNMI) function.	DSIPARM
DSICRTTD	same	CNM router initialization member	DSIPARM
DSIDB2DF	same	Sample initialization member for task DSIDB2MT	DSIPARM
DSIDMN	same	Provided for migration purposes. It includes DSIDMNU.	DSIPARM
DSIELMEM	same	Includes initialization parameters for task DSIELTSK	DSIPARM
DSIEX21	same	Assembler exit used with DSITCPRF for encryption	CNMSAMP
DSIGEMTR	same	Tivoli Ready for z/OS configuration sample	DSIPARM
DSIHINIT	same	Prototype of the LU 6.2 high performance definition statements	DSIPARM
DSIILGCF	same	Syslog task configuration	DSIPARM
DSIKINIT	same	Initializes values for the DSIKREM task of the command facility	DSIPARM
DSILOGBK	same	Includes initialization parameters for NetView disk log support	DSIPARM
DSILUCTD	same	Contains initialization parameters for the CNM data-transfer task for subarea 01 only (A01MPU in the sample network).	DSIPARM
DSINDEF	same	Contains the status monitor network node definition	DSIPARM
DSIOPF	same	Operator definitions and passwords	DSIPARM
DSIOPFAU	same	Operator definitions for existing AUTO1 and AUTO2 autotasks.	DSIPARM
DSIOPFEX	same	Example operator definitions and passwords that can be modified or deleted	DSIPARM
DSIOPFIX	same	Operator definitions that cannot be modified	DSIPARM
DSIOPFST	same	Operator definitions that can be modified	DSIPARM
DSIOPFU	same	Operator definitions and passwords for user-defined operators.	DSIPARM
DSIPRFRGR	same	Initializes the generic receiver task DSINVGR	DSIPRF
DSIPRFLC	same	Provides the DSILCOPR profile statements for an operator ID to run as an unattended operator	DSIPRF

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
DSIPROFA	same	Contains a profile for a network operator. An operator using this profile is not an authorized receiver and can enter class 2 commands only.	DSIPRF
DSIPROFB	same	Contains a profile for a network operator. An operator using this profile is an authorized receiver and can enter both class-1 and class 2 commands.	DSIPRF
DSIPROFC	same	Contains a profile for an automated operator (that is, an operator started with the AUTOTASK command to run as an unattended operator)	DSIPRF
DSIPROFD	same	Contains a profile for an automated operator (that is, an operator started with the AUTOTASK command to run as an unattended operator). This profile is used by unattended operators who want to run OVERRIDE CMD=LOW.	DSIPRF
DSIPROFE	same	Defines an automated operator profile for the NetView Bridge RTRINIT command This is a sample profile for an automated operator (that is, an operator started with the AUTOTASK command to run as an unattended operator). This profile is used by the BRIGOPER automated operator.	DSIPRF
DSIPROFF	same	Defines automated operator profile for the NetView Bridge REMOTE BR command This is a sample profile for an automated operator (that is, an operator started with the AUTOTASK command to run as an unattended operator). This profile is used by the REMOPER automated operator.	DSIPRF
DSIPROFG	same	Automated operator profile that is functionally equivalent to the DSIPROFD profile. It is provided for compatibility reasons.	DSIPRF
DSIPROFI	same	Automated operator profile for the event management autotask	DSIPRF
DSIPROFJ	same	Automated operator profile for the GMFHS common operations services (COS) gateway autotask	DSIPRF
DSIPROFN	same	Automated operator profile for the NetView agent data collection autotasks (AUTODCn)	DSIPRF
DSIPROFK	same	Automated operator profile for instrumentation autotask	DSIPRF
DSIPROFP	same	Automated operator profile for an autotask that has high CPU processor utilization and runs with SLOGCMDR=NO	DSIPRF

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
DSIQTSKI	same	Defines RODM and NetView tasks to the RODM access and control subtask (DSIQTSK) Refer to <i>IBM Z NetView Automation Guide</i> for an example of using the DSIQTSK task to manage your RODMs. The example illustrates using RODM to automate the recovery of a failed resource. For each RODM you specify in DSIQTSKI, you can define an initialization command processor to be called when the DSIQTSK subtask successfully connects to RODM. You can specify the name of this command processor within each RODM definition statement. DSIQTSKI also contains information about which RODMs you need to connect to.	DSIPARM
DSIREXCF	same	Rexec server task configuration	DSIPARM
DSIRHOST	same	RSH security file	DSIPARM
DSIROVSI	same	Defines the initialization values for the task DSIROVS	DSIPARM
DSIRSHCF	same	RSH server task configuration	DSIPARM
DSIRTTTD	same	TCP/IP alert receiver	DSIPARM
DSISCHED	same	CHRON command calendar schedule file	DSIPARM
DSISECUR	same	RMTCMD security table	DSIPARM
DSISVRTD	same	Initialization values for the save/restore (DSISVRT) task of the command facility	DSIPARM
DSITBL01	same	Automation definitions for subarea 01	DSIPARM
DSITCONM	same	Initialization values for DSITCONT	DSIPARM
DSITCPCF	same	Initialization values for DSITCPIP	DSIPARM
DSITCPRF	same	Defines the encryption keys for NETCONV sessions that use TCP/IP and for HTTP server sessions.	DSIPRF
DSITDSR	same	Application Management Instrumentation for NetView on a host not running the Event/ Automation Service	DSIPARM
DSITSK	same	Defines a list of tasks that are allowed for NetView System Services	DSIPARM
DSIUINIT	same	Defines the initialization values for the DSIUDST task. Values for this task are specified in the CNMSTYLE member.	DSIPARM
DSIVPARM	same	Initialization parameters for VPD task supporting Network Asset Management	DSIPARM
DSIWBMEM	same	Initialization parameters for the NetView Web Server	DSIPARM
DSIW3PRF	same	Properties definitions for 3270 Web sessions	DSIPARM

Table 20. List of Samples (continued)

Distributed As	Name	Description	Data Set Name
DSIZCECF	same	Initialization member for task DSICORSV	DSIPARM
DSIZCETB	same	Automation table sample for the event correlation engine	DSIPARM
EZLCMENT	same	NetView command definitions for base AON commands	DSIPARM
EZLSI300	same	IEBCOPY AON members to user DSIPARM data set	CNMSAMP
EZLSI301	same	IEBCOPY AON members to user CNMPNL1 data set	CNMSAMP
FKVCMENT	same	NetView command definitions for AON/SNA commands	DSIPARM
FKXCMENT	same	NetView command definitions for AON/TCP commands	DSIPARM
FKXIPMTB	same	Configuration data for NetView IP management automation	DSIPARM
FKXOPFIP	same	Operator definitions for NetView IP management automation	DSIPARM
FKXSI101	same	Allocates the VSAM Saved Packet Trace database	CNMSAMP
FKXSI201	same	REXX Exec that loads an initialization record into the Saved Packet Trace database	CNMSAMP
IHSAEVNT	same	Event/Automation Service startup procedure	CNMSAMP

Appendix D. Partitioned Data Sets and Members

This appendix lists some of the ddnames in the NetView start procedure CNMPROC and some of the data set members that the NetView program uses. The NetView program looks in the data sets specified on the DD statement for the given ddname for the members listed in [Table 21 on page 143](#).

Table 21. List of ddnames and Data Set Members

ddname	Member	Description
BNJPNL1	Various	Source panel definitions (hardware monitor)
BNJPNL2	Various	Color maps (hardware monitor)
CNMDVIPP	None	Primary data set for distributed DVIPA statistics
CNMDVIPS	None	Secondary data set for distributed DVIPA statistics
CNMPNL1	Various	Source panel definitions (NetView)
DSIARPT	Various	Stores the output reports produced from running tests of the automation table. A summary report and a detail report are produced.
DSIASRC	Various	Testing automation tables
DSICLD	Various	Command list definitions
DSILIST	Various	Automation table listings
DSIMSG	Various	Disk-resident messages
DSIOPEN	CNMKEYS	Key settings for components
	CNMKEY2	Plus any non-protected members
<i>User-defined</i>		
DSIPARM	AAUCNMTD	Session monitor CNMI details
	AAUKEEP1	Session monitor data wrap details
	AAUPRMLP	Session monitor initialization
	AAURTM1	Session monitor RTM classes
	CNMCMD	Command definitions
	CNMSTYLE	Initialization parameters
	DSIALATD	ALIAS initialization
	DSIALTAB	ALIAS name translations
	DSIAMLTD	Session monitor cross-domain details
	DSICNM	Status monitor initialization
	DSICPINT	NPS initialization
	DSICRTTD	CNM router task
	DSIELMEM	External log initialization (if no SMF)
	DSILOGBK	Network log initialization
	DSILUCTD	Session monitor data transfer task details

Table 21. List of ddnames and Data Set Members (continued)

ddname	Member	Description
	DSINDEF	Status monitor network database
	DSIOPF	Operator definitions
	DSISVRTD	Save/restore initialization
	DSITBL01	Sample automation table
	DSIVPARM	NAM initialization
	CNMBxxx	Sense code descriptors
	<i>User definitions</i>	User-defined members
DSIPRF	DSIPROFA	Sample operator profile
	DSIPROFB	Sample operator profile
	DSIPROFC	AUTOTASK profile (basic)
	DSIPROFD	AUTOTASK profile (MVS console)
	DSIPRFLC	Unattended operator profile
DSIVTAM	VTAM details	For span of control

Appendix E. Starting the NetView Program before VTAM or TCP/IP Network Services

If you start the NetView program before VTAM or TCP/IP network services, you receive error messages from enabled towers and other functions that rely on these network services. Note that these error messages might mask other error messages that are unrelated to the unavailability of the network services.

This appendix provides information for configuring the NetView program to minimize errors related to VTAM and TCP/IP network services during initialization. This information is followed by instructions for starting NetView components and related functions when the VTAM network services or the TCP/IP network services are available, so that the enabled towers and related functions are fully operational in your configuration.

The instructions for starting NetView program features are divided into two sets. The first set focuses on features of the NetView program that depend on other features, providing a list that begins with more atomic or independent features and progresses through other features that depend on them. Steps in the instructions about features that are not used can be skipped.

The second set shows how to start other NetView program features that depend on VTAM or TCP/IP network services. These features are not used by other features of the NetView program, so this list of instructions does not need to be followed sequentially. Again, steps in the instructions related to features that are not used can be skipped.

If you have other products or have created your own features that depend on NetView program features and VTAM or TCP/IP network services, you might consider adding instructions for those features to the instructions provided in this section so that error message traffic is minimized and, when VTAM or TCP/IP network services become available, these products and features can be made fully operational.

The information for configuring the NetView program and, on availability of VTAM or TCP/IP network services, starting features that rely on those network services is organized according to the following tasks.

- Enable towers that you want to use, but prevent initialization of their related functions until the required network services are available.

It is necessary to enable towers, in particular, before starting the NetView program, because towers cannot be dynamically enabled after the NetView program is started. For a number of features, tower selections determine definitions that are read during NetView program initialization.

- For each enabled tower or related function, configure the NetView program so that the related function of the enabled tower can be started when VTAM or TCP/IP network services are available.

Depending on how you detect and react to the availability of VTAM or TCP/IP network services, additional NetView program configuration might be required:

- NetView command security to accommodate any changes in tasks that execute commands that start NetView program features

Many commands that the NetView program normally runs to start towers and related functions have one of the following characteristics:

- Run on the primary program operator interface task (PPT)
- Scheduled by PPT to run on an autotask that is associated with the tower or related function, in addition to or instead of an initial command defined in a profile for such an autotask
- Run as an initial command on an autotask that is associated with the tower or related function, for example, the value of the IC keyword on a PROFILE statement in an operator profile used by the autotask

Any of these characteristics has the effect of running the commands either under the authority of the PPT or the authority of the autotasks that are used for the towers or related functions. You can adjust your command authority based on exactly how you choose to run the commands that start NetView program features.

- Operator profiles to set up initial commands that start NetView program features

One option for executing commands that are defined on `AUTOTASK.function.INITCMD` statements in the `CNMSTYLE` member and its included members would be to define them as initial commands in the operator profiles of autotasks that are started when VTAM or TCP/IP networking services become available.

- Automation statements for messages indicating the availability of VTAM or TCP/IP networking services that schedule commands that start NetView program features

VTAM issues the `IST020I` message when its initialization has completed. The TCP/IP program of the z/OS Communications Server issues the `EZAIN11I` message when its services are available.

The `DSITBL01` sample in the `DSIPARM` data set contains an example of the automation of the `IST020I` message to schedule the `CNMEVTAM CLIST`. The NetView program will also schedule the `CNMEVTAM CLIST` when the PPT successfully opens its Access Method Control Block (ACB) after NetView initialization has completed. Because its source code is provided with the NetView program, you can tailor the `CNMEVTAM CLIST` to perform other actions or you can provide your own procedures and automation to act when VTAM initialization completes.

The `CNMSEPTL` sample in the `DSIPARM` data set contains an example of the automation of the `EZAIN11I` message. This automation table entry is used by the Discovery Manager feature (DISCOVERY tower) of the NetView program. You could code automation to schedule your own procedures when this message is received. If you are using the Discovery Manager feature, ensure that your automation does not prevent the `EZAIN11I` message from also being processed by Discovery Manager.

The directions in this section assume that the NetView subsystem interface and the NetView program are configured to ensure that the `IST020I` and `EZAIN11I` messages are made available to NetView automation.

- Additional code to control the start of NetView program features that require VTAM or TCP/IP network services

The NetView program schedules the `CNMEVTAM CLIST` to run when VTAM initialization completes. There is no such procedure scheduled when the TCP/IP program of the z/OS Communications Server completes initialization. You might decide to create a procedure and use NetView automation to schedule it when the TCP/IP program completes initialization.

- Start tasks and run commands that make the selected NetView program features operational.

Preventing features that require network services from starting

You can change the definitions and code supplied in NetView samples. These changes minimize the number of error messages produced when NetView program features that require VTAM or TCP/IP network services are started before those network services are available. If all of the changes are implemented, the error messages are limited to these messages to indicate errors that are related to network services.

- `DSI064A OPENACB FAILED, ACBOFLG = X'02', ACBERROR = '5C', APPLID = xxxxxPPT, LUNAME = xxxxxPPT` where xxxxx is the NetView domain ID. This message is issued because the PPT cannot open its ACB due to the unavailability of VTAM.
- `BNH604I INIT REQUEST FAILED, ERRNO = nnn` by the execution of the `CNMERSYN CLIST` when there is a `RMTSYN` statement in the `CNMSTYLE` member and its included members that specifies a host name that must be resolved for IP-based (IPv4 or IPv6) communication between two instances of the NetView program. The NetView program runs the `CNMERSYN CLIST` every 30 minutes while there are host names to resolve.

Follow these steps to configure the NetView program so that features requiring VTAM or TCP/IP network services are not started before those services are available.

Assuming that the definitions of data services tasks (DSTs) and optional tasks (OPTs) are as they are displayed in the CNMSTYLE member and its included members that are included with the NetView program, add these statements to your CxxSTGEN member of the DSIPARM data set, where xx is the NV2I value with which you started the NetView program, to ensure that DSTs and OPTs requiring VTAM and TCP/IP network services, even indirectly, are not started:

```
TASK.CNMCALRT.INIT=N
TASK.CNMTAMEL.INIT=N
TASK.DSIHPDST.INIT=N
TASK.DSITCONT.INIT=N
TASK.DSIUDST.INIT=N
TASK.DSIXCFMT.INIT=N
TASK.DSI6DST.INIT=N
TASK.DUIDGHB.INIT=N
%>IF TOWER('GRAPHICS') THEN
TASK.DUIFSSCO.INIT=N
%>IF TOWER('AON') THEN DO;
TASK.EZLTSTS.INIT=N
TASK.EZLTLOG.INIT=N
* EZLTDDF is already INIT=N, since AON will eventually start the task itself
%> END
```

Precede these statements in your CxxSTGEN with this statement to enable data REXX.

```
/*%DATA REXX ...*/
```

An example is provided in the CNMSTGEN sample that is included with the NetView program. DSTs and OPTs that require VTAM or TCP/IP network services and that are not shown above are already defined with INIT=N in the CNMSTYLE member and its included members.

Although the DSIACBMT task is defined with INIT=Y and attempts to open the NetView main ACB (the application LU that matches the domain name), no message is issued, so you do not need to change the definition.

Add these statements to your CxxSTGEN member of the DSIPARM data set. These statements either prevent autotasks that are associated with towers and other NetView program features from being started by specifying LOGOFF for the initial command, or ensure that the NetView program starts the autotask and runs a command that does essentially nothing by specifying NODATA for the initial command. For each statement that specifies a tower or subtower in parentheses, you can omit that statement if you do not enable the tower or subtower.

```
AUTOTASK.?AUTOIP.INITCMD = NODATA
AUTOTASK.?Policy.INITCMD = NODATA
*
(AMI)AUTOTASK.?AMI.INITCMD = LOGOFF
*
(AON)AUTOTASK.?AON.INITCMD = NODATA
(AON.TCP.IDS)AUTOTASK.?IDS.INITCMD = LOGOFF
(AON.TCP.IDS)AUTOTASK.?IDS2.INITCMD = LOGOFF
*
(DISCOVERY)AUTOTASK.?COLTSK7.INITCMD = LOGOFF
(DISCOVERY.INTERFACES)AUTOTASK.?COLTSK5.INITCMD = LOGOFF
(DISCOVERY.TELNET)AUTOTASK.?COLTSK6.INITCMD = LOGOFF
*
(DVIPA)AUTOTASK.?DVIPSTAT.INITCMD = LOGOFF
(DVIPA)AUTOTASK.?DVIPAUTM.INITCMD = LOGOFF
(DVIPA)AUTOTASK.?COLTSK1.INITCMD = LOGOFF
(DVIPA.DVTAD)AUTOTASK.?COLTSK2.INITCMD = LOGOFF
(DVIPA.DVCONN)AUTOTASK.?COLTSK3.INITCMD = LOGOFF
(DVIPA.DVROUT)AUTOTASK.?COLTSK4.INITCMD = LOGOFF
*
(GRAPHICS)AUTOTASK.?NMCPOLICY.INITCMD = LOGOFF
(GRAPHICS)AUTOTASK.?EVENTMGR.INITCMD = LOGOFF
(GRAPHICS)AUTOTASK.?COSGATEWAY.INITCMD = LOGOFF
*
(IPMGT)AUTOTASK.?IPMGT.INITCMD = LOGOFF
(IPMGT.IDS)AUTOTASK.?IDS.INITCMD = LOGOFF
(IPMGT.IDS)AUTOTASK.?IDS2.INITCMD = LOGOFF
*
(MSM)AUTOTASK.?MSMALERTS.INITCMD = LOGOFF
```

```
(MSM)AUTOTASK.?VBVWORKER.INITCMD = LOGOFF
(MSM.OPN)AUTOTASK.?MSMOPENALERTS.INITCMD = LOGOFF
*
(NVSOA)AUTOTASK.?NVSOAPTSK.INITCMD = LOGOFF
*
(TEMA)AUTOTASK.?NALCLOP.INITCMD = LOGOFF
(TEMA)AUTOTASK.?EMAAUTOM.INITCMD = LOGOFF
(TEMA.HEALTH)AUTOTASK.?NAPOLTSK1.INITCMD = LOGOFF
(TEMA.CONNECT)AUTOTASK.?NAPOLTSK2.INITCMD = LOGOFF
(TEMA.CONINACT)AUTOTASK.?NAPOLTSK3.INITCMD = LOGOFF
(TEMA.SESSACT)AUTOTASK.?NAPOLTSK4.INITCMD = LOGOFF
*
```

If an autotask that is part of a tower or NetView program feature requiring VTAM or TCP/IP network services was not included, it is because the sample profile for the autotask either has no initial command or has an initial command that does not require the network services.

Add this statement to your CxxSTGEN member to ensure that Z NetView Enterprise Management Agent is not started.

```
(TEMA)INIT.EMAAUTO = NO
```

Other NetView program features have their INIT statements assigned the value No in the CNMSTYLE member.

```
(TCPIPCOLLECT.TCPCONN)INIT.TCPCONN = No
(TCPIPCOLLECT.PKTS)INIT.PKTS = No
(TCPIPCOLLECT.PKTS)INIT.OPKT = No
INIT.NRM = No
(DVIPA)Init.DVIPASTATS = No
```

The CNMSTYLE member and its included members contain examples of auxiliary commands, some of which start functions for enabled towers. To prevent these functions from being started, override the sample auxInitCmd statements by adding these statements to your CxxSTGEN member:

```
(NLDM)auxInitCmd.SNLDM = NODATA
(NPDA)auxInitCmd.SNPDA = NODATA
auxInitCmd.POLICY = NODATA
(DISCOVERY)auxInitCmd.ZDISC = NODATA
```

For each statement that specifies a tower in parentheses, you can omit that statement if you did not enable the tower.

If you have enabled the GRAPHICS tower, NetView program initialization schedules a STARTCNM GRAPHICS command to start tasks needed by that feature. The STARTCNM command, a synonym for CNME1015, whose source code is provided with the NetView program, does not check the availability of network services to ensure that they are available before starting the tasks associated with the GRAPHICS tower, so errors might result when the network services are unavailable.

One way to prevent the errors would be to replace the code in the DO group that follows this statement:

```
when parm1 = 'GRAPHICS' & TOWER('GRAPHICS')
```

For example, if you set a TCPIP_UP common global variable to Y when you determine that the TCP/IP program of the z/OS Communications Server completed initialization, you can choose to let the STARTCNM GRAPHICS command start its tasks only when both TCP/IP and VTAM are available. Do this by replacing the code in the DO group with coding similar to this example:

```
slist3.0 = 0
TCPIP_UP = ''
'GLOBALV GETC TCPIP_UP'
if ((TCPIP_UP = 'Y') &
    (VTAM() <> '')) then          /* TCP/IP and VTAM are up */
  do
    slist3.1 = 'START CNMTAMEL'   /* Might need VTAM, TCP/IP, or both*/
    slist3.2 = 'START DUIDGHB'    /* Needs TCP/IP */
    slist3.3 = 'START DUIFSSCO'   /* Needs neither VTAM nor TCP/IP directly */
    slist3.4 = 'AUTOTASK DUIFCSGW' /* Might need VTAM */
    slist3.0 = 4
    if (TOWER('GRAPHICS.SNATM')) then /* SNA Topology Manager subtower enabled */
```

```

do
    slist3.5 = 'AUTOTASK FLBTOPO' /* Needs VTAM */
    slist3.0 = 5
end
end

```

If you have enabled the AMI tower, NetView program initialization will insert automation definitions from the DSIAMIAT member of the DSIPARM data set, as well as its included members, after automation table definitions in the DSITBL01 member and its included members are loaded. Because some of the automation associated with the AMI tower attempts to schedule commands on the autotask defined for the AMI function and because an earlier AUTOTASK definition ensured that autotask was not started, an error message like this message might be displayed:

```
DW0032E AUTOMATION ACTION 'CNMETDAS' COULD NOT BE ROUTED TO TASK(S) AUTOAMI
```

One way to prevent the error message is to remove or comment out the following statements in the CNMSTASK member of the DSIPARM data set.

```

AUTOCMD.DSIAMIAT.list = *NONE*
AUTOCMD.DSIAMIAT.marker = AMI
AUTOCMD.DSIAMIAT.order = B

```

If the AMI tower is not enabled, it is not necessary to alter the AUTOCMD statements as they are not processed during NetView program initialization.

Preparing features that require network services

You can configure your system to start some NetView program features when VTAM or TCP/IP network services are available.

- For each `AUTOTASK.function.INITCMD` statement in the original CNMSTYLE member and its included members that had a command (that is, did not have the value `*NONE*`), ready that command for execution when the required network services become available.

One way to make the command ready for execution is to make each command an initial command in the operator profile of the applicable autotask. The command is executed when the autotask was started.

For example, if you defined `SECOPTS.OPERSEC = NETVPW` in your CNMSTYLE member and you provided the (MSM) `AUTOTASK.?MSMDEFAULT.INITCMD = LOGOFF` override for (MSM) `AUTOTASK.?MSMDEFAULT.INITCMD = INITTOPO` by following the instructions in the last section, then you can add `IC=INITTOPO` to the `PROFILE` statement in the operator profile used by the autotask defined as the value of the `function.autotask.MSMdefault` statement, which is `AUTOMSM` in the CNMSTYLE member.

The following table summarizes the information from the `AUTOTASK.function.INITCMD` statements in the CNMSTYLE member and its included members, listing commands for towers and related functions that rely on VTAM or TCP/IP network services. You can code the command as an initial command in the profile for the autotask that is defined for the function. This shows the name of the profile taken from the operator definition in the DSIOPF sample and its included members.

Table 22. Information from the AUTOTASK.function.INITCMD statements			
Command	Function	Autotask	Sample Profile
INITTOPO (FLCAINIT)	?MSMDEFAULT	AUTOMSM	FLCSPRFB
FKXEIPMI	?IPMGT	AUTIPMGT	DSIPROFC
NACMD (CNME8200)	?NALCLOP	AUTONALC	DSIPROFC
CNME8208 SOASVR1	?NVSOAPTSK	AUTONVSP	DSIPROFC

Several of the operators use the same sample profile, so an initial command can check the operator on which it is running and then choose the applicable command from the table.

If you coded `SECOPTS.OPERSEC = SAFDEF` in the `CNMSTYLE` member instead of updating the applicable members of the `DSIPRF` data set, update the `IC` keyword in the `NETVIEW` segment for each user profile in the System Authorization Facility (SAF) product you are using. You might have to update your command security configuration to accommodate the changes in the ways these commands are run.

- Comment out or remove the `POLICY` statement for each tower that you have not enabled. The `CNMSTYLE` member has a `POLICY` statement for each of the `AON` and `GRAPHICS` towers.

```
POLICY.AON = EZLCFG01
POLICY.GRAPHICS = DUIPOLCY
```

For each `POLICY` statement whose corresponding tower is not enabled, the NetView program issues the `EZL601I POLICY FILE xxxx NOT LOADED, yyy TOWER NOT ENABLED` message, where `xxxx` is the name of the policy member and `yyy` is the applicable tower.

- For the `auxInitCmd` definitions that were overridden by following the instructions in the previous section, provide some mechanism for executing the original commands when the required `VTAM` or `TCP/IP` services are available.

As an example, when `VTAM` issues the `IST020I` message to indicate that initialization is complete, a `REXX` procedure driven by NetView automation could run the following commands:

```
'STARTCNM NLDM'
'STARTCNM NPDA'
```

One method is to add these commands to a section of code in `CNMEVTAM`, whose source code is included with the NetView program. An automation table statement in the `DSITBL01` sample causes the NetView program to schedule the `CNMEVTAM MSG` command when the `IST020I` message is received.

The `CNME1015 (STARTCNM) CLIST` that is included with the NetView program ensures that the `NLDM` and `NPDA` towers are enabled before attempting to start the tasks associated with these towers.

- If you enabled the `AMI` tower, prepare the NetView automation for that tower.

The `CNMSTYLE` member and its included members provide for sample automation table `DSITBL01` and its included members to be loaded as the first automation table. If the `AMI` tower is enabled, then `DSIAMIAT` and its included members are inserted after that.

The `AUTOTBL MEMBER=DSIAMIAT INSERT AFTER=1` command can be used to insert `DSIAMIAT` and its included members after `DSITBL01` and its included members. If you want to insert your own automation tables before `DSIAMIAT` and its included members, you can change the applicable `AUTOTBL` keywords to suit your installation.

The `AUTOTBL` command can be added to the `CNMEVTAM CLIST` or a command procedure that you create and schedule to be driven when `VTAM` and `TCP/IP` network services become available.

- If `CNME1015` was changed to prevent the `STARTCNM GRAPHICS` command from starting tasks while the `TCP/IP` program was inactive, then some mechanisms should allow the `STARTCNM GRAPHICS` command to start tasks again when `TCP/IP` completes initialization.

The previous section described the addition of code to the `CNME1015` sample to prevent the starting of tasks that implement the functions of the `GRAPHICS` tower unless a `TCPIP_UP` common global variable had the value `Y` and the `VTAM()` function returned a non-null value, indicating that the NetView program detected that `VTAM` initialization was complete. You must also create a command procedure to assign the value `Y` to the `TCPIP_UP` common global variable when `TCP/IP` initialization completes, as indicated by the `EZAIN11I` message for `z/OS Communications Server TCP/IP`.

Starting features that require network services

This section provides two lists of instructions for starting NetView program features when `VTAM` or `TCP/IP` services become available. The first list must be completed in the order in which it is written because it includes the starts of NetView program features that depend on other features that are started first. When applicable, each entry in the list identifies the network services on which it depends and also

describes interrelationships with other NetView program features. These instructions might include notes about other NetView program features that do not have VTAM or TCP/IP network service dependencies if those other NetView program features are required by the NetView program feature.

The second list of instructions is for starting NetView program features that do not have interdependencies beyond the VTAM or TCP/IP network services. The NetView program features in this list can be started in any order.

Start instructions for interdependent features

The features listed in this section must be completed in the order listed because it includes the starts of NetView program features that depend on other features that are started first. When applicable, each entry in the list identifies the network services on which it depends and also describes interrelationships with other NetView program features. These instructions might include notes about other NetView program features that do not have VTAM or TCP/IP network service dependencies if those other NetView program features are required by the NetView program feature.

DSIHPDST, SNA LU6.2 High Performance Transport DST

Network service:

VTAM

Used by these NetView program features:

Remote operations (RMTCMD) over SNA
NetView bridge

Other product or user applications using the DSIHSNDS macro (assembly language), calling the CNMHSMU service routine or calling the CNMHSENDMU macro (PL/I), and calling the Cnmhsmu() function (C/C++) also require DSIHPDST.

Command to execute:

START TASK=DSIHPDST

DSI6DST, SNA Management Services (MS) Transport DST

Network Service:

VTAM

Used by these NetView program features:

NetView generic receiver
Alert forwarding over LU6.2
Focal point feature
Common operations services (RUNCMD over LU6.2)
NETCONV over LU6.2

Other product or user applications using the DSI6SNDS macro (assembly language), calling the CNMSMU service routine or calling the CNMSENDMU macro (PL/I), and calling the Cnmsmu() function (C/C++) also require DSI6DST.

Command to execute:

START TASK=DSI6DST

Note: If you have VTAMCP .USE = Yes coded in the CNMSTYLE member, then a NetView subsystem interface address space with program-to-program interface (PPI) function must be started.

DSIGDS, Network Product Support DST (including Common Operations Services, RUNCMD)

Network Service

VTAM

Used by these NetView program features:

MSM tower (communication with service points via SNA LU6.2)

GRAPHICS tower (communication with network management gateways using common operations services)

Other product or user applications using common operations services (RUNCMD) also require DSIGDS.

Command to execute:

START TASK=DSIGDS

DUIDGHB, resolver DST

Network Service

TCP/IP

Used by these NetView program features:

Remote operations (RMTCMD) over IP
NETCONV over IP

Command to execute:

START TASK=DUIDGHB

DSIRTTR, IP alert CP-MSU router DST

Network Service

TCP/IP

Used by these NetView program features:

NetView Resource Manager
CNMSMSIP sample

Command to execute:

START TASK=DSIRTTR

NetView generic receiver

Network Service

VTAM

Command to execute:

AUTOTASK OPID=DSINVGR

NetView bridge

Network Service:

VTAM

Commands to execute:

- For NetView bridge

```
AUTOTASK OPID=BRIGOPER
```

- For remote NetView bridge

```
AUTOTASK OPID=REMOPER
```

Note: NetView bridge also requires a NetView subsystem address space to be started with PPI function.

Autotask defined as the value of the function.autotask.policy statement

AUTOAON is also the autotask defined as the value of the function.autotask.AON and function.autotask.autoip statements in the CNMSTYLE member, and dependencies listed are based on these sample definitions.

Network Service:

VTAM or TCP/IP

Used by these NetView program features:

AON tower
DISCOVERY tower
IPMGT tower
DVIPA tower
Policy

Commands to execute:

AUTOTASK OPID=AUTOAON

Session monitor: NLDM tower**Network Service:**

VTAM

Used by these NetView program features:

TEMA tower

Command to execute:

STARTCNM NLDM

Remote operations, RMTCMD**Network Service:**

VTAM or TCP/IP

Used by these NetView program features:

AMI tower
AON tower
DISCOVERY tower
MSM tower
GRAPHICS tower
TIMER command
ROUTE stage of PIPE
ACTIVEACTIVE tower

Uses of RMTCMD by AMI and MSM are SNA LU6.2 (VTAM) only.

Command to execute:

START TASK=DSIUDST

Note: If you want the NetView program to be able to send or receive remote commands over IP, the RMTINIT.IP statement must be set to YES in the CNMSTYLE member. YES is the initial setting.

TCP/IP Data Collection, TCPIPCOLLECT tower**Network Service:**

TCP/IP

Used by these NetView program features:

DISCOVERY tower

Commands to execute:

Using the operator IDs defined in function .autotask examples for the TCPIPCOLLECT tower and its subtowers in the CNMSTYLE member and assuming a single TCP/IP stack named TCPIP

For the CONNSEC subtower (on z/OS 2.3 or later):

```
AUTOTASK OPID=AUTOCSEC  
CONNSEC DEFINE OPID=AUTOCSEC,TCPNAME=TCPIP  
CONNSEC START,TCPNAME=TCPIP
```

For the TCPCONN subtower:

```
AUTOTASK OPID=AUTOTCPC
TCPCONN DEFINE OPID=AUTOTCPC TCPNAME=TCPIP
TCPCONN START TCPNAME=TCPIP
```

For the PKTS subtower and packet trace data:

```
AUTOTASK OPID=AUTOPKTS
PKTS DEFINE OPID=AUTOPKTS,TCPNAME=TCPIP
PKTS START TCPNAME=TCPIP
```

For the PKTS subtower and OSA packet data

```
AUTOTASK OPID=AUTOOPKT
PKTS DEFINE OPID=AUTOOPKT,
      TCPNAME=TCPIP,
      PSOURCE=OSA
PKTS START PSOURCE=OSA
      TCPNAME=TCPIP
```

Hardware monitor, NPDA tower**Network Service:**

VTAM

Used by these NetView program features:

- AON tower
- MSM tower
- GRAPHICS tower
- DISCOVERY tower
- NetView Resource Manager
- SNA/6000 workstation alerts dynamic feature
- Event/Automation Service (E/AS)

Other product or user applications that depend on management services unit (MSU) automation, other than those applications that are provided by an SNMP trap automation task, or that run installation exits; (XITCI, XITVI, XITVO, XITVN) on the BNJDSERV task also require hardware monitor.

Command to execute:

STARTCNM NPDA

After the hardware monitor and DSICRTR task complete initialization, the NetView alert receiver (CNMCALRT task) can complete initialization.

DSIXCFMT, XCF services optional task**Network Service:**

TCP/IP

Used by these NetView program features:

DISCOVERY tower

Command to execute:

START TASK=DSIXCFMT

Policy and AON tower**Network Service:**

None directly for Policy; however, other NetView program features that are dependent on network services also require Policy to be initialized.

AON might require VTAM or TCP/IP through commands it uses, as well as other NetView program features on which it relies, such as remote operations (RMTCMD).

Used by these NetView program features:

For Policy, users are:

AON tower
DISCOVERY tower
GRAPHICS tower
IPMGT tower

Commands to execute:

If the GRAPHICS tower is enabled, then start the autotask named in the value of the `function.autotask.NMCpolicy` statement, DUIFPOLI in the CNMSTYLE member. The following command can be used to start DUIFPOLI:

```
AUTOTASK OPID=DUIFPOLI
```

If the GRAPHICS tower is enabled, ensure that RODM is started and that the GMFHS data model is loaded.

If RODM is started at this point in the process, the TEMA and DISCOVERY towers are enabled, and if the Z NetView Enterprise Management Agent is not running, you might receive error messages such as the following messages:

```
BNH638I 'WRITE TO NETVIEW ENTERPRISE MANAGEMENT AGENT DATA SPACE'  
REQUEST FAILED, RETURN CODE 808  
DW0050E FOR PROBLEM DEBUG: COMPONENT: TEMA MODULE: DSITEPFT RC: 808  
Stack_Configuration_and_Status
```

The types of data for which you receive these messages depend on which subtowers are enabled for the DISCOVERY and TEMA towers.

These instructions defer the start of the Z NetView Enterprise Management Agent to prevent requests for data that the NetView program has not yet collected.

Regardless of whether the GRAPHICS tower is enabled, run a command to initialize NetView policy on the autotask named in the value of the `function.autotask.Policy` statement in the CNMSTYLE member. This autotask was started in an earlier step. The following command can be used to initialize NetView policy:

```
EXCMD AUTOAON,EZLEANTL
```

If other products or user applications load their own policies, you might reconsider the position of policy initialization in this list of start instructions. If the AON tower is enabled, the EZLEANTL command also initializes AON.

IP Management, IPMGT tower**Network Service:**

TCP/IP

Commands to execute:

Assuming that AUTIPMGT is the operator ID defined on the (IPMGT) `FUNCTION.AUTOTASK.IPMGT` statement in the CNMSTYLE member, run this command:

```
AUTOTASK OPID=AUTIPMGT
```

If you did not define the FKXEIPMI command as an initial command in the operator profile, then schedule it to run. One method to start the command is as follows:

```
EXCMD AUTIPMGT,FKXEIPMI
```

GRAPHICS tower**Network Service:**

VTAM or TCP/IP

Used by these NetView program features:

DISCOVERY tower
MSM tower
NetView Resource Manager
AMI tower

Commands to execute:

To begin, the autotask assigned as the value of the `function.autotask.NMCpolicy` statement, `DUIFPOLI` in the `CNMSTYLE` member, was started in an earlier step in this list of instructions.

Start the autotask assigned as the value of the `function.autotask.eventMGR` statement, `DUIFEAUT` in the `CNMSTYLE` member, using a command like the following command:

```
AUTOTASK OPID=DUIFEAUT
```

Start the autotask assigned as the value of the `function.autotask.COS` gateway statement, `DUIFCSGW` in the `CNMSTYLE` member, using a command such as the following command:

```
AUTOTASK OPID=DUIFCSGW
```

To start the other components of the GRAPHICS tower, run the following command:

```
STARTCNM GRAPHICS
```

Network Resource Manager

Network Service:

VTAM or TCP/IP, depending on configured communication modes

Command to execute:

```
INITNRM (CNME8601)
```

Multisystem Manager, MSM tower

Network Service:

VTAM or TCP/IP

Commands to execute:

Start the autotask assigned as the value of the `function.autotask.MSMdefault` statement, `AUTOMSMD`, in the `CNMSTYLE` member:

```
AUTOTASK OPID=AUTOMSMD
```

Start the autotask assigned as the value of the `function.autotask.MSMalerts` statement, `AUTOSMSM`, in the `CNMSTYLE` member:

```
AUTOTASK OPID=AUTOSMSM
```

If you are using the OPN subtower of MSM, then start the autotask assigned as the value of the `function.autotask.MSMOpenAlerts` statement, `AUTOOPNA`, in the `CNMSTYLE` member:

```
AUTOTASK OPID=AUTOOPNA
```

Initialize MSM as follows:

- Specify the MSM autotask name (for example `AUTOMSMD`) on the `MSMdefault` statement:

```
(MSM)function.autotask.MSMdefault = AUTOMSMD
```

- Specify the initial command name as `INITTOPT`:

```
(MSM)AUTOTASK.?MSMdefault.InitCmd = INITTOPT
```

Event/Automation Service (E/AS)

Network Service:

TCP/IP

Used by these NetView program features:

AMI tower

Commands to execute:

To start the Event/Automation Service (E/AS), use the z/OS START (S) command, as follows.

```
S IHSAEVNT
```

(using the name of the sample E/AS start procedure that is provided with the NetView program)

The IHSAINIT sample that is provided with the NetView program is configured for E/AS to automatically start the alert adapter task (ALERTA), the message adapter task (MESSAGEA), and the event receiver task (EVENTRCV). If you must start them, you can use z/OS MODIFY (F) commands like these:

```
F IHSAEVNT,START TASK=ALERTA
F IHSAEVNT,START TASK=MESSAGEA
F IHSAEVNT,START TASK=EVENTRCV
```

To start the trap-to-alert conversion task (TRAPALRT), which might receive SNMP traps that can indicate changes to IP resources, use the z/OS MODIFY (F) command, as follows:

```
F IHSAEVNT,START TASK=TRAPALRT
```

If you are using the alert-to-trap conversion task (ALRTTRAP) to convert alerts to SNMP traps and pass them to the z/OS Communications Server SNMP agent for forwarding to SNMP managers, use the z/OS MODIFY (F) command to start the task as follows:

```
F IHSAEVNT,START TASK=ALRTTRAP
```

If you are using the confirmed alert adapter task (ALERTC) or confirmed message adapter task (MESSAGEC) to send EIF events confirmed at the application level, you can use z/OS MODIFY (F) commands to start the tasks as follows:

```
F IHSAEVNT,START TASK=ALERTC
F IHSAEVNT,START TASK=MESSAGEC
```

Application Management Instrumentation, AMI tower

Network Service:

VTAM

Commands to execute:

```
AUTOTASK OPID=AUTOAMI
AUTOTBL MEMBER=DSIAMIAI INSERT AFTER=1
INITAMI
```

If you are using the ACB monitoring feature, then run the following command:

```
INITAMON
```

If you are using the ACB monitoring feature, ensure that a NetView subsystem address space with PPI function is active. If you enabled the ACB monitoring feature that saves data to a DB2 database, ensure that the DSIDB2MT task is active.

Discovery Manager, DISCOVERY tower

Network Service:

VTAM or TCP/IP (through remote operations)

Used by these NetView program features:

TEMA tower

Commands to execute:

Start the autotask that is defined on the `function.autotask.COLTSK7 CNMSTYLE` statement. For example, for the `function.autotask.COLTSK7=AUTOCT7` statement, you can use the following command:

```
AUTOTASK OPID=AUTOCT7
```

If you are using the DISCOVERY INTERFACES subtower, start the autotask that is defined on the `function.autotask.COLTSK5 CNMSTYLE` statement. For example, for the `function.autotask.COLTSK5=AUTOCT5` statement, you can use the following command:

```
AUTOTASK OPID=AUTOCT5
```

If you are using the DISCOVERY TELNET subtower, start the autotask that is defined on the `function.autotask.COLTSK6 CNMSTYLE` statement. For example, for the `function.autotask.COLTSK6=AUTOCT6` statement, you can use the following command:

```
AUTOTASK OPID=AUTOCT6
```

The autotask that is defined on the `function.autotask.Policy` statement (for example, `AUTOAON`) was started in an earlier step in this list of instructions. Start the Discovery Manager. For example, you can use the following command:

```
EXCMD AUTOAON,CNMEERSC
```

Note: If you enabled a TEMA subtower that corresponds to an enabled DISCOVERY subtower and the Z NetView Enterprise Management Agent did not start or did not complete initialization, you might receive BNH638I and DWO050E messages similar to those described in [“Policy and AON tower” on page 154](#).

DVIPA tower**Network Service:**

TCP/IP

Used by these NetView program features:

TEMA tower

Commands to execute:

Start the autotask defined as the value of the `function.autotask.DVIPAUTM` statement, `DVIPAUTO` in the `CNMSTYLE` member, using a command such as the following command:

```
AUTOTASK OPID=DVIPAUTO
```

If you are going to log distributed DVIPA statistics, then start the autotask defined as the value of the `function.autotask.DVIPSTAT` statement, `DVIPSTAT` in the `CNMSTYLE` member, using a command such as the following command:

```
AUTOTASK OPID=DVIPSTAT
```

To initialize the distributed DVIPA statistics logging function, execute a command like the following command:

```
DVIPALOG START
```

Start the collection autotask defined as the value of the `function.autotask.COLTSK1` statement, `AUTOCT1` in the `CNMSTYLE` member, using a command such as the following command:

```
AUTOTASK OPID=AUTOCT1
```

If you are using the DVTAD subtower of DVIPA, start the collection autotask defined as the value of the function .autotask.COLTSK2 statement, AUTOCT2 in the CNMSTYLE member, using a command such as the following command:

```
AUTOTASK OPID=AUTOCT2
```

If you are using the DVCONN subtower of DVIPA, start the collection autotask defined as the value of the function .autotask.COLTSK3 statement, AUTOCT3 in the CNMSTYLE member, using a command such as the following command:

```
AUTOTASK OPID=AUTOCT3
```

If you are using the DVROUT subtower of DVIPA, start the collection autotask defined as the value of the function .autotask.COLTSK4 statement, AUTOCT4 in the CNMSTYLE member, using a command such as the following command:

```
AUTOTASK OPID=AUTOCT4
```

Note: If you have enabled a TEMA subtower that corresponds to an enabled DVIPA subtower and the Z NetView Enterprise Management Agent has not been started or has not completed initialization, you might receive BNH638I and DWO050E messages similar to those described in [“Policy and AON tower”](#) on page 154.

Z NetView Enterprise Management Agent, TEMA tower

Network Service:

VTAM

Used by these NetView program features:

VTAM or TCP/IP (communication between the Z NetView Enterprise Management Agent and a Tivoli Enterprise Management Server)

Commands to execute:

Start the autotask assigned as the value of the function.autotask.NALCLOP statement AUTONALC in the CNMSTYLE member, using a command such as the following command:

```
AUTOTASK OPID=AUTONALC
```

If you are using Tivoli Enterprise Monitoring Server on the same instance of z/OS on which the NetView program is running, start the Tivoli Enterprise Monitoring Server.

If you have defined Z NetView Enterprise Management Agent start parameters using the EMAAUTO.TEMA.procStr statement in the CNMSTYLE member, the NetView program attempts to use that to start the Z NetView Enterprise Management Agent. If you have not provided the EMAAUTO.TEMA.procStr statement or gave it the value *NONE*, then you must start the Z NetView Enterprise Management Agent.

Start the rest of the features of the tower by running a command on the autotask assigned as the value of the function.autotask.NALCLOP statement, AUTONALC in the CNMSTYLE member, as in the following example:

```
EXCMD AUTONALC,NACMD
```

NACMD starts the polling tasks for the following TEMA subtowers, if they are enabled:

- CONINACT
- CONNACT
- HEALTH
- SESSACT

NetView Discovery Library Adapter (DLA)

Network Service:

TCP/IP

Command to execute:

The autotask defined as the value of the `function.autotask.DLAAUTO` statement, `AUTO2` in the `CNMSTYLE` member, was allowed to start.

An example of a command to schedule the DLA is the following command:

```
EXCMD AUTO2 CHRON AT=(02:15:00),EVERY=(INTERVAL=()),  
DAYSWEEK=(MON),ID=CNMDLA,COMMAND='CNMDLAR'
```

The DLA expects to be able to start or use a NetView TSO server that can use TCP/IP services.

SNMP trap automation DST

Network Service:

TCP/IP

Used by the NetView program feature:

DVIPA

Command to execute:

If you have configured an SNMP trap automation task named `CNMTRAPD` but have not added `TASK` statements to your `CNMSTYLE` member and its included members, you can start the task using this command:

```
START TASK=CNMTRAPD,MOD=DSIZDST,MEM=CNMTRAPI,PRI=5
```

DSICORSV, Correlation service optional task

Network Service:

TCP/IP

Command to execute:

`START TASK=DSICORSV`

Although no tower or related function relies on this feature, it does connect to a correlation engine server running outside of the NetView program.

DSIAL2WS, SNA/6000 workstation alerts dynamic DST

Network Service:

VTAM

Command to execute:

`START TASK=DSIAL2WS`

Start instructions for independent features

The NetView program features that are included in this section do not have interdependencies beyond the VTAM or TCP/IP network services. The NetView program features in this list can be started in any order.

Status monitor (xxxxxVMT optional task, in which xxxxx is the NetView domain ID)

Network service:

VTAM

Command to execute:

`START TASK=xxxxxVMT` (where xxxxx is the NetView domain ID)

ALIASAPL, Alias name translation DST

Network service:

VTAM

Command to execute:

START TASK=ALIASAPL

DSIIPLOG, SYSLOGD server optional task**Network service:**

TCP/IP

Command to execute:

START TASK=DSIIPLOG

DSIKREM, Central Site Control Facility (CSCF) DST**Network service:**

VTAM

Command to execute:

START TASK=DSIKREM

DSIRSH, RSH server optional task**Network service:**

TCP/IP

Command to execute:

START TASK=DSIRSH

DSIRXEXC, REXEC server optional task**Network service:**

TCP/IP

Command to execute:

START TASK=DSIRXEXC

DSIWBTSK, NetView web application DST**Network service:**

TCP/IP

Command to execute:

START TASK=DSIWBTSK

VPDTASK, Vital product data optional task**Network service:**

VTAM

Command to execute:

START TASK=VPDTASK

Web Services Gateway, NVSOA tower**Network service:**

TCP/IP

Command to execute:

Using the operator ID defined on the (NVSOA) `function.autotask.NVSOAPTSK` statement in the CNMSTYLE member, execute this command:

```
AUTOTASK OPID=AUTONVSP
```

IP Port Monitoring**Network service:**

TCP/IP

Commands to execute:

Start the autotask defined as the value of the function.autotask.IPPORTMON statement, AUTOPORT in the CNMSTYLE member. An example of a command for starting the autotask follows:

```
AUTOTASK OPID=AUTOPORT
```

To start IP port monitoring, execute a command such as the following example:

```
EXCMD AUTOPORT,CNMETSTL
```

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